Topics in Causal Inference

S&DS 6165, Yale University, Fall 2026

Johan Ugander

Table of contents

Topics in Causal Inference **3**
Course description . . . 3
Schedule . . . 3

How These Notes Are Made **5**
What guides a lecture . . . 5
Drafting and revision process . . . 5
Responsibility and limitations . . . 6

L01: Introduction **7**

A course about choices **8**

The comparison hidden in every unit **9**

What the two-column table assumes **11**

Neyman's move: model the assignment, not the outcomes **12**

A randomized design is a distribution **13**
Complete randomization . . . 13
Bernoulli randomization . . . 14

Why randomization identifies an average effect **15**

Estimand, estimator, estimate—and population **17**

What to carry forward **18**

Advanced material **19**
Potential outcomes as missing data . . . 19
Random arm sizes under Bernoulli assignment . . . 19
Exact randomization variance under complete randomization . . . 20
Center the science table and recall the design moments . . . 20
Variance of each arm mean . . . 21
Covariance between the arm means . . . 22
Assemble the pieces . . . 23

Further reading **25**

References **26**

L02: Basic Estimators and Experimental Pitfalls **27**

From an unbiased estimator to an uncertainty statement **28**

A four-unit experiment by hand **30**

Inverse-probability weighting **31**
The Bernoulli subtlety . . . 32

Hájek normalization **33**
A first variance comparison: HT versus Hájek . . . 33
Ratio bias from normalization . . . 35
HT versus Hájek in a finite population . . . 35

Normalization is a continuum **37**

Regression adjustment **40**
GREG from a known baseline total . . . 41
CUPED: choosing the control coefficient . . . 42

Calibration: make the weighted arm reproduce its target **45**

What to carry forward **48**

Advanced material **49**
Practical experimentation: errors larger than the standard error . . . 49
The propensity-score paradox . . . 49
A short history of inverse-probability normalization . . . 50
Choosing estimators across families of experiments . . . 50

Further reading **51**

References **52**

L03: Experimental Design, Blocking, Balancing **53**

Ten coin flips disguised as one thousand **54**

Blocking: randomize among comparable units **56**
Matched pairs . . . 57

Design balance versus analysis adjustment **59**

AIPW: prediction plus randomized correction **60**

Pair matching in cluster-randomized experiments **62**

Rerandomization **64**
What rerandomization buys . . . 65
The robustness tradeoff . . . 65

What to carry forward **66**

Advanced material **67**
Gram–Schmidt walk designs . . . 67
Correlated cluster designs without rich covariates . . . 68
Optimization-based designs and the value of randomness . . . 68
Efron's biased coin . . . 68
D-, A-, and E-optimal design . . . 69
Staggered rollouts beyond SUTVA . . . 69

Further reading **70**

References **71**

L04: Positivity, Trimming, Target Populations, and Partial Identification **72**

A randomized experiment can still lack a comparison **73**

Positivity in a finite randomized design **74**

What the inverse probability is doing **76**
Weak probabilities enter the variance . . . 77

Four responses, four different claims **78**

Clipping: regularize an estimator **79**

Trimming: change the target explicitly **81**
The superpopulation counterpart . . . 82
A numerical precision–relevance tradeoff . . . 83

Equipoise: change the target smoothly **84**
What changes in an observational study? . . . 85

When the target cannot be changed **86**

Manski's accounting: bound what the design cannot expose **87**
The familiar observational form . . . 88

Lee bounds: use randomized selection structure **89**

Smoothness: controlled borrowing across support **91**
A one-dimensional example . . . 92

Identification is not inference **93**

What to carry forward **94**

Advanced material **95**
Why a variance-optimal trimming rule has a threshold form . . . 95
Adaptive trimming changes the inferential problem . . . 96
Extrapolation as a bias–bias–variance tradeoff . . . 96

Further reading **98**

References **99**

L05: Interference **106**
A treatment changes an environment . . . 106
What SUTVA lets us leave out . . . 106
Four mechanisms that create interference . . . 108
Which treatment effect? . . . 109
Universal adoption and individual treatment . . . 109
Exposure mappings compress the science table . . . 110
Moving SUTVA to clusters . . . 111
From observed exposures to an estimator . . . 112
The inverse probability is now an exposure probability . . . 112
Positivity and joint observability . . . 113
Testing whether interference is present . . . 113
Make the null sharp on a conditional experiment . . . 113
Randomize the randomization mechanism . . . 114
Use randomized increases in treatment allocation . . . 115
Designing for informative exposures . . . 116
Graph cluster randomization . . . 116
Randomize the clustering too . . . 117
A clustering is optimal for an objective . . . 117
Advanced material: designs built around incompatible exposures . . . 119
Returning to the deployment question . . . 120
Further reading . . . 120
References . . . 121

L06: Estimation with Structural Models **128**
Learning about a rollout before everyone is treated . . . 128
A model of interactions, rather than a model of identical units . . . 129
Setup and target . . . 129
Every binary response surface has a polynomial representation . . . 129
Low order is the substantive restriction . . . 130
The global effect is a sum of coefficients . . . 131
Start with additive spillovers . . . 131
Randomization isolates a neighbor's contribution . . . 131
An explicit four-assignment calculation . . . 132
From coefficient estimation to the pseudoinverse . . . 132
The known design replaces a missing regression sample . . . 132
What singularity means . . . 134
The geometry behind the weights . . . 135
Higher-order interactions under independent assignment . . . 135
Centering provides an orthogonal basis . . . 135
A pair interaction and a revealing cancellation . . . 136
Precision requires information about shared assignments . . . 137
Interaction order controls a local difficulty . . . 137
Shared neighbors generate covariance . . . 137

Inference is a further problem . . . 138
Changing the design while retaining the model . . . 138
Covariate adjustment should target the estimator's variance . . . 139
Other structural information: monotone interference . . . 140
Restricting directions instead of interaction order . . . 140
What monotonicity can provide . . . 140
Advanced material: model error and the cost of extrapolation . . . 141
An exact bias decomposition . . . 141
Choosing an order and extending the experiment . . . 142
Further reading . . . 142
References . . . 143

L07: Policy Evaluation and Panel Data **150**
Evaluate a rule, not just a treatment label . . . 150
Potential outcomes for whole seed sets . . . 151
A four-person example . . . 152
Evaluate a policy using the experiment we actually ran . . . 153
The design and the target need not be the same distribution . . . 153
Why retain seed identities when strategies were randomized? . . . 154
Normalization, support, and uncertainty . . . 155
Policy evaluation when we observe a panel . . . 156
What a before-and-after comparison needs . . . 157
Low rank means shared patterns with different responses . . . 158
Which missing entries are determined? . . . 159
Estimating a mean matrix from noisy untreated cells . . . 160
What the penalty does computationally . . . 160
Turning predictions into an effect estimate . . . 161
Choosing complexity and checking the comparison . . . 161
Other ways to construct the missing comparison . . . 162
Weighted differences rather than an entire completed matrix . . . 162
What must be checked before calling the result causal? . . . 163
Synthesis . . . 164
Further reading . . . 164
References . . . 165

L08: Advanced Estimation Topics **172**

L09: AI for Causal Inference **173**

Notation **174**

Course notation **175**
Units, samples, and populations . . . 175
Treatments, assignments, and exposures . . . 175
Potential outcomes and observed outcomes . . . 175
Estimands and estimators . . . 176
Assignment and sampling probabilities . . . 176
Finite-population and superpopulation operators . . . 176

Networks and interference . . . 177
Stochastic policy evaluation . . . 177
Panel data and matrices . . . 178
Probability, convergence, and norms . . . 178

References **179**

Problem Sets **186**

Projects **187**
Project 1 . . . 187
Possible papers . . . 187
Project 2 . . . 188
Possible papers . . . 189
Project 3: Open-ended research question . . . 190

Topics in Causal Inference

S&DS 6165, Yale University, Fall 2026

Course description

This course explores modern topics in causal inference with a focus on randomized experiments and experimental design. The course develops the potential outcomes framework and builds toward advanced topics including covariate balancing, overlap conditions and positivity violations, causal inference under interference, and policy evaluation. The emphasis is on both theoretical foundations and applied practice. Causal inference from observational data is de-emphasized. Students will complete presentation and project work including both applied and theoretical replication studies.

Instructor: Prof. Johan Ugander · **Meetings:** Wednesdays, 9:25a–11:20a

Schedule

Individual lectures are under development through an instructor-directed, AI-assisted process. How these notes are made. The guest lecture and presentation weeks are shown but do not have lecture-note pages; break weeks are shown as non-meeting weeks.

Lecture	Date	Topic
L01	Sep 2	Introduction
L02	Sep 9	Basic Estimators and Experimental Pitfalls
—	Sep 16	**Guest lecture — Prof. Harsh Parikh**
L03	Sep 23	Experimental Design, Blocking, Balancing
—	Sep 30	**Presentation I**
L04	Oct 7	Positivity, Trimming, Target Populations, and Partial Identification
L05	Oct 14	Interference
—	Oct 21	**Fall break — no class**
L06	Oct 28	Estimation with Structural Models

Lecture	Date	Topic
—	Nov 4	**Presentation II**
L07	Nov 11	Policy Evaluation and Panel Data
L08	Nov 18	Advanced Estimation Topics
—	Nov 25	**Thanksgiving break — no class**
L09	Dec 2	AI for Causal Inference
—	Dec 9	**Presentation III**

How These Notes Are Made

These lecture notes for the Yale S&DS graduate course "Topics in Causal Inference" are being developed through an instructor-directed, AI-assisted workflow. This page describes the generation process.

What guides a lecture

Every lecture begins with a public `brief.md`, written by me. The brief is a concise content outline: it records the lecture's scope and emphases, a minimal literature list, and any examples, demonstrations, constraints, or open questions I choose to specify. It does not prescribe the section structure of the finished notes or pre-write assessments of the cited literature. For transparency, each lecture page links to a minimally formatted HTML rendering of the exact `brief.md` associated with that lecture.

Each lecture folder begins with only two files: the brief and a canonical `index.qmd` containing the lecture page. Assets are added only when the lecture actually needs them.

The brief is not the only instruction supplied during drafting. Four course-wide documents establish shared expectations:

- `LECTURE_SPEC.md` defines the audience, mathematical depth, and expository style.
- `WRITING_STYLE.md` distills recurring principles from my research writing and adapts them to lecture exposition, with particular emphasis on problem-driven explanation and technical intuition.
- `NOTATION.md` records notation that should remain consistent across lectures.
- `DEMO_SPEC.md` requires interactive demonstrations to have accessible, deterministic fallbacks for PDF readers.

These public files expose the durable course-specific instructions. They do not reproduce the model provider's system instructions or chain of thought during drafting. The current plan uses GPT 5.6 Sol (Extra High), with plans to experiment with the possible benefits of 5.6 Sol (Pro).

Drafting and revision process

The intended workflow is:

1. **Scope.** The instructor drafts a concise content outline and addresses lecture-specific scope, emphasis, and constraints.

2. **Literature.** The brief retains concise citations rather than extensive source commentary. Bibliographic details and technical claims are checked rather than inferred from memory. Primary papers are preferred for foundational claims and historical attributions; reviews and textbooks are used selectively when their synthesis or clarification is especially useful. Historical background and additional pointers that would interrupt the main lecture path are collected in a concise **Further reading** section.
3. **Initial generation.** OpenAI GPT-5.6 Sol, normally in either the Extra High or Pro configuration, writes an initial version directly into the lecture's `index.qmd`. Its metadata records the model and date and sets `generation-locked: true`. This flag prevents accidental wholesale regeneration; it does not make the file read-only. The model and configuration may change as the course develops, and the metadata will identify the system actually used.
4. **Brief-driven iteration.** During early drafting, the instructor may review the result, revise the brief, and explicitly request another full generation. That request temporarily clears the generation lock; the newly generated version is then locked again. A changed brief by itself never overwrites lecture text.
5. **Instructor revision.** Once the lecture's structure is satisfactory, the instructor edits the canonical lecture file directly and may substantially revise the generated material. After those edits begin, revisions are targeted so that instructor-authored text is not overwritten.
6. **Publication.** Quarto renders the same source to a web page and to PDF. Interactive web demonstrations are checked alongside their static PDF fallbacks.

Responsibility and limitations

AI assistance can accelerate outlining, exposition, examples, code, and editorial checking. It does not establish authority, verify a citation merely by naming it, or replace instructor judgment. The instructor remains responsible for what appears in the published notes.

The linked brief is best understood as the lecture's starting specification, not a complete audit log. A lecture may change after source review, classroom experience, or correction of an error. Substantive departures from the brief reflect the instructor's continuing authorship of the notes.

L01: Introduction

Potential outcomes and design-based causal inference

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Separate the causal target from choices made during design and analysis.
- Use a science table to define potential outcomes, SUTVA, and the finite-population average treatment effect.
- Define randomized designs and show how complete randomization identifies average potential outcomes and yields an unbiased estimator.

A course about choices

Causal inference begins with a comparison: what would happen under one course of action rather than another? Data do not answer that question on their own. We must decide what comparison matters, how treatment will be assigned, and how the resulting observations will be analyzed.

This course emphasizes **design-based inference**. We will treat the units and their potential outcomes as fixed and locate randomness in an assignment mechanism that we know because we chose it. That perspective keeps two kinds of choices visibly separate:

- **Design choices** determine the probability distribution used to assign treatment. Complete randomization, blocking, rerandomization, and clustered assignment are different designs.
- **Analysis choices** determine how the realized data are turned into an estimate. Difference in means, inverse-probability weighting, regression adjustment, and calibration are different analyses.

The causal target—the **estimand**—usually comes first and remains fixed while we compare designs and analyses. Sometimes the target itself changes. Trimming units with poor overlap in L04, for example, may improve the statistical problem by changing the population about which we ask the question. That is not merely a better estimator of the old target.

This distinction is a recurring discipline: **what is the question, what was randomized, and what was calculated?**

The comparison hidden in every unit

Consider six plots in an agricultural experiment. Label the plots $i = 1, \dots, 6$, and let the treatment label z equal 1 for fertilizer and 0 for control. For plot i, define $Y_i(z)$ to be the yield that would be observed if that plot were assigned treatment z. The plot therefore has two potential outcomes, $Y_i(1)$ and $Y_i(0)$, and their difference

$$\tau_i = Y_i(1) - Y_i(0)$$

is its **individual treatment effect**. Before discussing how treatment is assigned, imagine that both potential outcomes are written down for every plot.

Plot i	Yield under control $Y_i(0)$	Yield under treatment $Y_i(1)$	Individual effect $\tau_i = Y_i(1) - Y_i(0)$
1	10	13	3
2	12	12	0
3	8	12	4
4	15	14	-1
5	9	11	2
6	11	14	3

This is often called the **science table**. It is not a table we observe. Rather, it represents the fixed causal quantities that an experiment is meant to teach us about. To connect the table to observed data, we require a well-defined treatment, no interference between units, and consistency between assignment and observation; these requirements are stated more precisely below.

Let $Z_i \in \{0, 1\}$ denote plot i's actual assignment. We use an uppercase Z_i because, before randomization, the assignment is a random variable. Once the design draws its assignment, only

$$Y_i^{\text{obs}} = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$$

is observed. If $Z_i = 1$, the formula selects $Y_i(1)$; if $Z_i = 0$, it selects $Y_i(0)$. The other entry in the row remains missing. This is the fundamental obstacle: the unit-level comparison is scientifically meaningful but cannot be observed jointly for the same unit.

The table also forces a substantive question. Why summarize the six effects by their average? The average effect is often a natural policy or welfare summary, and randomization makes it especially tractable. But the table may also contain harm, unequal gains, or effects concentrated in a subgroup. An average is a target, not a law of nature.

For this course, the default target is the **finite-population average treatment effect**

$$\tau = \frac{1}{N}\sum_{i=1}^{N} \tau_i = \overline{Y}(1) - \overline{Y}(0), \tag{0.1}$$

where N is the number of experimental units and

$$\overline{Y}(z) = \frac{1}{N}\sum_{i=1}^{N} Y_i(z)$$

is the finite-population mean under treatment level z. In the table, $\overline{Y}(1) = 76/6$, $\overline{Y}(0) = 65/6$, and therefore $\tau = 11/6$. These are features of these six plots; at this point no larger population or probability model for their yields has been invoked.

What the two-column table assumes

Writing $Y_i(z)$ assumes that the single label z contains all the treatment information needed to specify unit i's potential outcome. This bundles two ideas commonly called the **stable unit treatment value assumption** (SUTVA) (Rubin 1980):

1. **No multiple versions:** the treatment labeled z has no causally distinct, unrecorded versions. If several fertilizer formulations or application protocols matter for yield, the binary label must be refined or the versions must be declared irrelevant to the question.
2. **No interference:** unit i's outcome does not depend on other units' assignments. If fertilizer runs off from a neighboring plot, $Y_i(z)$ is generally not enough; the neighbors' assignments also matter.

Consistency then links the potential outcome to observation: if the realized assignment is $Z_i = z$, then $Y_i^{\text{obs}} = Y_i(z)$. SUTVA is not a metaphysical truth. It is a simplification whose credibility depends on how units and treatments are defined. L05 will replace $Y_i(z)$ with potential outcomes indexed by richer assignment or exposure patterns. The corresponding science table becomes much larger, but the logic—define the possible interventions before asking what is identified—survives.

Neyman's move: model the assignment, not the outcomes

One response to the missing entry is to model it. We might fit a linear model, posit a distribution for outcomes, or impute each unobserved potential outcome. Such strategies can be useful, but their conclusions depend on the outcome model.

The design-based alternative starts elsewhere. Hold the entire science table fixed and introduce a known random mechanism that reveals selected entries. The probability calculation then concerns which assignment vector was drawn, not how the potential outcomes were generated.

Fisher made randomization central to experimental inference, while Neyman's agricultural experiments and survey-sampling work developed the finite-population, design-based argument (Fisher 1935; Splawa-Neyman, Dabrowska, and Speed 1990; Neyman 1934). In survey sampling, known inclusion probabilities let us learn about a fixed collection of units (Horvitz and Thompson 1952). In a randomized experiment, known treatment probabilities let us learn about a fixed collection of causal contrasts. Rubin's later formulation made the potential-outcomes language central across randomized and observational studies (Rubin 1974, 2005).

The contrast with an outcome model is therefore not "statistics versus no statistics." It is a choice about what supplies the probability statement:

design-based:	condition on the potential outcomes and average over $\mathbf{Z}$,
model-based:	posit a probability law for potential outcomes or observed responses, possibly conditional on treatment and pretreatment covariates.

In the first line, repeated-assignment statements concern the same plots and the same science table under different possible treatment assignments. In the second, probability also describes how outcomes vary across units or hypothetical samples. We will often combine the two—using regression, for example, to improve precision—but the assignment mechanism will remain the foundation for causal comparisons in randomized experiments.

A randomized design is a distribution

Let

$$\mathbf{Z} = (Z_1, \dots, Z_N) \in \{0,1\}^N$$

be the assignment vector. Uppercase $\mathbf{Z}$ denotes the random vector before the design is realized; lowercase $\mathbf{z}$ denotes one particular assignment it could produce. A **randomized experimental design** is a probability distribution

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z})$$

over possible assignment vectors $\mathbf{z}$. The subscript Z emphasizes that this probability comes from the assignment mechanism, with the potential outcomes held fixed. The support

$$\mathcal{Z} = \{\mathbf{z} : \mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) > 0\}$$

records exactly which assignments the design can produce.

This definition matters. Saying "the study is randomized" is incomplete until we know both the support and the probabilities placed on it. Those choices determine which causal comparisons are possible and how estimators behave under repeated assignment.

Complete randomization

Under complete randomization, the experimenter fixes the treated count N_1 in advance and therefore also fixes the control count $N_0 = N - N_1$. Assume $0 < N_1 < N$, so both arms are nonempty. Every vector containing exactly N_1 ones is equally likely:

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) = \begin{cases} \binom{N}{N_1}^{-1}, & \sum_{i=1}^N z_i = N_1, \\ 0, & \text{otherwise.} \end{cases}$$

There are $\binom{N}{N_1}$ ways to choose the treated units, which explains the probability in the first line. Each unit has marginal treatment probability N_1/N, but the assignments are not independent. If one unit is treated, one fewer treatment slot remains. In fact, for $i \neq j$,

$$\mathbb{E}_Z(Z_i Z_j) = \mathbb{P}_Z(Z_i = 1, Z_j = 1) = \frac{N_1(N_1 - 1)}{N(N-1)},$$

and therefore

$$\mathrm{Cov}_Z(Z_i, Z_j) = -\frac{(N_1/N)(N_0/N)}{N-1}.$$

That weak negative dependence is precisely what fixes the arm sizes.

Bernoulli randomization

Under an independent Bernoulli design with common probability p,

$$Z_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p), \qquad \mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) = p^{\sum_i z_i}(1-p)^{N-\sum_i z_i}.$$

Now each unit receives an independent random draw, so the treated count $N_1 = \sum_i Z_i$ is random rather than fixed by the design. Throughout the course, N_1 and N_0 denote realized arm sizes: under complete randomization their values are fixed in advance, while under Bernoulli randomization they vary across possible assignments. The Bernoulli design is operationally simple and sometimes convenient for sequential enrollment. It can also produce imbalance. The probability of an empty treatment arm is $(1-p)^N$, and the probability of an empty control arm is p^N. We return to that complication in L02.

> **i** Design comes before positivity
>
> Write $\pi_i = \mathbb{P}_Z(Z_i = 1)$ for unit i's marginal treatment probability. **Positivity** is a property of the assignment mechanism relative to the causal comparison. For both arm-specific finite-population means to be learned without extrapolation, every unit's two relevant potential outcomes must each have a chance to be observed. Thus, for the treatment-control comparison,
>
> $$0 < \pi_i < 1 \qquad \text{for every } i.$$
>
> Under nondegenerate complete randomization, $\pi_i = N_1/N$; under the common-probability Bernoulli design, $\pi_i = p$. Notice that complete randomization satisfies this unit-level requirement even though it assigns probability zero to every vector with the wrong treated count. For more complicated exposures, the relevant requirement concerns the exposure contrasts defining the estimand—not an abstract rule that every assignment vector must be possible.

Why randomization identifies an average effect

Return to complete randomization. Here N_1 and N_0 are fixed, positive constants chosen by the design. The observed treated and control means are

$$\overline{Y}_1^{\text{obs}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i^{\text{obs}}, \qquad \overline{Y}_0^{\text{obs}} = \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i^{\text{obs}}.$$

Consistency lets us rewrite these observable quantities in terms of the science table:

$$\overline{Y}_1^{\text{obs}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i(1), \qquad \overline{Y}_0^{\text{obs}} = \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i(0).$$

For example, when $Z_i = 1$, the factor Z_i selects $Y_i(1)$; when $Z_i = 0$, that unit contributes zero to the treated total. Under complete randomization, $\mathbb{E}_Z(Z_i) = N_1/N$. The notation $\mathbb{E}_Z$ means that we average only over assignments produced by the design, holding all potential outcomes fixed. Because the arm sizes are fixed, their denominators can be taken outside the expectation:

$$\mathbb{E}_Z\left(\overline{Y}_1^{\text{obs}}\right) = \frac{1}{N_1}\sum_i \mathbb{E}_Z(Z_i)Y_i(1) = \overline{Y}(1),$$
$$\mathbb{E}_Z\left(\overline{Y}_0^{\text{obs}}\right) = \frac{1}{N_0}\sum_i \mathbb{E}_Z(1-Z_i)Y_i(0) = \overline{Y}(0).$$

The cancellation is simple but important: each unit enters the treated average with probability N_1/N, and division by the fixed treated count turns those inclusion probabilities into equal weight $1/N$ on every $Y_i(1)$. The same argument applies to control.

These equalities are the design-based **identification** step: each fixed arm-specific mean is represented as the expectation of an observable random variable under the known assignment distribution. They do not identify either missing potential outcome for an individual unit. They also show why the distinction between complete and Bernoulli randomization matters. Under Bernoulli assignment, the denominators N_1 and N_0 are random and may equal zero, so one cannot repeat this calculation by simply moving them outside the expectation. L02 handles that case directly.

The corresponding **estimator** in the realized experiment is the difference in observed means,

$$\hat{\tau}_{\text{DM}} = \overline{Y}_1^{\text{obs}} - \overline{Y}_0^{\text{obs}}. \tag{0.1}$$

Linearity of design expectation then establishes its design unbiasedness:

$$\mathbb{E}_Z(\hat{\tau}_{\text{DM}}) = \tau. \tag{0.2}$$

The equality is a fact about the design. It does not require linearity, normal outcomes, constant effects, or a random sample from a larger population. It says that if we repeatedly reassigned these same N units while leaving their science table unchanged, the estimator would average to the fixed finite-population effect.

Randomization does not literally reveal both potential outcomes for any unit. It makes the observed treatment group representative of the finite population in expectation, and likewise for the control group. Identification concerns the average; unbiasedness concerns the repeated-assignment behavior of the estimator used to estimate it.

Estimand, estimator, estimate—and population

Three words prevent a great deal of confusion:

- The **estimand** is the fixed target, here τ in Equation 0.1.
- The **estimator** is the rule that maps the random assignment and resulting observed outcomes to $\hat{\tau}_{\mathrm{DM}}$. Before assignment, its value is random under the design.
- The **estimate** is the numerical value produced by the realized assignment and outcomes.

The target population must be stated just as carefully. Our τ averages over the N experimental units. It is a **finite-population** estimand. Even if those units were sampled from a larger population, treatment randomization by itself justifies the repeated-assignment statement for this realized collection of units.

We might instead posit a population distribution P for a generic unit's potential outcomes and target the **superpopulation** ATE

$$\tau_P = \mathbb{E}_P\{Y(1) - Y(0)\}.$$

Here $\mathbb{E}_P$ averages over units generated by P, rather than over alternative treatment assignments for the same units. Moving from the realized finite-population ATE to τ_P requires a sampling, transport, or superpopulation argument in addition to treatment randomization. A conditional average treatment effect,

$$\tau(x) = \mathbb{E}_P\{Y(1) - Y(0) \mid X = x\},$$

targets units with a particular value x of pretreatment covariates X. Under interference, one may instead define a global average treatment effect comparing two population-wide assignment regimes—for example, the average outcome if everyone were assigned treatment versus the average outcome if everyone were assigned control. These targets answer different questions; random assignment in one experimental sample does not automatically identify all of them.

What to carry forward

The science table states the causal comparisons. The design states which entries can be revealed and with what probabilities. The analysis states how the revealed entries are combined. Under complete randomization, the difference in observed means is design-unbiased for the finite-population average treatment effect. Every later refinement in the course should be locatable in that three-part structure.

Advanced material

Potential outcomes as missing data

Potential outcomes invite a missing-data reading: after assignment, one entry in each row of the science table is observed and the other is missing. More explicitly, the indicator that $Y_i(1)$ is observed is Z_i, while the indicator that $Y_i(0)$ is observed is $1 - Z_i$. Rubin's missing-data framework distinguishes mechanisms according to whether missingness depends on observed or unobserved values (Rubin 1976; Seaman et al. 2013). Very roughly:

- **MCAR**: missingness is independent of observed and missing values.
- **MAR**: conditional on observed information, missingness does not depend on the missing values.
- **MNAR**: dependence on missing values remains after conditioning.

Under complete randomization, assignment probabilities do not depend on the values in the fixed science table. In that limited, design-based sense, the missingness pattern for each treatment-specific outcome resembles MCAR: which entry is revealed is randomized independently of the values themselves. The assignment indicators remain dependent across units because the treated count is fixed, so the analogy should not be read as a claim that every missingness indicator is independent. More importantly, causal inference requires the interventions and their potential outcomes to be well defined; generic missing-data notation does not supply that scientific content.

Random arm sizes under Bernoulli assignment

Under i.i.d. Bernoulli assignment, $N_1 \sim \text{Binomial}(N, p)$. The usual difference in means is undefined when $N_1 = 0$ or $N_1 = N$. Conditional on any nondegenerate realized count $N_1 = n_1$, however,

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z} \mid N_1 = n_1) = \binom{N}{n_1}^{-1} \qquad \text{when } \sum_i z_i = n_1.$$

The conditional design is therefore complete randomization with n_1 treated units, and the difference in means is conditionally design-unbiased. This does not make the ordinary difference in means unconditionally unbiased as a universally defined estimator: it still has no value on the two empty-arm events unless an extra convention is imposed. L02 introduces Horvitz–Thompson estimation, which uses the known assignment probabilities and remains defined and unbiased without conditioning on nondegeneracy.

Exact randomization variance under complete randomization

We now derive Neyman's exact variance formula rather than simply state it. The calculation has three parts: find the randomization variance of each observed arm mean, find their covariance, and then combine those pieces. The covariance matters because complete randomization selects complementary treatment and control groups rather than two independent samples.

The difference in means varies across assignments because different units enter each arm. To describe that variation, define the finite-population variances

$$S_z^2 = \frac{1}{N-1}\sum_{i=1}^{N}\{Y_i(z) - \overline{Y}(z)\}^2, \qquad S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i - \tau)^2.$$

Here S_1^2 and S_0^2 are S_z^2 evaluated at $z = 1$ and $z = 0$. All three variances are fixed features of the science table. The only randomness below comes from the assignment.

Center the science table and recall the design moments

It will help to work with outcomes centered around their finite-population means. For this derivation only, write

$$a_i = Y_i(1) - \overline{Y}(1), \qquad b_i = Y_i(0) - \overline{Y}(0),$$

so that $\sum_i a_i = \sum_i b_i = 0$. Also define the finite-population covariance between the two columns of the science table,

$$S_{10} = \frac{1}{N-1}\sum_{i=1}^{N} a_i b_i.$$

Under complete randomization, exactly N_1 of the N units are treated. A specified unit is treated with probability N_1/N, so

$$\mathrm{Var}_Z(Z_i) = \frac{N_1}{N}\left(1 - \frac{N_1}{N}\right) = \frac{N_1 N_0}{N^2}.$$

For two distinct units i and j, the probability that both are treated is

$$\mathbb{E}_Z(Z_i Z_j) = \frac{N_1}{N}\frac{N_1 - 1}{N-1} = \frac{N_1(N_1-1)}{N(N-1)}.$$

Subtracting $\mathbb{E}_Z(Z_i)\mathbb{E}_Z(Z_j)$ gives

$$\mathrm{Cov}_Z(Z_i, Z_j) = -\frac{N_1 N_0}{N^2(N-1)}, \qquad i \neq j.$$

This negative covariance is the algebraic expression of sampling without replacement: once unit i occupies one of the fixed N_1 treatment slots, there is one fewer slot available to unit j. Treating the assignment indicators as independent would discard exactly the feature of complete randomization that produces the finite-population correction.

Variance of each arm mean

Centering the treated mean gives

$$\overline{Y}_1^{\text{obs}} - \overline{Y}(1) = \frac{1}{N_1} \sum_{i=1}^{N} Z_i a_i.$$

The finite-population mean $\overline{Y}(1)$ is fixed, so it does not affect the variance. Expanding the variance of the sum into its diagonal and off-diagonal pieces yields

$$\begin{aligned}
\operatorname{Var}_Z(\overline{Y}_1^{\text{obs}}) &= \frac{1}{N_1^2} \operatorname{Var}_Z\left(\sum_i Z_i a_i \right) \\
&= \frac{1}{N_1^2} \left\{ \sum_i \operatorname{Var}_Z(Z_i) a_i^2 + \sum_{i \neq j} \operatorname{Cov}_Z(Z_i, Z_j) a_i a_j \right\} \\
&= \frac{1}{N_1^2} \left\{ \frac{N_1 N_0}{N^2} \sum_i a_i^2 - \frac{N_1 N_0}{N^2(N-1)} \sum_{i \neq j} a_i a_j \right\}.
\end{aligned}$$

Here and below, $\sum_{i \neq j}$ runs over ordered pairs of distinct units. Thus both (i, j) and (j, i) appear.

The off-diagonal sum is not a new unknown. Because the a_i are centered,

$$0 = \left(\sum_i a_i \right)^2 = \sum_i a_i^2 + \sum_{i \neq j} a_i a_j,$$

and hence $\sum_{i \neq j} a_i a_j = -\sum_i a_i^2$. Substituting this identity, and using $\sum_i a_i^2 = (N-1) S_1^2$, gives

$$\begin{aligned}
\operatorname{Var}_Z(\overline{Y}_1^{\text{obs}}) &= \frac{1}{N_1^2} \frac{N_1 N_0}{N^2} \left(1 + \frac{1}{N-1} \right) \sum_i a_i^2 \\
&= \frac{N_0}{N N_1} S_1^2 = \left(\frac{1}{N_1} - \frac{1}{N} \right) S_1^2.
\end{aligned}$$

The factor $1/N_1 - 1/N$ is the usual sampling-without-replacement correction. By the same calculation, now treating the N_0 control units as a simple random sample from the fixed $Y_i(0)$ values,

$$\operatorname{Var}_Z(\overline{Y}_0^{\text{obs}}) = \left(\frac{1}{N_0} - \frac{1}{N} \right) S_0^2.$$

Covariance between the arm means

The two arm means are built from complementary groups, so their covariance must be included. Centering the control mean makes that dependence visible:

$$\begin{aligned}\overline{Y}_0^{\text{obs}} - \overline{Y}(0) &= \frac{1}{N_0}\sum_i (1 - Z_i) b_i \\ &= -\frac{1}{N_0}\sum_i Z_i b_i.\end{aligned}$$

The second equality uses $\sum_i b_i = 0$. Therefore,

$$\text{Cov}_Z(\overline{Y}_1^{\text{obs}}, \overline{Y}_0^{\text{obs}}) = -\frac{1}{N_1 N_0}\,\text{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right).$$

Expand the covariance inside the last expression, again separating the terms with $i = j$ from those with $i \neq j$:

$$\text{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right) = \frac{N_1 N_0}{N^2}\sum_i a_i b_i - \frac{N_1 N_0}{N^2(N-1)}\sum_{i \neq j} a_i b_j.$$

Because both sets of outcomes are centered,

$$0 = \left(\sum_i a_i\right)\left(\sum_j b_j\right) = \sum_i a_i b_i + \sum_{i \neq j} a_i b_j.$$

Thus the off-diagonal sum is $-\sum_i a_i b_i$. Using the definition of S_{10} then gives

$$\text{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right) = \frac{N_1 N_0}{N} S_{10},$$

and consequently

$$\text{Cov}_Z(\overline{Y}_1^{\text{obs}}, \overline{Y}_0^{\text{obs}}) = -\frac{S_{10}}{N}.$$

The minus sign comes from writing the centered control mean as a negative sum over treatment indicators. It does not imply that this covariance must be negative: its sign also depends on the finite-population covariance S_{10} between the two potential-outcome columns.

Assemble the pieces

Since $\hat{\tau}_{\mathrm{DM}} = \overline{Y}_1^{\mathrm{obs}} - \overline{Y}_0^{\mathrm{obs}}$,

$$\begin{aligned}\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) &= \mathrm{Var}_Z(\overline{Y}_1^{\mathrm{obs}}) + \mathrm{Var}_Z(\overline{Y}_0^{\mathrm{obs}}) - 2\,\mathrm{Cov}_Z(\overline{Y}_1^{\mathrm{obs}}, \overline{Y}_0^{\mathrm{obs}}) \\ &= \left(\frac{1}{N_1} - \frac{1}{N}\right) S_1^2 + \left(\frac{1}{N_0} - \frac{1}{N}\right) S_0^2 + \frac{2S_{10}}{N}.\end{aligned}$$

The remaining step rewrites the covariance between the two columns in terms of treatment-effect heterogeneity. Since

$$\tau_i - \tau = \{Y_i(1) - \overline{Y}(1)\} - \{Y_i(0) - \overline{Y}(0)\} = a_i - b_i,$$

we have

$$\begin{aligned}S_\tau^2 &= \frac{1}{N-1}\sum_i (a_i - b_i)^2 \\ &= S_1^2 + S_0^2 - 2S_{10}.\end{aligned}$$

Substituting $2S_{10} = S_1^2 + S_0^2 - S_\tau^2$ into the preceding variance expression produces Neyman's exact formula (Splawa-Neyman, Dabrowska, and Speed 1990):

$$\boxed{\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}}. \tag{0.1}$$

The first two terms are the arm-specific variance contributions before accounting for how the two potential outcomes are paired within units. The subtraction by S_τ^2/N records that pairing. If every unit has the same treatment effect, then $S_\tau^2 = 0$. With heterogeneous effects, $S_\tau^2 > 0$, so the exact variance is smaller than $S_1^2/N_1 + S_0^2/N_0$. This comparison describes the terms in the displayed decomposition; it does not say that changing a science table to create more heterogeneous effects must improve precision, since S_1^2 and S_0^2 may change as well.

A check using the six-plot science table. Suppose three of the six plots are assigned to treatment. From the complete table,

$$S_1^2 = \frac{22}{15}, \qquad S_0^2 = \frac{37}{6}, \qquad S_\tau^2 = \frac{113}{30}.$$

Neyman's formula therefore gives

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{22/15}{3} + \frac{37/6}{3} - \frac{113/30}{6} = \frac{23}{12}.$$

There are $\binom{6}{3} = 20$ possible balanced assignments. Computing the difference in means under each one and averaging its squared deviation from $\tau = 11/6$ gives the same $23/12$.

The numerical agreement is a useful check on the algebra; the formula itself applies to any fixed science table under complete randomization.

We cannot generally identify S_τ^2 from the observed experiment. Its calculation requires knowing each pair $(Y_i(1), Y_i(0))$, yet randomization reveals only one member of each pair. Equivalently, the cross-column covariance S_{10} is unknown. By contrast, provided each arm contains at least two units, its observed outcomes let us estimate S_1^2 and S_0^2. Omitting the nonnegative S_τ^2/N term therefore gives an upper bound on the exact variance; replacing S_1^2 and S_0^2 with observed within-arm sample variances produces Neyman's conservative variance estimator, developed in L02.

Further reading

For the foundations of randomized experiments, read Fisher's *Design of Experiments* alongside Neyman's agricultural-experiment analysis (Fisher 1935; Splawa-Neyman, Dabrowska, and Speed 1990). Neyman's representative-method essay and Horvitz–Thompson estimation make the survey-sampling lineage explicit (Neyman 1934; Horvitz and Thompson 1952). Rubin developed the modern potential-outcomes and missing-data presentation (Rubin 1974, 1976, 2005), while his comment on Fisher randomization connects potential outcomes to sharp-null randomization tests (Rubin 1980). Holland sharpened the distinction between causal effects and statistical association (Holland 1986).

Pearl's structural approach encodes causal claims through equations and graphs rather than beginning from a randomized assignment distribution (Pearl 1995, 2009). The two frameworks overlap in many applications, but this course develops the potential-outcomes, design-based route. For later examples of targets changing with the population or treatment received, see overlap-based trimming (Crump et al. 2009) and the local average treatment effect (Angrist, Imbens, and Rubin 1996).

Lecture brief: View the source `brief.md` · How these notes are made

References

L02: Basic Estimators and Experimental Pitfalls

Weighting, normalization, regression adjustment, and calibration

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Use a four-unit experiment to separate an estimator's randomization variance from an estimated standard error.
- Introduce inverse-probability weighting and see why random total weight can make an unbiased estimator unstable.
- Compare the HT and Hájek variances, then develop adaptive normalization as a more flexible response to random total weight.
- Reinterpret regression adjustment, CUPED, GREG, and calibration as related control-variate strategies.

From an unbiased estimator to an uncertainty statement

Under complete randomization with fixed arm sizes N_1 and N_0, L01 established that

$$\hat{\tau}_{\mathrm{DM}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i^{\mathrm{obs}} - \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i^{\mathrm{obs}}$$

is design-unbiased for the finite-population average treatment effect τ. Unbiasedness is a statement about the center of the estimator's randomization distribution. It does not say how dispersed that distribution is.

For fixed potential outcomes, define the finite-population variances

$$S_z^2 = \frac{1}{N-1}\sum_{i=1}^{N}\{Y_i(z) - \overline{Y}(z)\}^2, \qquad S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i - \tau)^2,$$

where $\tau_i = Y_i(1) - Y_i(0)$. These are fixed features of the science table, not variances over a model for the outcomes. Neyman's exact randomization variance is (Splawa-Neyman, Dabrowska, and Speed 1990)

$$V_Z := \mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}. \tag{0.1}$$

After assignment, the observed sample variance in arm z, when $N_z > 1$, is

$$s_z^2 = \frac{1}{N_z - 1}\sum_{i:Z_i=z}\{Y_i^{\mathrm{obs}} - \overline{Y}_z^{\mathrm{obs}}\}^2.$$

Keep three objects separate:

1. V_Z is a fixed but generally unknown property of the design and science table.
2. $\widehat{V}_N = s_1^2/N_1 + s_0^2/N_0$ is a random **variance estimator**, where s_z^2 is the sample variance of the observed outcomes in arm z.
3. After assignment, $\widehat{V}_N^{\mathrm{obs}}$ is the realized **variance estimate**, and $\sqrt{\widehat{V}_N^{\mathrm{obs}}}$ is the reported standard error.

Because the individual effects are not jointly observed, S_τ^2 is generally unidentified. The Neyman estimator drops its negative contribution. Under complete randomization,

$$\mathbb{E}_Z(\hat{V}_N) - V_Z = \frac{S_\tau^2}{N} \geq 0,$$

so it is conservative. When treatment effects are constant, $S_\tau^2 = 0$ and $\hat{V}_N$ is unbiased for V_Z; this does not mean that every realized variance estimate equals the true randomization variance.

A four-unit experiment by hand

Consider the fixed science table

Unit	$Y_i(0)$	$Y_i(1)$	τ_i
1	1	3	2
2	2	3	1
3	4	5	1
4	5	9	4

The finite-population ATE is $\tau = 2$. Treat exactly two units. The six equally likely assignments give

Treated units	$\hat{\tau}_{\mathrm{DM}}$
$\{1, 2\}$	-1.5
$\{1, 3\}$	0.5
$\{1, 4\}$	3.0
$\{2, 3\}$	1.0
$\{2, 4\}$	3.5
$\{3, 4\}$	5.5

For example, if units 1 and 4 are treated, their observed treated outcomes are 3 and 9, while the observed control outcomes for units 2 and 3 are 2 and 4. The treated and control means are therefore 6 and 3, giving $\hat{\tau}_{\mathrm{DM}} = 3$. The calculation for every row uses the same fixed science table but reveals a different pair of entries from each potential-outcome column.

The six estimates average to 2. Their variance, using the randomization probabilities rather than a sample-variance denominator, is

$$\frac{1}{6}\sum_{\mathbf{z}\in\mathcal{Z}}\{\hat{\tau}_{\mathrm{DM}}(\mathbf{z}) - 2\}^2 = \frac{31}{6}.$$

The same answer follows from Equation 0.1: $S_1^2 = 8$, $S_0^2 = 10/3$, and $S_\tau^2 = 2$. If units 1 and 4 are treated, the realized within-arm variances are $s_1^2 = 18$ and $s_0^2 = 2$, so $\hat{V}_N^{\mathrm{obs}} = 10$ and the reported standard error is $\sqrt{10}$. The true randomization variance remains $31/6$; one realized variance estimate need not equal it.

This tiny example is worth reconstructing at the board because it makes the source of every probability statement visible. There is one science table, six possible assignments, and a different observed estimate under each assignment.

Inverse-probability weighting

Complete randomization gave every unit the same treatment probability. More general designs need not. An experiment might deliberately assign a scarce treatment less often to expensive units, or use strata with different treatment fractions. The units whose potential outcomes are revealed under a given treatment state would then represent unequal amounts of the full experimental population. Inverse-probability weighting corrects that unequal representation.

Let

$$\pi_i = \mathbb{P}_Z(Z_i = 1), \qquad 0 < \pi_i < 1.$$

The **Horvitz–Thompson** (HT) estimators of the two finite-population means are

$$\hat{\mu}_{1,\mathrm{HT}} = \frac{1}{N}\sum_{i=1}^{N} \frac{Z_i Y_i^{\mathrm{obs}}}{\pi_i}, \qquad \hat{\mu}_{0,\mathrm{HT}} = \frac{1}{N}\sum_{i=1}^{N} \frac{(1-Z_i) Y_i^{\mathrm{obs}}}{1-\pi_i},$$

and $\hat{\tau}_{\mathrm{HT}} = \hat{\mu}_{1,\mathrm{HT}} - \hat{\mu}_{0,\mathrm{HT}}$ (Horvitz and Thompson 1952). A treated unit with $\pi_i = 0.2$, for example, receives weight $1/0.2 = 5$: it is observed under treatment only one fifth of the time across repeated assignments, so each such observation represents five assignment opportunities. Formally, each unit's weighted treatment indicator has expectation one,

$$\mathbb{E}_Z\left(\frac{Z_i}{\pi_i}\right) = 1.$$

Linearity of expectation then gives

$$\mathbb{E}_Z(\hat{\mu}_{1,\mathrm{HT}}) = \frac{1}{N}\sum_i Y_i(1) = \overline{Y}(1),$$

and similarly for control. This argument uses only the marginal inclusion probabilities; the Z_i need not be independent. Positivity is essential, because a potential outcome assigned probability zero can never receive an inverse-probability correction. The conclusion is **unconditional** design unbiasedness: it averages over every assignment in the design, including the all-treated and all-control assignments that a Bernoulli design permits.

Under complete randomization, $\pi_i = N_1/N$, so

$$\hat{\mu}_{1,\mathrm{HT}} = \frac{1}{N_1}\sum_i Z_i Y_i^{\mathrm{obs}}.$$

Thus HT and difference in means coincide. Under i.i.d. Bernoulli assignment, they do not.

The Bernoulli subtlety

With common Bernoulli probability p, the usual difference in means divides by the random counts N_1 and N_0. It is undefined when either count is zero. Conditional on any $N_1 = n_1 \in \{1, \dots, N-1\}$, however, the assignment is a complete randomization with n_1 treated units. Hence

$$\mathbb{E}_Z(\hat{\tau}_{\mathrm{DM}} \mid N_1 = n_1) = \tau.$$

It is therefore conditionally unbiased on every nondegenerate arm size, but it is not an everywhere-defined, unconditionally unbiased estimator over the original Bernoulli design. By contrast,

$$\hat{\tau}_{\mathrm{HT}} = \frac{1}{N} \sum_i \left\{ \frac{Z_i Y_i^{\mathrm{obs}}}{p} - \frac{(1 - Z_i) Y_i^{\mathrm{obs}}}{1 - p} \right\}$$

is defined and design-unbiased on the full assignment space.

> **What HT is correcting**
>
> Inverse-probability weighting gives a unit that is rarely assigned to a particular treatment state more representation when that state occurs. The identity $\mathbb{E}_Z(Z_i/\pi_i) = 1$ is the entire unbiasedness argument for the treated potential-outcome mean. The cost is that small probabilities create large and noisy weights.

Hájek normalization

HT weights have the right total **in expectation**, not in every assignment. For the treated mean, their realized sum is

$$\widehat{N}_1^w = \sum_i \frac{Z_i}{\pi_i}, \qquad \mathbb{E}_Z(\widehat{N}_1^w) = N.$$

The notation $\widehat{N}_1^w$ does not denote the number of treated units. It is the population size represented by the treated units after weighting. HT divides its weighted outcome total by the known N even when $\widehat{N}_1^w \neq N$. The Hájek estimator instead divides by the realized total weight (Hájek 1971):

$$\hat{\mu}_{1,\mathrm{H}} = \frac{\sum_i Z_i Y_i^{\mathrm{obs}}/\pi_i}{\sum_i Z_i/\pi_i}, \tag{0.1}$$

with an analogous control estimator. On an empty arm its denominator is zero. Under common-probability Bernoulli assignment, the Hájek arm means are the ordinary arm means. Under complete randomization, HT, Hájek, and difference in means all coincide.

With unequal probabilities, normalization can alter both finite-sample bias and variance; neither change has a universal sign. We compare assignment variances first, then turn to ratio bias from the random denominator.

A first variance comparison: HT versus Hájek

To see what normalization changes without carrying treatment and control notation through every line, isolate one generic finite-population mean under independent Bernoulli sampling. Let $I_i \sim \mathrm{Bernoulli}(\pi_i)$ independently reveal fixed values Y_i. The target total is $T_Y = \sum_i Y_i = N\mu$. Its inverse-probability estimate and the corresponding estimated population size are

$$\hat{T}_Y = \sum_i \frac{I_i Y_i}{\pi_i}, \qquad \widehat{N} = \sum_i \frac{I_i}{\pi_i}, \qquad \mu = \frac{1}{N}\sum_i Y_i.$$

This is the treated-arm problem with $I_i = Z_i$ and $Y_i = Y_i(1)$. For the control arm, take $I_i = 1 - Z_i$, $Y_i = Y_i(0)$, and replace π_i by $1 - \pi_i$. HT is $\hat{T}_Y/N$, whereas Hájek is $\hat{T}_Y/\widehat{N}$. Both numerators estimate the same total; only the denominator differs.

For HT, the estimation error has the exact representation

$$\hat{\mu}_{\mathrm{HT}} - \mu = \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right) Y_i.$$

Independence makes the cross-unit covariance terms vanish, giving the exact variance

$$\mathrm{Var}_Z(\hat{\mu}_{\mathrm{HT}}) = \frac{1}{N^2}\sum_i \frac{1-\pi_i}{\pi_i} Y_i^2. \tag{0.2}$$

Writing $Y_i = (Y_i - \mu) + \mu$ separates two sources of error:

$$\hat{\mu}_{\mathrm{HT}} - \mu = \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right)(Y_i - \mu) + \mu\left(\frac{\widehat{N}}{N} - 1\right). \tag{0.3}$$

The second term is produced by the random total weight. If the same constant a were added to every Y_i, the target mean would increase by a, but the HT estimation error would also acquire $a(\widehat{N}/N - 1)$. Thus HT variance can depend on the common additive baseline of the outcomes, not only on how dispersed they are around their mean.

When $\widehat{N} > 0$, Hájek divides by the realized total weight, so the same additive shift simply adds a to both the estimate and its target. Its estimation error is unchanged. The price is a random denominator, which makes an exact finite-sample variance less transparent.

The Taylor calculation shows how normalization removes the random-baseline term. Apply $f(t, n) = t/n$ to $(\hat{T}_Y, \widehat{N})$ and expand around the expectations $(T_Y, N) = (N\mu, N)$. Because the two derivatives at that point are $1/N$ and $-\mu/N$,

$$\frac{\hat{T}_Y}{\widehat{N}} \approx \mu + \frac{\hat{T}_Y - T_Y}{N} - \frac{\mu}{N}(\widehat{N} - N).$$

Substituting the two weighted totals and collecting each unit's contribution gives

$$\hat{\mu}_{\mathrm{H}} - \mu \approx \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right)(Y_i - \mu). \tag{0.4}$$

Consequently, its first-order variance is

$$\mathrm{Var}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{1}{N^2}\sum_i \frac{1-\pi_i}{\pi_i}(Y_i - \mu)^2. \tag{0.5}$$

The contrast between Equation 0.2 and Equation 0.5 is now explicit: HT's exact variance involves the squared outcomes Y_i^2, whereas the first-order Hájek variance involves squared deviations $(Y_i - \mu)^2$. With common $\pi_i = p$,

$$\mathrm{Var}_Z(\hat{\mu}_{\mathrm{HT}}) - \mathrm{Var}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{1-p}{pN}\mu^2.$$

The difference is large when the outcome mean is far from zero relative to the outcomes' dispersion. Equivalently, a common additive shift changes HT's variance because random total weight multiplies that shift, while it leaves the first-order Hájek variance unchanged. This is the classic survey-sampling linearization logic (Särndal, Swensson, and Wretman 1989; Särndal, Swensson, and Wretman 1992). The calculation isolates one arm; for the treatment-effect estimator, the treated- and control-arm variances must also be combined with their design covariance.

Ratio bias from normalization

Variance is not the entire comparison. A second-order expansion gives the approximate ratio bias

$$\mathrm{Bias}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{\mu \,\mathrm{Var}_Z(\widehat{N}) - \mathrm{Cov}_Z(\hat{T}_Y, \widehat{N})}{N^2}.$$

Here

$$\mathrm{Var}_Z(\widehat{N}) = \sum_i \frac{1-\pi_i}{\pi_i}, \qquad \mathrm{Cov}_Z(\hat{T}_Y, \widehat{N}) = \sum_i \frac{1-\pi_i}{\pi_i} Y_i.$$

The expression is informative even before it is evaluated: ratio bias appears when fluctuations in the estimated outcome total and the estimated population size do not line up as μ times one another. For common $\pi_i = p$, the two terms cancel. This agrees with the exact fact that, conditional on any positive realized sample size, the Hájek mean is the ordinary sample mean under simple random sampling. The Taylor calculations nevertheless remain approximations to an unconditional ratio problem whose denominator can be zero. They are most informative when $\widehat{N}$ is concentrated near N. With small effective samples, extreme probabilities, or influential units, higher-order terms can matter and neither estimator has a universal MSE advantage.

HT versus Hájek in a finite population

The demonstration uses

$$Y_i(0) = b + \sigma x_i, \qquad Y_i(1) = b + \sigma x_i + \tau,$$

with finite-population x_i centered at zero. The common baseline b shifts every potential outcome by the same amount, so it changes neither the individual treatment effects nor the ATE. For an HT arm mean, however, that shift adds the random error term $b(\widehat{N}/N - 1)$. Because Hájek divides by $\widehat{N}$, the same shift simply adds b to its arm-mean estimate and does not change that estimate's variance. When $b = 0$, this first-order shift-cancellation advantage disappears; the random Hájek denominator can then produce more conditional MSE than HT's fixed denominator.

The left panel uses centered outcomes with no effect; the right introduces a large common baseline. The reversal isolates the random-total-weight mechanism in Equation 0.3.

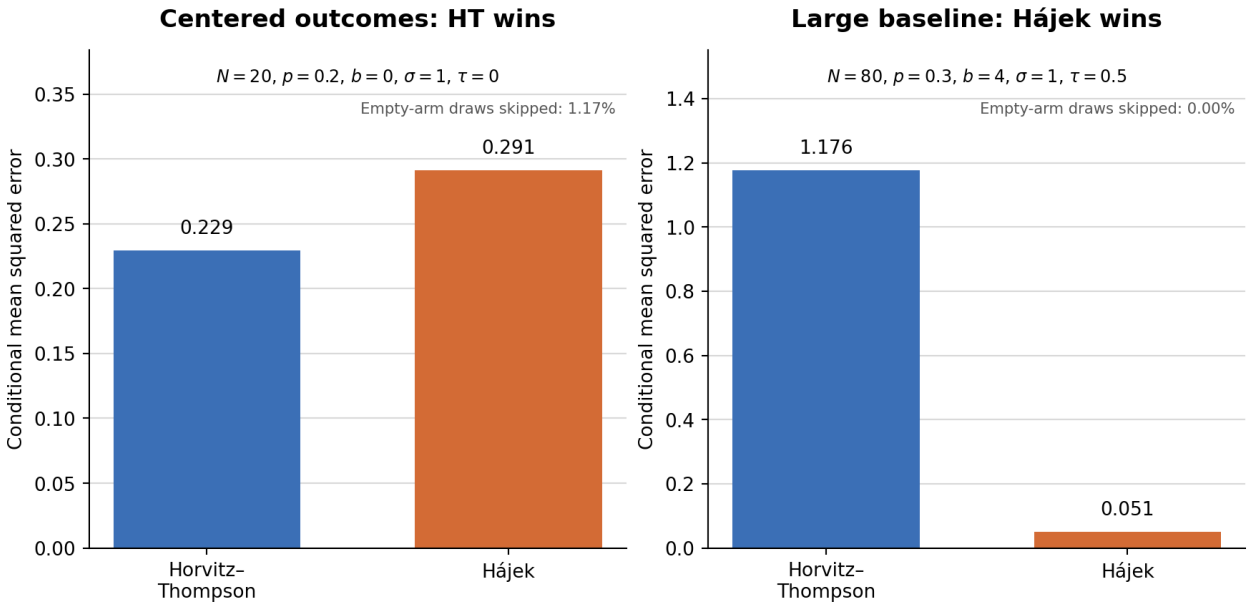


Figure 1: Two fixed simulation settings show that neither estimator uniformly minimizes finite-sample MSE. HT wins in the centered setting; Hájek wins when outcomes have a large common baseline.

Normalization is a continuum

HT leaves the realized total-weight error uncorrected; Hájek removes it completely. That contrast suggests a less binary question: how much of the denominator fluctuation should an estimator use? Trotter and Tukey introduced an affine family that makes this amount explicit; Khan and Ugander (2023) develop the family for inverse-probability weighting. For the same generic mean problem, define

$$\hat{\mu}_\lambda = \frac{\hat{T}_Y}{(1-\lambda)N + \lambda\widehat{N}} = \frac{\hat{T}_Y}{N + \lambda(\widehat{N} - N)}. \tag{0.1}$$

$\lambda = 0$ gives HT and $\lambda = 1$ gives Hájek. The second expression makes the role of λ visible: it controls how strongly the random error $\widehat{N} - N$ enters the denominator. Values outside $[0, 1]$ are allowed; the family is an affine normalization family, not a convex interpolation. In a realized finite sample, however, a chosen value is usable only when $N + \lambda(\widehat{N} - N) \neq 0$. Extrapolated values can drive this denominator near zero or make it negative, so a first-order variance optimum is not by itself a finite-sample stability guarantee.

The same first-order expansion used for Hájek yields

$$\hat{\mu}_\lambda - \mu \approx \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right)(Y_i - \lambda\mu). \tag{0.2}$$

This equation turns the choice of denominator into the choice of a centering constant. HT uses no centering, because $\lambda\mu = 0$. Hájek centers at the target mean, because $\lambda\mu = \mu$. An intermediate or extrapolated value uses a different constant.

Before changing probability frameworks, the optimal first-order choice can be written entirely in finite-population notation. Let $q_i = (1 - \pi_i)/\pi_i = \mathrm{Var}_Z(I_i/\pi_i)$. Under independent Bernoulli inclusion, Equation 0.2 gives

$$\mathrm{Var}_Z(\hat{\mu}_\lambda) \approx \frac{1}{N^2}\sum_i q_i(Y_i - \lambda\mu)^2.$$

When $\mu \neq 0$ and at least one $q_i > 0$, minimizing this quadratic yields

$$\lambda_N^* = \frac{\sum_i q_i Y_i}{\mu \sum_i q_i}, \qquad c_N^* := \lambda_N^*\mu = \frac{\sum_i q_i Y_i}{\sum_i q_i}. \tag{0.3}$$

Thus the useful object is not fundamentally a fraction λ. It is the centering constant c_N^*, a weighted mean of the fixed Y_i that emphasizes units with noisier inverse-probability weights.

To state the adaptive choice used by Khan and Ugander (2023), we now make an explicit change of framework. The preceding variance calculations held a finite population fixed. Their asymptotic analysis instead treats the pairs (Y_i, π_i) as i.i.d. draws of a generic pair (Y, P), where P is a random inclusion probability. In this paragraph, $\mu = \mathbb{E}(Y)$ is the superpopulation mean rather than the mean of one fixed finite population. Define the weight-noise factor $Q = (1 - P)/P$ and the two population moments

$$\Pi = \mathbb{E}(Q), \qquad A = \mathbb{E}(YQ).$$

These expectations are the superpopulation analogues of $N^{-1} \sum_i q_i$ and $N^{-1} \sum_i q_i Y_i$ in Equation 0.3.

For the usual $\sqrt{N}$ scaling, the part of the first-order asymptotic variance that depends on λ is

$$\sigma_\lambda^2 = \mathbb{E}\{Q(Y - \lambda\mu)^2\}. \tag{0.4}$$

Superpopulation sampling also contributes variation in Y itself, but that term does not depend on λ and therefore does not affect which member of the normalization family minimizes variance.

This is a quadratic in λ. When $\mu \neq 0$ and $\Pi > 0$, differentiating it gives

$$\frac{d\sigma_\lambda^2}{d\lambda} = -2\mu(A - \lambda\mu\Pi),$$

and hence the minimizer

$$\lambda^* = \frac{A}{\Pi\mu}. \tag{0.5}$$

The numerator and denominator become easier to interpret after absorbing the factor μ. Define the optimal centering constant

$$c^* = \frac{A}{\Pi}$$

so that $\lambda^* = c^*/\mu$. The constant c^* is an average of Y under weights proportional to Q: it emphasizes outcomes attached to the inclusion probabilities that create the most inverse-weight noise. If the inclusion probability is common, Q is constant, $c^* = \mu$, and Hájek is optimal within this first-order family. With heterogeneous inclusion probabilities, the high-weight portion of the population can have a mean different from μ; then neither HT nor Hájek need use the best centering constant, and λ^* need not lie between 0 and 1.

When $\mu = 0$, the product $\lambda\mu$ vanishes in Equation 0.2, so the first-order variance does not depend on λ. This does not mean that normalization is irrelevant in every finite sample. It means that λ, which divides a useful centering constant by μ, is a poor coordinate near zero. The constant c^* remains well defined.

The population moments A and Π are unknown in the generic inclusion problem. The identity $\mathbb{E}(I_i Y_i/\pi_i \mid Y_i, \pi_i) = Y_i$ motivates the IPW sample analogues

$$\widehat{A} = \frac{1}{N}\sum_i \frac{1-\pi_i}{\pi_i}\frac{I_i Y_i}{\pi_i}, \qquad \widehat{\Pi} = \frac{1}{N}\sum_i \frac{1-\pi_i}{\pi_i}\frac{I_i}{\pi_i}, \qquad \hat{c} = \frac{\widehat{A}}{\widehat{\Pi}}.$$

The remaining obstacle is μ in $\lambda^* = c^*/\mu$. Adaptive normalization uses the current mean estimate in that denominator, updates the normalization, and repeats. Starting, for example, at $\hat{\mu}^{(0)} = \hat{\mu}_{\mathrm{HT}}$, the iteration sets

$$\hat{\lambda}^{(t)} = \frac{\hat{c}}{\hat{\mu}^{(t-1)}}, \qquad \hat{\mu}^{(t)} = \frac{\hat{T}_Y}{(1-\hat{\lambda}^{(t)})N + \hat{\lambda}^{(t)}\widehat{N}}.$$

The two equations form a feedback loop: the current mean estimate supplies the denominator of $\hat{\lambda}^{(t)}$, and the resulting normalization supplies the next mean estimate. At a nonzero fixed point, substitute $\hat{\lambda} = \hat{c}/\hat{\mu}$ into the denominator and multiply through:

$$N\hat{\mu} + \hat{c}(\widehat{N} - N) = \hat{T}_Y.$$

Solving for $\hat{\mu}$ gives the closed form

$$\hat{\mu}_{\mathrm{AN}} = \frac{\hat{T}_Y}{N} + \hat{c}\left(1 - \frac{\widehat{N}}{N}\right) = \hat{\mu}_{\mathrm{HT}} - \hat{c}\left(\frac{\widehat{N}}{N} - 1\right). \tag{0.6}$$

The fixed point is therefore a **control-variate**, or regression-control, estimator. The quantity $D = \widehat{N}/N - 1$ has design expectation zero and is correlated with the part of HT error generated by total-weight fluctuation. For any *fixed* coefficient c, subtracting cD preserves exact design unbiasedness; choosing the population coefficient c^* minimizes the first-order variance in this one-dimensional control family. HT uses $c = 0$, Hájek corresponds at first order to $c = \mu$, and adaptive normalization plugs in $\hat{c}$.

Because $\hat{c}$ is estimated from the same realization as D, the mean-zero property of D alone does not prove exact unbiasedness of the adaptive estimator. The guarantee in Khan and Ugander (2023) is asymptotic: under their moment and consistency conditions, the adaptive variance is no larger than the corresponding HT or Hájek variance. Unstable estimated moments or a nearly empty effective sample can still matter in finite samples. The closed form in Equation 0.6 is also preferable if an iterate passes near zero, where the λ parameterization becomes unstable.

Regression adjustment

The weighted-population-size error is one random quantity whose expectation is known from the design. Pretreatment covariates provide many more. Suppose, for example, that the treated arm happens to contain users who were more active before an experiment. That difference cannot have been caused by treatment, but if prior activity predicts the outcome, it can make the unadjusted outcome difference unusually large or small.

Under complete randomization with fixed positive arm sizes, define the realized covariate means

$$\overline{X}_1^{\text{obs}} = \frac{1}{N_1}\sum_i Z_i X_i, \qquad \overline{X}_0^{\text{obs}} = \frac{1}{N_0}\sum_i (1-Z_i) X_i.$$

The covariates are vectors, so these are componentwise arm averages. Their difference is observable after assignment. The superscript obs refers to the realized treatment and control groups; it does not suggest that X_i was measured after treatment. We write the difference as

$$\hat{\tau}_X = \overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}$$

This difference has design expectation zero: randomization does not guarantee balance in the realized experiment, but it makes the imbalance average to zero across assignments. If β maps a one-unit covariate imbalance into predicted outcome imbalance, the natural correction is

$$\hat{\tau}_{\text{adj}}(\beta) = \hat{\tau}_{\text{DM}} - \beta^\top \hat{\tau}_X. \tag{0.1}$$

For any coefficient β fixed before assignment, $\mathbb{E}_Z\{\hat{\tau}_{\text{adj}}(\beta)\} = \tau$ follows immediately: the adjustment subtracts a fixed multiple of a mean-zero random quantity. When X predicts outcomes, this can remove chance imbalance that would otherwise masquerade as treatment effect. An arbitrary coefficient can instead add noise; design unbiasedness and variance reduction are separate claims.

A common slope is not required. If $\overline{X} = N^{-1}\sum_i X_i$ is known, fixed treatment- and control-arm coefficients give the following estimator, where $\overline{Y}_z^{\text{obs}}$ denotes the observed outcome mean in arm z:

$$\begin{aligned}\hat{\tau}_{\text{adj}}(\beta_1, \beta_0) = &\left\{\overline{Y}_1^{\text{obs}} - (\overline{X}_1^{\text{obs}} - \overline{X})^\top \beta_1\right\} \\ &- \left\{\overline{Y}_0^{\text{obs}} - (\overline{X}_0^{\text{obs}} - \overline{X})^\top \beta_0\right\}.\end{aligned} \tag{0.2}$$

For fixed coefficients, each bracket is design-unbiased for its corresponding finite-population potential-outcome mean. Taking $\beta_1 = \beta_0 = \beta$ recovers Equation 0.1. In practice, the coefficients are usually fitted from the experimental outcomes. They are then assignment-dependent, so the exact finite-sample mean-zero argument no longer follows merely from $\mathbb{E}_Z(\hat{\tau}_X) = 0$. The regression should

allow treatment-specific slopes—equivalently, include treatment-by-centered-covariate interactions—because the two potential-outcome surfaces need not be parallel (Freedman 2008; Lin 2013). Randomization protects the large-sample causal interpretation under suitable regularity conditions, but careless specifications or standard errors can lose efficiency or misstate uncertainty.

This returns us to normalization. In Equation 0.6, $\widehat{N}/N - 1$ is the inverse-weighted imbalance in the constant feature 1: its target mean is exactly one, but its realized weighted mean need not be. Adaptive normalization uses that scalar imbalance as a control variable. Conventional regression adjustment applies the same logic to informative pretreatment features. GREG makes the connection especially explicit.

GREG from a known baseline total

Suppose an online platform fixes a population of N accounts before an experiment. For every account it already knows X_i, the number of active days during the preceding four weeks. The finite-population mean

$$\overline{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$$

is therefore known exactly before assignment. By contrast, $Y_i(1)$, the experiment-period outcome under treatment, is observed only for treated accounts.

The treated accounts may happen to include unusually active users. Under complete randomization, for example, their pre-period mean might be 4.8 active days even though the full roster mean is 4.0. Randomization makes this discrepancy zero on average across assignments, but it is present in the experiment that actually occurred. If a prespecified relationship says that one additional pre-period active day predicts roughly two units of the experiment-period outcome, the 0.8-day imbalance suggests about 1.6 units of chance inflation in the treated outcome mean.

The same idea extends to a vector of pretreatment covariates. Define its HT estimate within the treated arm,

$$\widehat{\overline{X}}_{1,\mathrm{HT}} = \frac{1}{N}\sum_i \frac{Z_i X_i}{\pi_i}, \qquad \mathbb{E}_Z\left(\widehat{\overline{X}}_{1,\mathrm{HT}}\right) = \overline{X}.$$

The discrepancy $\overline{X} - \widehat{\overline{X}}_{1,\mathrm{HT}}$ is observable and has design expectation zero. If β converts covariate imbalance into predicted outcome imbalance, the natural correction is

$$\hat{\mu}_{1,\mathrm{GREG}}(\beta) = \hat{\mu}_{1,\mathrm{HT}} + (\overline{X} - \widehat{\overline{X}}_{1,\mathrm{HT}})^\top \beta. \tag{0.3}$$

Under complete randomization, $\widehat{\overline{X}}_{1,\mathrm{HT}}$ is simply the ordinary treated-arm covariate mean, so it is the 4.8 in the example. For a coefficient β fixed relative to assignment, Equation 0.3 is the **generalized-difference**, or regression-control, construction at the heart of GREG. Replacing β by a weighted regression estimate gives the usual **generalized regression estimator** (GREG) (Cassel, Särndal, and Wretman 1976). In the scalar example, $\beta = 2$ makes the correction $(4.0 - 4.8)2 = -1.6$.

Writing $m_1(x) = x^\top \beta$ gives an equivalent prediction-plus-residual form:

$$\hat{\mu}_{1,\mathrm{GREG}}(m_1) = \frac{1}{N}\sum_i m_1(X_i) + \frac{1}{N}\sum_i \frac{Z_i}{\pi_i}\{Y_i^{\mathrm{obs}} - m_1(X_i)\}. \tag{0.4}$$

The two terms divide the work. The full roster supplies the average prediction, which can be computed without observing any experiment-period outcomes. Inverse-probability weighting is then applied only to the prediction errors among treated units. If an intercept is desired, augment X_i with a leading 1. Under unequal probabilities, correcting imbalance in that intercept is exactly a correction for $\widehat{N}/N - 1$; under complete randomization, the intercept is already balanced. For β fixed before assignment,

$$\hat{\mu}_{1,\mathrm{GREG}}(\beta) - \overline{Y}(1) = \frac{1}{N}\sum_i \left(\frac{Z_i}{\pi_i} - 1\right)\{Y_i(1) - X_i^\top \beta\}.$$

Exact design unbiasedness therefore does not require the linear prediction to be correct: misspecification changes the residuals, but the IPW correction still centers their contribution correctly. Under independent Bernoulli assignment, the exact variance is

$$\mathrm{Var}_Z\{\hat{\mu}_{1,\mathrm{GREG}}(\beta)\} = \frac{1}{N^2}\sum_i \frac{1-\pi_i}{\pi_i}\{Y_i(1) - X_i^\top \beta\}^2.$$

HT is the case $\beta = 0$. At the other extreme, if the prediction were exact for every unit, every residual would be zero and the treated potential-outcome mean could be recovered from full-roster predictions without assignment noise. Between those extremes, the fixed-coefficient estimator reduces variance when the chosen coefficient leaves less design-weighted residual variation than the raw outcomes; an arbitrary or poorly transported coefficient need not help.

For control, replace Z_i/π_i by $(1-Z_i)/(1-\pi_i)$ and use a control-outcome prediction m_0. Subtracting the two prediction-plus-residual arm means estimates the ATE. This is the separate-arm version of the adjustment in Equation 0.2.

In practice, β is often fitted from the realized outcomes. Then the fixed-coefficient argument no longer establishes exact finite-sample unbiasedness. A coefficient fitted on historical data can be frozen before assignment; a coefficient fitted within the experiment needs the regression-adjustment qualifications discussed above. L03 uses the prediction-plus-IPW construction for both arms and names the resulting general form AIPW (Robins, Rotnitzky, and Zhao 1994).

CUPED: choosing the control coefficient

The same online setting provides a natural scalar control variable: the outcome metric measured before the experiment. In this subsection, take X_i to be that single pre-period metric rather than a vector. **CUPED**, short for controlled-experiment using pre-experiment data, transforms the observed outcome to

$$Y_i^{\mathrm{cuped}}(\theta) = Y_i^{\mathrm{obs}} - \theta(X_i - \overline{X}).$$

Subtracting the same pretreatment quantity from $Y_i(1)$ and $Y_i(0)$ does not change unit i's treatment effect. Centering at $\overline{X}$ makes the transformed outcome retain the original finite-population mean within each treatment state. Taking the difference in arm means therefore gives, equivalently,

$$\hat{\tau}_{\mathrm{CUPED}}(\theta) = \hat{\tau}_{\mathrm{DM}} - \theta\hat{\tau}_X. \tag{0.5}$$

Here θ has the units of outcome per unit of the pre-period metric: it says how much outcome difference should be removed for one unit of chance covariate imbalance. For every θ fixed before assignment, this estimator remains design-unbiased because $\mathbb{E}_Z(\hat{\tau}_X) = 0$. The choice of θ controls precision. Expanding the variance of $\hat{\tau}_{\mathrm{DM}} - \theta\hat{\tau}_X$ gives

$$\begin{aligned}\mathrm{Var}_Z\{\hat{\tau}_{\mathrm{CUPED}}(\theta)\} = {} & \mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) + \theta^2\,\mathrm{Var}_Z(\hat{\tau}_X) \\ & - 2\theta\,\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X).\end{aligned} \tag{0.6}$$

When $\mathrm{Var}_Z(\hat{\tau}_X) > 0$, this quadratic is minimized at

$$\theta^* = \frac{\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X)}{\mathrm{Var}_Z(\hat{\tau}_X)}. \tag{0.7}$$

This is an oracle coefficient: its covariance with the treatment-effect estimator depends on the potential-outcome schedule and is not known before the experiment. At this value, the variance reduction relative to unadjusted difference in means is $\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X)^2 / \mathrm{Var}_Z(\hat{\tau}_X)$. Thus θ should be near zero when the pre-period measure does not predict experiment-period outcomes, and a poor coefficient can increase rather than decrease variance.

Under complete randomization, define the finite-population variance and arm-specific outcome–covariate covariances by

$$S_X^2 = \frac{1}{N-1}\sum_i (X_i - \overline{X})^2, \qquad S_{zX} = \frac{1}{N-1}\sum_i \{Y_i(z) - \overline{Y}(z)\}(X_i - \overline{X}).$$

The complete-randomization covariance identities are

$$\mathrm{Var}_Z(\hat{\tau}_X) = S_X^2\left(\frac{1}{N_1} + \frac{1}{N_0}\right),$$

and

$$\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X) = \frac{S_{1X}}{N_1} + \frac{S_{0X}}{N_0}.$$

The second expression has a plus sign because both $\hat{\tau}_{\mathrm{DM}}$ and $\hat{\tau}_X$ subtract their control-arm means. Substituting these identities into Equation 0.7 gives

$$\theta^* = \frac{S_{1X}/N_1 + S_{0X}/N_0}{S_X^2(1/N_1 + 1/N_0)} = \frac{N_0}{N}\frac{S_{1X}}{S_X^2} + \frac{N_1}{N}\frac{S_{0X}}{S_X^2}.$$

The oracle coefficient is therefore an allocation-weighted average of the two arm-specific slopes, with each slope weighted by the size of the opposite arm. Under equal allocation it is their simple average; if the two slopes coincide, their common value is optimal for any allocation. This calculation explains both why a pooled slope can work well and why treatment-specific slopes can improve on it.

The original CUPED recommendation is to use the same metric from a nearby pre-experiment period: it is often strongly correlated with the experiment-period metric while remaining unaffected by treatment (Deng et al. 2013). There are several practical ways to choose θ:

- **Estimate it before the experiment.** Use comparable historical experiments, a separate cohort, or relationships between earlier pretreatment windows, then freeze the coefficient before assignment. This preserves the exact fixed-coefficient design argument, but relies on stability across periods.
- **Estimate the oracle combination in the experiment.** Estimate a slope within each arm, say $\hat{\beta}_1$ and $\hat{\beta}_0$, and combine them as $(N_0/N)\hat{\beta}_1 + (N_1/N)\hat{\beta}_0$, mirroring the last expression for θ^*. In a balanced experiment this is the simple average. Regressing Y_i^{obs} on an intercept, Z_i, and $X_i - \overline{X}$ gives a familiar pooled common slope; including Z_i means that the slope is learned from within-arm outcome–covariate association rather than from the treatment/control mean difference. In a large balanced experiment it provides a convenient estimate of the same average when both arms inherit similar covariate dispersion; ordinary pooled regression need not reproduce the oracle weighting in an unbalanced experiment. Because the same outcomes choose the coefficient and estimate the effect, exact finite-sample unbiasedness is no longer automatic; inference should account for the fitted adjustment.
- **Allow treatment-specific slopes when warranted.** Include $Z_i(X_i - \overline{X})$, or fit the two arm-specific slopes directly. A single common θ can leave precision unused when the prognostic relationship differs by arm (Lin 2013).
- **Regularize or cross-fit richer adjustments.** With many baseline variables or a flexible predictor, assess predictive performance out of sample and use sample splitting or cross-fitting to limit overfitting. Cross-fitting does not by itself guarantee exact design unbiasedness, especially under fixed-total assignment. Only pretreatment variables belong in the adjustment.

Calibration: make the weighted arm reproduce its target

Return first to the completely randomized account experiment. Its treated-arm design weights are $d_i = 1/\pi_i = N/N_1$. They sum to N automatically, but if the treated users average 4.8 pre-period active days, their weighted covariate total is $N(4.8)$ rather than the known roster total $N(4.0)$. With unequal treatment probabilities, the same issue can affect both the total weight and the covariate total: design weights reproduce the full-roster targets in expectation over assignments, not necessarily in the realized assignment.

Calibration replaces the base weights by nearby weights that make selected weighted facts about the observed arm equal known full-roster targets. Its goal is not to make the treated arm resemble the roster in every respect. It enforces the particular comparisons encoded by the analyst's chosen features. In the scalar example, it chooses weights w_i satisfying

$$\sum_{i:Z_i=1} w_i = N, \qquad \sum_{i:Z_i=1} w_i X_i = N\overline{X}. \tag{0.1}$$

The first equation makes the weighted treated arm represent N accounts. The second makes its weighted pre-period total equal the known full-roster total. Under complete randomization the base weights already satisfy the first equation, but adjusting them to satisfy the second can disturb that total, so both constraints are imposed together.

For several covariates, first choose K scalar calibration features $h_1(X_i), \dots, h_K(X_i)$. These can be individual coordinates of X_i, indicators for categories, or prespecified transformations and interactions. They express which roster summaries the weighted arm should match. Collect them with an intercept into the $(K+1)$-dimensional column vector

$$g_i = \begin{pmatrix} 1 \\ h_1(X_i) \\ \vdots \\ h_K(X_i) \end{pmatrix}, \qquad t_g = \sum_{i=1}^{N} g_i.$$

The leading 1 is an intercept coordinate; g_i is an ordinary vector, not a binomial coefficient. Because the pretreatment covariates are known for the full experimental roster, the target total t_g is known. The calibration equation is the vector equality

$$\sum_{i:Z_i=1} w_i g_i = t_g. \tag{0.2}$$

Written component by component, it requires

$$\sum_{i:Z_i=1} w_i = N, \qquad \sum_{i:Z_i=1} w_i h_k(X_i) = \sum_{i=1}^{N} h_k(X_i), \quad k = 1, \ldots, K.$$

No binary-covariate assumption is involved. For a binary feature, the equation matches a count; for a continuous feature, it matches a weighted sum, and therefore its mean once the intercept constraint holds.

The constraints usually admit many weight vectors: matching $K+1$ totals does not by itself say which unit should absorb each adjustment. Let $\mathbf{w}$ collect the treated-arm weights. Calibration resolves that ambiguity by starting from the design weights d_i and moving them as little as possible while hitting the targets:

$$\min_{\mathbf{w}} \sum_{i:Z_i=1} d_i \, \phi\left(\frac{w_i}{d_i}\right) \quad \text{subject to} \quad \sum_{i:Z_i=1} w_i g_i = t_g, \tag{0.3}$$

where the convex function ϕ is minimized at 1. Thus the objective favors w_i/d_i near one, while the constraints force exactly as much movement as is needed for balance. With a strictly convex distance and feasible constraints, the solution is unique. Quadratic distance, $\phi(r) = (r-1)^2/2$, gives affine adjustments

$$w_i = d_i(1 + \eta^\top g_i),$$

whereas relative-entropy distance, $\phi(r) = r \log r - r + 1$, gives the positive exponential tilt

$$w_i = d_i \exp(\eta^\top g_i).$$

In each case $\eta \in \mathbb{R}^{K+1}$ is a vector of Lagrange-multiplier parameters chosen so that Equation 0.2 holds. Entropy calibration becomes classical raking when the features are indicators for categorical margins (Deville and Särndal 1992; Deville, Särndal, and Sautory 1993).

The resulting treated-mean estimator is

$$\hat{\mu}_{1,\text{cal}} = \frac{1}{N} \sum_{i:Z_i=1} w_i Y_i^{\text{obs}}.$$

Its potential advantage follows directly from the balance equations. This does not require assuming a linear outcome model. For any chosen β, define the residuals algebraically by

$$Y_i(1) = g_i^\top \beta + e_i.$$

Then

$$\hat{\mu}_{1,\text{cal}} - \overline{Y}(1) = \frac{1}{N} \left\{ \sum_{i:Z_i=1} w_i e_i - \sum_{i=1}^{N} e_i \right\}. \tag{0.4}$$

To obtain Equation 0.4, substitute $Y_i(1) = g_i^\top \beta + e_i$ into both the weighted observed total and the full-roster total. The two $g_i^\top \beta$ contributions are equal because of Equation 0.2, so they cancel. The only remaining error is imbalance in the residuals.

This identity gives the calibration objective its statistical meaning. If the chosen features predict outcomes well, their exact balance removes a systematic source of assignment noise. If $Y_i(1) = g_i^\top \beta$ exactly for every unit, all residuals are zero and the calibrated treated mean equals $\overline{Y}(1)$ exactly. With weakly predictive or overly numerous features, the benefit can be small or outweighed by unstable weights.

When the calibration problem is feasible and the relevant weighted feature matrix has full rank, quadratic distance gives the calibrated estimator an exact GREG representation with a corresponding weighted least-squares coefficient. Other smooth calibration distances generally produce different finite-sample estimators, although they can share a first-order GREG approximation under regularity conditions (Deville and Särndal 1992). Unlike the generalized-difference construction with a coefficient fixed before assignment, calibrated weights depend on the realized assignment, so exact design unbiasedness is not automatic.

The control arm can be calibrated separately to the same roster targets, beginning from $d_i = 1/(1-\pi_i)$. Subtracting the calibrated arm means estimates the ATE. More constraints are not always better: they can force extreme weights and inflate the residual variance. With nonnegative weights and an intercept constraint, the target covariate mean must lie in the observed arm's convex hull. Signed quadratic weights can reach targets outside that hull only when the target remains in the affine span of the observed feature vectors.

Diagnostics should therefore include residual balance, weight dispersion, and sensitivity to the feature set. For positive weights, one useful summary is the Kish effective sample size

$$N_{\text{eff}} = \frac{(\sum_{i:Z_i=1} w_i)^2}{\sum_{i:Z_i=1} w_i^2}.$$

It equals the observed arm size under equal weights and decreases as a small number of units carry more of the weighted total. It is a warning signal about weight concentration, not by itself a complete variance calculation.

What to carry forward

Many estimators in this lecture share one architecture, either exactly or through a first-order approximation:

$$\text{unbiased or identifying core} + \text{ a correction with known or approximately known mean zero.}$$

HT corrects unequal observation probabilities. Hájek stabilizes random total weight. Trotter–Tukey chooses how much to normalize. Regression, CUPED, GREG, and calibration remove predictable imbalance. Some corrections preserve exact unbiasedness only when their coefficients are fixed relative to assignment; fitted corrections generally require additional finite-sample or asymptotic analysis. Their differences matter, but seeing the common control-variate structure makes the menu navigable.

Advanced material

Practical experimentation: errors larger than the standard error

Large samples do not rescue a broken experiment. The practical program begins with Kohavi, Henne, and Sommerfield (2007); Crook et al. (2009) then documents recurring failures in online controlled experiments: incorrect variance and power calculations, misleading percent-change intervals, bot traffic, instrumentation failures, ramp-up effects, and treatment/control populations that are not actually comparable. These are failures of measurement and implementation before they are failures of asymptotic approximation.

The replication evidence in Kohavi et al. (2026) adds a selection lesson. Eight large-scale tests of previously reported experimentation "patterns" had median samples in the millions and small minimum detectable effects, yet only two produced significant movement in the expected direction and one moved significantly in the opposite direction. The discrepancy is consistent with underpowered discovery studies, selective reporting, and the **winner's curse**: conditional on being selected for a large estimate, a noisy estimate is likely exaggerated.

Practical discipline follows: establish sample-ratio and instrumentation checks, specify the primary metric and analysis before looking, ramp cautiously, report uncertainty in interpretable units, and replicate claims that will guide many future experiments.

The propensity-score paradox

Why estimate an assignment probability that the experimenter already knows? The efficiency-bound analysis of Hahn (1998) sets up the issue. In a semiparametric superpopulation setting, Hirano, Imbens, and Ridder (2003) show that inverse-probability estimators using an estimated propensity score can be more efficient than otherwise analogous estimators using the true propensity.

There is no information paradox. Common likelihood and series estimators of the propensity score satisfy empirical fitting equations that can also impose useful covariate-balance conditions. Those conditions create a projection or control-variate correction, much like Equation 0.1 and Equation 0.2; arbitrary propensity estimation need not impose the same balance. The estimated propensity is not better knowledge of the randomization device. It is a route to an estimator that exploits realized covariate balance. In a randomized experiment, it is clearer to describe the gain directly as covariate adjustment or calibration, while retaining the known design probabilities as the basis for identification and inference.

A short history of inverse-probability normalization

Horvitz and Thompson used inverse inclusion probabilities to estimate finite-population totals exactly unbiasedly (Horvitz and Thompson 1952). Ratio estimation and Hájek normalization can trade finite-sample bias for stability; whether that trade is favorable depends on the design and finite population. Trotter and Tukey placed normalization on a continuum (Trotter and Tukey 1956), while Basu's survey-sampling arguments exposed how formally unbiased estimators can behave poorly when they ignore structural information (Basu 1971).

The same ratio appears in importance sampling: self-normalized importance sampling is stable and invariant to an unknown normalizing constant, but biased in finite samples (Owen 2013). Counterfactual learning rediscovered the same practical tradeoff when propensities are highly variable (Swaminathan and Joachims 2015). Adaptive normalization makes the amount of normalization estimable and connects its fixed point to regression control (Khan and Ugander 2023). The recurring problem is not domain-specific: random weights carry useful correction and avoidable noise at the same time.

Choosing estimators across families of experiments

Parikh et al. (2026) argue against choosing whichever estimator looks best inside a single realized trial. That practice turns estimator selection into another opportunity for overfitting. They instead use sample splitting across a **family** of randomized trials to compare estimators under a prespecified objective.

Their applications make the objective dependence concrete: weighted least squares performed best for inference-oriented criteria, while difference in means performed best for decision regret in the examined settings. The point is not a new universal winner. It is that "best estimator" is incomplete without a loss function, a relevant family of experiments, and a selection procedure separated from final evaluation.

Further reading

For the survey-sampling foundations, begin with Horvitz and Thompson (1952) and the exchange between Basu (1971) and Hájek (1971), then use Särndal, Swensson, and Wretman (1992) for the systematic development. The GREG and calibration line runs through Cassel, Särndal, and Wretman (1976), Deville and Särndal (1992), and Deville, Särndal, and Sautory (1993). For regression adjustment in experiments, reading Freedman (2008) together with Lin (2013) shows both the finite-sample concern and the modern interacted-regression resolution.

The normalization sequence from classical conditional Monte Carlo to adaptive IPW runs through Trotter and Tukey (1956) and Khan and Ugander (2023); Owen (2013) and Swaminathan and Joachims (2015) show the same self-normalization tradeoff in importance sampling and counterfactual learning. Deng et al. (2013) gives the original CUPED presentation in online experimentation.

For extensions beyond the main lecture path, read Hahn (1998) with Hirano, Imbens, and Ridder (2003) for the semiparametric efficiency logic behind estimated propensities. Kohavi, Henne, and Sommerfield (2007), Crook et al. (2009), and Kohavi et al. (2026) connect estimator validity to experimentation practice and selective replication, while Parikh et al. (2026) treats estimator choice itself as a problem to be evaluated across a family of experiments.

Lecture brief: View the source `brief.md` · How these notes are made

References

L03: Experimental Design, Blocking, Balancing

Balance, rerandomization, and covariate-aware designs

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Use a grouped-assignment example to show why marginal treatment probabilities do not determine precision.
- Develop blocking and matched pairs as design-stage uses of pretreatment covariates.
- Contrast design-stage balance with regression and AIPW corrections made during analysis.
- Treat rerandomization as a complete design whose balance restrictions must also govern inference.

Ten coin flips disguised as one thousand

An experimental design is a probability distribution over complete assignment vectors, not merely a list of unit-level treatment probabilities. The distinction matters whenever assignments are dependent. As an extreme example, suppose $N = 1000$ units each have marginal treatment probability $1/2$. Partition them into ten groups of one hundred and flip one fair coin per group. Every unit in a group receives the same assignment.

The marginal probabilities match i.i.d. Bernoulli assignment, but the design contains only ten independent random decisions. Let $g = 1, \ldots, 10$ index the groups, let $\mathcal{J}_g$ be the set of units in group g, and let $A_g \in \{0, 1\}$ be the result of its coin flip. Thus $Z_i = A_g$ whenever $i \in \mathcal{J}_g$. With treatment probability $1/2$, the observable HT estimator (Horvitz and Thompson 1952) can be written first using observed outcomes and then, by consistency, using the corresponding potential outcomes:

$$\hat{\tau}_{\text{HT}} = \frac{1}{N}\sum_{i=1}^{N}\left\{2Z_i Y_i^{\text{obs}} - 2(1-Z_i)Y_i^{\text{obs}}\right\} = \frac{1}{N}\sum_{i=1}^{N}\left\{2Z_i Y_i(1) - 2(1-Z_i)Y_i(0)\right\}.$$

The second expression may look as though it uses both potential outcomes, but its assignment indicators select only the observed one: the $Y_i(0)$ term vanishes for a treated unit and the $Y_i(1)$ term vanishes for a control unit. Subtracting the fixed estimand and collecting units that share a coin gives the exact error representation

$$\hat{\tau}_{\text{HT}} - \tau = \frac{1}{N}\sum_{g=1}^{10}(2A_g - 1)\sum_{i\in\mathcal{J}_g}\{Y_i(1) + Y_i(0)\}.$$

The inner sum is a fixed group total; all assignment randomness in that group is the single sign $2A_g - 1$. Independence of the ten signs therefore gives

$$\text{Var}_Z(\hat{\tau}_{\text{HT}}) = \frac{1}{N^2}\sum_{g=1}^{10}\left[\sum_{i\in\mathcal{J}_g}\{Y_i(1) + Y_i(0)\}\right]^2. \tag{0.1}$$

Under i.i.d. unit-level Bernoulli assignment, the corresponding variance is

$$\text{Var}_Z(\hat{\tau}_{\text{HT}}) = \frac{1}{N^2}\sum_{i=1}^{N}\{Y_i(1) + Y_i(0)\}^2. \tag{0.2}$$

The contrast between Equation 0.1 and Equation 0.2 is the location of the square. In the grouped design, outcomes are summed within a group and then squared, so the one hundred units sharing a coin move together. Under unit-level assignment, each unit contributes its own squared term.

Take the stark special case $Y_i(1) = a$ and $Y_i(0) = b$ for every unit. The grouped design has variance $(a+b)^2/10$. That is exactly the i.i.d. Bernoulli variance for **ten** units, not one thousand. I.i.d. Bernoulli assignment of all one thousand units has variance $(a+b)^2/1000$.

This exact equivalence is specific to HT in the example with constant potential outcomes. It partly reflects HT's random total weight: conditional on at least one treated and one control group, difference in means is exactly $a-b$ and has zero variance. With heterogeneous group totals, however, every estimator's assignment variation still originates in only ten coin flips. The example therefore combines two lessons from L02 and L03: normalization matters, and dependence determines the effective source of randomization.

The point is not that dependence is always harmful. The positive dependence in this example makes many units move together. Negative dependence, by contrast, can force treatment and control counts or covariate totals to balance and thereby improve precision. What determines these properties is the **joint distribution of the whole assignment vector Z**, not the marginal probabilities considered one unit at a time.

Blocking: randomize among comparable units

The grouped example used dependence badly: it made every unit in a group receive the same treatment. Blocking uses dependence for the opposite purpose. Imagine a student experiment in which prior achievement is strongly predictive of the outcome. A simple complete randomization can, by chance, place more high-achieving students in one arm. Before assigning treatment, we can divide students into groups with similar prior achievement and require both treatment arms to appear within every group. The comparison is then made among students who were similar before treatment.

Formally, suppose pretreatment covariates divide the finite population into H nonoverlapping **blocks**, also called strata. Let $\mathcal{I}_h$ denote the units in block h and $N_h = |\mathcal{I}_h|$ its size. We choose in advance to treat exactly N_{1h} units and control $N_{0h} = N_h - N_{1h}$ units in that block. **Stratified complete randomization**, or blocking, performs complete randomization independently within each $\mathcal{I}_h$. Thus units in different blocks are randomized separately, while assignments inside a block are negatively dependent because their treated count is fixed.

For arm $z \in \{0, 1\}$, define the observed mean within block h by

$$\overline{Y}_{zh}^{\text{obs}} = \frac{1}{N_{zh}} \sum_{i \in \mathcal{I}_h} \mathbf{1}\{Z_i = z\} Y_i^{\text{obs}},$$

where $\mathbf{1}\{Z_i = z\}$ is the indicator that unit i was assigned to arm z, and N_{zh} is the fixed number assigned to that arm in block h. The within-block treatment-control contrast is

$$\hat{\tau}_h = \overline{Y}_{1h}^{\text{obs}} - \overline{Y}_{0h}^{\text{obs}}$$

Define the block-specific finite-population effect

$$\tau_h = \frac{1}{N_h} \sum_{i \in \mathcal{I}_h} \{Y_i(1) - Y_i(0)\}.$$

Complete randomization within the block makes $\hat{\tau}_h$ design-unbiased for τ_h. The population ATE is not an equally weighted average of the block effects unless all blocks have the same size: every unit, rather than every block, must receive equal weight. This yields

$$\hat{\tau}_{\text{block}} = \sum_{h=1}^{H} \frac{N_h}{N} \hat{\tau}_h. \tag{0.1}$$

The weights N_h/N are the fractions of experimental units in the blocks, so $\tau = \sum_h (N_h/N)\tau_h$ and Equation 0.1 is design-unbiased for τ. To describe its precision, let S_{zh}^2 be the finite-population

variance of $Y_i(z)$ among units in block h, and let $S^2_{\tau h}$ be the within-block variance of the individual effects τ_i. Applying Neyman's finite-population variance calculation inside each block and then adding across the independently randomized blocks gives (Splawa-Neyman, Dabrowska, and Speed 1990)

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{block}}) = \sum_{h=1}^{H} \left(\frac{N_h}{N}\right)^2 \left(\frac{S^2_{1h}}{N_{1h}} + \frac{S^2_{0h}}{N_{0h}} - \frac{S^2_{\tau h}}{N_h}\right). \tag{0.2}$$

The factor $(N_h/N)^2$ appears because each random within-block contrast enters the overall estimator multiplied by N_h/N. As in L02, $S^2_{\tau h}$ is generally unidentified: observing only one potential outcome per unit does not reveal the within-block variance of individual treatment effects. A usable conservative estimator drops these nonnegative terms and substitutes the observed within-block sample variances s^2_{zh} for S^2_{zh}:

$$\hat{V}_{\mathrm{Neyman}} = \sum_{h=1}^{H} \left(\frac{N_h}{N}\right)^2 \left(\frac{s^2_{1h}}{N_{1h}} + \frac{s^2_{0h}}{N_{0h}}\right).$$

This plug-in estimator requires enough observed units to estimate both within-block arm variances. In a matched-pair design there is only one observed outcome per arm in each pair, so uncertainty must instead be estimated from the collection of randomized pair contributions.

Blocking helps when the blocks capture pretreatment differences that predict outcomes. For example, if outcomes contain a large block-specific baseline component, that component is shared by the treated and control units compared within the block and cancels from their contrast. Between-block baseline differences therefore do not inflate the treatment comparison. Blocking also fixes arm counts within every block. With a common treatment fraction, it exactly balances the categorical block indicators; continuous covariates used to construct the blocks are balanced only approximately, to the extent that units placed together are similar.

Blocking is not free. Many fine blocks can be operationally difficult; attrition can break their intended structure; and a block must contain enough units to support the planned arm counts. Blocking on weakly prognostic variables can provide little gain. The design should also be preserved in the analysis: estimate effects within blocks and aggregate using the target-population weights.

Matched pairs

Matched pairs are the limiting case $N_h = 2$ and $N_{1h} = 1$. Suppose there are $H = N/2$ pairs, and label the two units in pair h by a and b. Let $A_h = 1$ when unit a is treated and $A_h = 0$ when unit b is treated. The signed observed difference

$$D_h = (2A_h - 1)\,(Y^{\mathrm{obs}}_{ha} - Y^{\mathrm{obs}}_{hb})$$

always means "treated outcome minus control outcome": when $A_h = 1$, it is $Y^{\mathrm{obs}}_{ha} - Y^{\mathrm{obs}}_{hb}$; when $A_h = 0$, the sign reverses. Its design expectation is the average of the two unit-level effects in pair h. Consequently,

$$\hat{\tau}_{\text{pair}} = \frac{1}{H} \sum_{h=1}^{H} D_h$$

is design-unbiased for the finite-population ATE. Because each pair supplies only one treated and one control observation, there is no within-arm variance estimate inside a pair. Precision is instead learned from how the randomized contributions D_h vary across pairs.

Pairing can be powerful because assignment only decides which of two deliberately similar units is treated. If the matched units have similar potential outcomes under each arm, the two possible values of D_h are close, reducing its assignment variance. Pairing concentrates randomization locally: each pair has only two possible assignments, although assignments remain independent across pairs. Y. Bai (2022) characterizes precision-optimal matched-pair designs, while Y. Bai, Romano, and Shaikh (2022) develops inference that respects the pairing. Poor matches can lock in bad comparisons, and missing outcomes are especially damaging. Matching should therefore be based on pretreatment information chosen for prognostic value, not on an urge to balance every measured variable.

Design balance versus analysis adjustment

Return to the prior-achievement example. We can use the baseline score **before assignment**, by blocking students with similar scores, or **after assignment**, by statistically correcting whatever baseline-score difference happened to arise. These are different interventions on the experiment. The first changes the assignment distribution; the second leaves the design alone and changes the estimator. The broader design-first principle is articulated by Rubin (2008), while Bruhn and McKenzie (2009) compares common randomization procedures in field-experiment practice:

- **At design:** restrict or structure assignments so covariate imbalance is small by construction.
- **At analysis:** allow a simpler assignment design and subtract the part of the outcome difference predicted by realized covariate imbalance.

To make the analysis-stage option precise, suppose X_i is a column vector of pretreatment covariates, $\overline{X}_z^{\text{obs}}$ is its observed sample mean in arm z, and β is a coefficient vector of the same dimension. Under two-arm complete randomization, consider the regression-adjusted form

$$\hat{\tau}_{\text{adj}}(\beta) = \hat{\tau}_{\text{DM}} - \beta^{\top}(\overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}).$$

The vector $\overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}$ is the realized covariate imbalance. Its inner product with β is the part of the raw outcome difference attributed to that imbalance, so the estimator subtracts it. If β is fixed with respect to assignment, the correction has design expectation zero because complete randomization gives each arm the same expected covariate mean. A coefficient aligned with the outcome–covariate relationship can therefore reduce assignment variance without changing the design expectation; a poorly chosen coefficient can instead increase variance.

A coefficient fitted from the experiment's outcomes is assignment-dependent and needs the finite-sample or asymptotic justification discussed in L02 (Freedman 2008; Lin 2013). Design balance is visible before outcomes are observed and protects a range of later analyses. Analysis adjustment can use continuous, high-dimensional, or nonlinear predictions more flexibly, but it requires appropriate estimation and uncertainty calculations. The two are complements: one can block on a few decisive variables and adjust for remaining prognostic variation.

This comparison also reveals a general principle. Design validity does not require a correct outcome model relating Y to X; its precision gains do. Balancing weak or wrongly represented features may provide little benefit, while concentrating the design too aggressively around one criterion can reduce robustness to outcome structure that criterion missed.

AIPW: prediction plus randomized correction

Covariate adjustment presents a useful tension. An outcome model can predict a missing potential outcome for every unit, but a misspecified model need not average to the correct finite-population mean. An HT estimator has the opposite strength: known assignment probabilities make it design-unbiased, but it may be noisy because it inverse-weights the full observed outcomes. **Augmented inverse probability weighting** (AIPW) combines these two starting points. It uses predictions to remove explainable variation and uses randomization to correct the predictions rather than trusting them outright.

The target for arm z is the finite-population mean $\overline{Y}(z)$. Let $m_z(x)$ predict the potential outcome under arm z from pretreatment covariates. For the moment, treat the N numbers $m_z(X_i)$ as fixed before assignment; they need not equal the actual potential outcomes. Also assume the relevant positivity condition

$$\pi_i(z) = \mathbb{P}_Z(Z_i = z) > 0 \qquad \text{for every unit and arm under study.}$$

Here $\pi_i(z)$ is the known probability that the design assigns unit i to arm z. Positivity ensures that every unit whose potential outcome contributes to $\overline{Y}(z)$ has some chance of revealing that outcome.

The population average of the predictions, $N^{-1}\sum_i m_z(X_i)$, is available because the covariates are observed for all experimental units. Correct it using the prediction errors observed in arm z:

$$\hat{\mu}_{z,\text{AIPW}} = \frac{1}{N}\sum_{i=1}^{N}\left[m_z(X_i) + \frac{\mathbf{1}\{Z_i = z\}}{\pi_i(z)}\{Y_i^{\text{obs}} - m_z(X_i)\}\right], \tag{0.1}$$

The indicator $\mathbf{1}\{Z_i = z\}$ is one exactly when unit i is assigned to arm z. Thus the second term is zero for units outside that arm and equals the observed residual divided by its assignment probability for units inside it. Subtracting the two arm-specific mean estimators gives

$$\hat{\tau}_{\text{AIPW}} = \hat{\mu}_{1,\text{AIPW}} - \hat{\mu}_{0,\text{AIPW}}$$

as an estimator of $\tau = \overline{Y}(1) - \overline{Y}(0)$ (Robins, Rotnitzky, and Zhao 1994; Tsiatis et al. 2008). The GREG construction in L02 has the same prediction-plus-weighted-residual architecture; AIPW writes it arm by arm and allows general prediction functions.

The construction can be read in three steps:

1. predict every unit's potential outcome under arm z;
2. among units actually observed in arm z, inverse-weight the prediction error;
3. add that correction to the population average prediction.

The correction works because, for each unit separately, inverse-probability weighting recovers the full prediction error. Since $Y_i^{\text{obs}} = Y_i(z)$ on the event $Z_i = z$,

$$\mathbb{E}_Z\left[\frac{\mathbf{1}\{Z_i = z\}}{\pi_i(z)}\{Y_i^{\text{obs}} - m_z(X_i)\}\right] = Y_i(z) - m_z(X_i).$$

Adding the fixed prediction back therefore gives design unbiasedness without assuming that the prediction is correct:

$$\begin{aligned}\mathbb{E}_Z(\hat{\mu}_{z,\text{AIPW}}) &= \frac{1}{N}\sum_i [m_z(X_i) + Y_i(z) - m_z(X_i)] \\ &= \overline{Y}(z).\end{aligned}$$

This is the central division of labor. The model supplies a precision device; the randomization correction supplies validity for the finite-population target. When $m_z(X_i)$ is close to $Y_i(z)$, the residuals passed through the random inverse-probability weights are smaller and the estimator can be more precise than HT. A poor fixed predictor does not destroy design unbiasedness, though it can increase variance.

> ⚠ Fitted predictions are part of the estimator
>
> The exact argument above treats m_z as fixed with respect to assignment. An externally trained or otherwise preassignment predictor preserves that exact conditional design-unbiasedness. If the same randomized outcomes are used to fit m_z, exact finite-sample unbiasedness does not follow automatically. Under independent assignment, carefully constructed leave-one-out or cross-fitted predictions can separate a unit's assignment from its prediction. Under complete, blocked, or rerandomized designs, assignments are dependent, so cross-fitting alone is not a universal finite-sample fix; one needs a design-specific argument or suitable asymptotics. Randomization remains the foundation, but "the model may be wrong" does not mean "the fitting procedure is irrelevant."

In observational causal inference, AIPW is often introduced through double robustness. That is not the focus here: the assignment probabilities in a randomized experiment are known. Our emphasis is the control-variate interpretation—prediction for precision, inverse-probability correction for the design-based target.

Pair matching in cluster-randomized experiments

Sometimes treatment is assigned to schools, villages, or clinics while outcomes are measured on individuals. With few clusters, ordinary cluster randomization can leave large baseline differences. Imai, King, and Nall (2009) argue for pairing similar clusters and randomizing one cluster within each pair; Middleton and Aronow (2015) develops the design-based estimand and weighting issues for cluster-randomized experiments.

The distinction between the unit of assignment and the unit of measurement is essential. Suppose there are K pairs indexed by $k = 1, \ldots, K$, and pair k contains clusters $j = 1, 2$. Cluster (k, j) contains n_{kj} individuals, and $N = \sum_{k=1}^{K} \sum_{j=1}^{2} n_{kj}$ is the total number of individuals. Under cluster-level treatment and no interference between clusters, define the cluster potential-outcome total

$$T_{kj}(z) = \sum_{\ell=1}^{n_{kj}} Y_{kj\ell}(z),$$

where ℓ indexes individuals within a cluster. The individual-weighted finite-population ATE is then

$$\tau = \frac{1}{N} \sum_{k=1}^{K} \sum_{j=1}^{2} \{T_{kj}(1) - T_{kj}(0)\}.$$

Let $A_k = 1$ mean that cluster 1 in pair k is treated; when $A_k = 0$, cluster 2 is treated. The two possibilities occur with probability $1/2$, independently across pairs. Write T_{kj}^{obs} for the total under the assignment cluster (k, j) actually receives. The HT estimator of the individual-weighted ATE is

$$\hat{\tau}_{\text{HT}} = \frac{2}{N} \sum_{k=1}^{K} (2A_k - 1)\{T_{k1}^{\text{obs}} - T_{k2}^{\text{obs}}\}.$$

The sign $2A_k - 1$ always orients the expression as treated-cluster total minus control-cluster total. The leading factor 2 is the inverse of each cluster's treatment or control probability. Averaging over the two assignments in pair k recovers the sum of the treatment effects for all individuals in both clusters; dividing by N therefore targets the individual-weighted ATE above.

Each pair contributes one independent randomized contrast, so design-based uncertainty is built from variation across those pair contributions. Two details are easy to miss:

- The effective randomization units are clusters, not the individuals nested within them.
- Weighting determines the estimand. The total-based HT expression targets an average over individuals. Averaging cluster means equally instead targets a cluster-weighted or pair-weighted effect; unequal cluster sizes make the distinction consequential.

Pair matching can substantially improve precision when matches are prognostic, but the analysis must honor both pairing and unequal cluster sizes. Treating all individual observations as independently randomized produces fictitious precision.

Rerandomization

Blocking is attractive when units fall naturally into small groups within which fixed treatment counts can be enforced. With several continuous covariates, however, useful blocks may be difficult to define. **Rerandomization** takes a different route: repeatedly draw an assignment from a familiar proposal design, measure its global covariate imbalance, and retain the draw only if that imbalance meets a prespecified standard (Morgan and Rubin 2012). It changes the design before outcomes are seen; it is not post hoc deletion of an inconvenient randomization.

Write $\mathbb{P}_{\mathrm{CR}}$ for the proposal complete-randomization design and assume equal-sized treatment arms. For the pretreatment covariate vector X_i, let

$$\hat{\tau}_X = \overline{X}_1^{\mathrm{obs}} - \overline{X}_0^{\mathrm{obs}}$$

be the vector of treated-minus-control covariate mean differences. Zero is perfect mean balance. Different coordinates may have different units and may be strongly correlated, so an ordinary Euclidean norm is not a meaningful common scale. Assuming the covariance matrix is nonsingular, the Mahalanobis balance score (Mahalanobis 1936) standardizes the imbalance by its covariance under the proposal design:

$$M(\mathbf{Z}) = \hat{\tau}_X^{\top} \left\{ \mathrm{Cov}_{\mathrm{CR}}(\hat{\tau}_X) \right\}^{-1} \hat{\tau}_X. \tag{0.1}$$

The covariance subscript "CR" matters: $\mathrm{Cov}_{\mathrm{CR}}(\hat{\tau}_X)$ describes how covariate imbalance would vary over repeated proposal assignments, with the covariates held fixed. Consequently, $M(\mathbf{Z})$ is a scalar standardized squared distance from exact balance. With a single covariate, it is simply the squared mean difference divided by its complete-randomization variance.

Choose a threshold a before assignment so that the acceptance event has positive proposal probability. Repeatedly draw a complete randomization and accept the first draw satisfying $M(\mathbf{Z}) \le a$. The actual rerandomization design, denoted $\mathbb{P}_{\mathrm{ReM}}$, is the proposal distribution conditional on acceptance:

$$\begin{aligned} \mathbb{P}_{\mathrm{ReM}}(\mathbf{Z} = \mathbf{z}) &= \mathbb{P}_{\mathrm{CR}}\{\mathbf{Z} = \mathbf{z} \mid M(\mathbf{Z}) \le a\} \\ &= \frac{\mathbb{P}_{\mathrm{CR}}(\mathbf{Z} = \mathbf{z}) \mathbf{1}\{M(\mathbf{z}) \le a\}}{\mathbb{P}_{\mathrm{CR}}\{M(\mathbf{Z}) \le a\}}. \end{aligned} \tag{0.2}$$

If complete randomization is uniform over assignments with the prescribed arm sizes, rerandomization is uniform over the subset that passes the balance rule. This is still a randomized design, provided the covariates, score, and threshold are fixed in advance and the acceptance procedure is recorded. Ad hoc rejection until the groups merely "look nice" does not define a reproducible assignment law.

What rerandomization buys

Let $\mathbf{1}_N$ denote the length-N vector of ones, so the treatment-control complement of $\mathbf{z}$ is $\mathbf{1}_N - \mathbf{z}$. If the acceptance rule is symmetric—$\mathbf{z}$ is accepted exactly when its complement is—then every accepted assignment has an equally likely accepted partner that reverses all arm labels. With equal arm sizes, each unit therefore retains marginal treatment probability 1/2, and difference in means remains design-unbiased. The joint assignment distribution has nevertheless changed: assignments with large $M(\mathbf{Z})$ now have probability zero. When the balanced covariates predict outcomes, removing those assignments can substantially reduce variance.

But the analysis must use the design that was actually run. An exact Fisher randomization test of a sharp null should sample from accepted assignments, not from all complete randomizations. Under the regularity conditions studied by X. Li, Ding, and Rubin (2018), the usual Neyman normal interval calculated as if complete randomization had been used is asymptotically conservative; this is not a universal finite-sample statement.

The large-sample result of X. Li, Ding, and Rubin (2018) explains the mechanism. Decompose the outcome variation into a component linearly associated with the balanced covariates and a residual component orthogonal to them. Under their regularity conditions, the difference-in-means fluctuation likewise separates into two independent pieces. The residual piece remains asymptotically Gaussian because the acceptance rule does not directly constrain it. The covariate-associated piece has a Gaussian limit truncated to the accepted balance region. Rerandomization therefore shrinks precisely the part of the estimator's variation aligned with X. The precision gain is larger when that part explains more of the outcome and when the threshold is more stringent. X. Li and Ding (2020) then studies how regression adjustment and rerandomization interact rather than treating design- and analysis-stage balance as substitutes.

The robustness tradeoff

Rerandomization has an appealing operational story and preserves multiple acceptable assignments. Its weaknesses are equally instructive:

- a stringent threshold lowers the acceptance probability and concentrates the assignment distribution;
- Mahalanobis balance is only as relevant as the chosen covariates and feature representation; in high dimensions, covariance regularization also changes the metric;
- for a fixed acceptance probability, increasing dimension weakens the attainable improvement in each direction and makes covariance estimation or regularization more consequential;
- unrecorded changes to the acceptance rule invalidate the stated design;
- analysis that pretends complete randomization was used mischaracterizes uncertainty.

The threshold therefore mediates balance against robustness. Making a smaller improves balance along the chosen covariate directions, but it leaves fewer possible assignments and makes the design more dependent on whether those directions capture the outcome structure that matters. Here robustness includes both protection against a misspecified balance criterion and enough randomness to support credible, reproducible inference. Perfect observed balance is not the only goal of a randomized experiment.

What to carry forward

Experimental design controls the dependence among assignments. Blocking uses scientifically meaningful local comparisons; matching makes those blocks as small as possible; rerandomization conditions on acceptable global balance. Analysis adjustment and AIPW can recover additional precision after a simpler design. In every case, the estimator and its uncertainty calculation must correspond to the assignment distribution that was actually used.

Advanced material

Gram–Schmidt walk designs

The word **discrepancy** describes a signed imbalance. To see the connection to experiments, suppose the treatment and control arms have equal size and define the assignment sign $s_i = 2Z_i - 1$, which is $+1$ for treatment and -1 for control. Then

$$\hat{\tau}_X = \overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}} = \frac{2}{N} \sum_{i=1}^{N} s_i X_i.$$

Thus choosing signs that make the vector sum $\sum_i s_i X_i$ small is exactly choosing an assignment with small covariate mean imbalance. Discrepancy theory studies how signs can be chosen so that many coordinates or other linear summaries remain small simultaneously. The norm used to judge the signed sum specifies which kinds of imbalance matter.

A purely discrepancy-minimizing algorithm might return one deterministic assignment. The Gram–Schmidt walk (GSW) instead turns a discrepancy-control procedure into a randomized assignment design (Harshaw et al. 2024). It generates the signs through a random walk while steering their covariate sum toward balance.

The design has a tuning parameter that governs how aggressively the walk prioritizes covariate balance relative to unconstrained randomization. At one extreme it behaves more like conventional random assignment; at the other it pushes harder to keep the linear discrepancy small. Crucially, it retains prescribed unit-level marginals, such as $\mathbb{P}_Z(Z_i = 1) = 1/2$, while inducing dependence among assignments. The resulting guarantees concern design-based mean squared error over specified classes of outcome structure. This separates GSW from selecting a single best-looking assignment: its balance comes from a probability distribution over assignments, not from abandoning randomization.

The guarantee also states its own limitation: linear balance protects outcomes whose conditional mean is well represented by the chosen linear features. Cytrynbaum (2026) shows that uniformly approaching classical efficiency over broad smooth outcome classes becomes impossible once dimension grows too quickly. The paper develops a nonparametric, kernelized GSW in a reproducing-kernel Hilbert space and identifies more favorable rates under additive structure. Kernelization expands the function class being balanced; it does not eliminate the need to say which structure is plausible.

Correlated cluster designs without rich covariates

The paired-cluster discussion used covariates to decide which clusters should be compared. Candogan, Chen, and Niazadeh (2024) ask a different question: can dependence be designed robustly when cluster size is the principal available structure? They study correlated cluster assignments with equal marginal treatment probabilities and a cluster-aware HT estimator, choosing dependence to minimize a worst-case variance criterion.

Their imbalance-reducing construction uses cluster sizes rather than a rich outcome-prediction model. The asymptotic guarantees require regularity, including that no single cluster dominate the population. The result is therefore neither covariate matching nor a claim that size is always the most prognostic variable. It shows that assignment correlation can be optimized against a stated worst-case class even when conventional matching information is sparse.

Optimization-based designs and the value of randomness

Once discrepancy has been written as a function of the assignment signs, one can search directly for the assignment minimizing it—for example, minimizing a norm of $\sum_i s_i X_i$ subject to fixed arm sizes. Mixed-integer optimization can produce striking balance in small samples (Bertsimas, Johnson, and Kallus 2015), and minimax formulations connect classes of plausible outcome functions to the imbalance metric that should be optimized (Kallus 2018). The objective is never neutral: choosing a norm, feature map, or function class says which outcome patterns the design is intended to protect.

Selecting one minimizing assignment eliminates design-based randomization. Randomizing between that assignment and its treatment-control complement restores unit-level marginals of 1/2, but leaves only two assignments in the support. Either construction can be brittle to aspects of the outcome surface omitted from the objective. More diffuse randomization distributes that risk. Kallus (2021) formalizes this through mixed-strategy optimal designs: with no informative structure, complete randomization is minimax; gains require restrictions on plausible outcome functions, and inference supplies a reason to retain sufficiently diffuse randomization.

GSW, rerandomization, and mixed-strategy optimization occupy different points on the same frontier. Each asks how much imbalance to remove while preserving enough randomness for robustness and inference.

Efron's biased coin

Efron's biased-coin design addresses a sequential problem: how can an experiment resist persistent arm-size imbalance without making its next assignment completely predictable? (Efron 1971) After n assignments, let $D_n = N_{1n} - N_{0n}$ be the treated count minus the control count. If $D_n = 0$, assign the next unit by a fair coin. If $D_n > 0$, treatment is overrepresented, so assign the next unit to control with a chosen probability $q > 1/2$; if $D_n < 0$, assign it to treatment with that probability. The parameter q controls the pull back toward equal counts.

At $q = 1/2$, the feedback disappears and the rule is ordinary i.i.d. Bernoulli assignment. As q rises toward one, balance is restored more forcefully but future assignments become easier to predict. The

design therefore makes the balance–randomness tradeoff explicit. Covariate-adaptive procedures such as Pocock and Simon (1975) generalize the same feedback idea to prognostic-factor imbalance, but the dependence created by that feedback must be reflected in the analysis.

D-, A-, and E-optimal design

Section 7.5 of Boyd and Vandenberghe (2004) gives a modern convex-optimization treatment of a classical model-based design problem whose general theory traces to Kiefer (1959). Suppose measurement option r has a known regressor vector v_r, and let $\lambda_r \geq 0$ be the fraction of measurement effort allocated to that option, with $\sum_r \lambda_r = 1$. In a homoskedastic linear model, the normalized information matrix is

$$\mathcal{I}(\lambda) = \sum_r \lambda_r v_r v_r^\top.$$

When this matrix is nonsingular, the coefficient covariance is proportional to $\mathcal{I}(\lambda)^{-1}$. The design problem chooses the allocation λ to make that covariance small, but a matrix is not ordered by a single universally best scalar. Common criteria emphasize different geometries:

- **D-optimality** maximizes $\det\{\mathcal{I}(\lambda)\}$, equivalently shrinking the volume of the coefficient confidence ellipsoid.
- **A-optimality** minimizes $\mathrm{tr}\{\mathcal{I}(\lambda)^{-1}\}$, the sum of the marginal coefficient-variance proxies.
- **E-optimality** maximizes the smallest eigenvalue of $\mathcal{I}(\lambda)$, protecting the least-informed coefficient direction.

These criteria are connected to covariate-balanced treatment design through convex optimization and worst-direction reasoning. They are not the same problem. Classical optimal design begins from a response model and chooses where to measure parameters; GSW chooses a randomized treatment allocation and seeks design-based protection over outcome classes. The analogy is most useful when it makes the assumed geometry explicit.

Staggered rollouts beyond SUTVA

In a staggered rollout, clusters or units adopt treatment at different assigned times. The stepped-wedge formulation of Hussey and Hughes (2007) is a canonical example. The design creates comparisons across adoption cohorts and calendar periods. Operational constraints often motivate it, but the schedule can also generate deliberate variation in treatment saturation and exposure duration.

That variation is valuable when no-interference or no-carryover assumptions are implausible: outcomes can be indexed by neighbors' adoption, local saturation, or time since treatment rather than by a single binary label. The cost is that calendar-time trends, anticipation, and dynamic effects become part of the estimand and design. A rollout is therefore not merely a delayed two-arm experiment; it requires a potential-outcome table indexed by assignment histories.

Further reading

For historical orientation, Fisher (1935) develops randomization and blocking, while Rubin (2008) states the modern design-before-analysis principle. Ding (2023) provides a concise contemporary treatment of finite-population experiments and rerandomization. For matched designs, read Y. Bai (2022) and Y. Bai, Romano, and Shaikh (2022) alongside the cluster-randomized perspective of Imai, King, and Nall (2009) and Middleton and Aronow (2015).

For rerandomization, begin with Morgan and Rubin (2012) for the design principle, then read X. Li, Ding, and Rubin (2018) for the non-Gaussian asymptotic distribution induced by acceptance and X. Li and Ding (2020) for its interaction with regression adjustment. Candogan, Chen, and Niazadeh (2024) shows how cluster-assignment dependence can instead be optimized from cluster-size information and a worst-case criterion.

The modern balance–robustness sequence runs from optimization-based designs (Bertsimas, Johnson, and Kallus 2015; Kallus 2018, 2021) to randomized discrepancy control (Harshaw et al. 2024) and its recent nonlinear extension (Cytrynbaum 2026). Efron (1971) and Pocock and Simon (1975) provide sequential counterparts: use randomization feedback to resist arm-size or covariate imbalance without becoming deterministic. For classical information-matrix optimality, pair Kiefer (1959) with the computational treatment in Boyd and Vandenberghe (2004); for staggered cluster rollouts, see Hussey and Hughes (2007).

Lecture brief: View the source `brief.md` · How these notes are made

References

L04: Positivity, Trimming, Target Populations, and Partial Identification

What can a design learn when some comparisons are rare or absent?

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Begin with randomized designs that make some treatment comparisons rare or impossible.
- Separate positivity, population overlap, and overlap in one realized experiment.
- Compare clipping with two explicit target changes: trimming and weighting toward equipoise.
- When the original target must remain, replace an unsupported point claim with bounds justified by boundedness, monotonicity, or smoothness.

A randomized experiment can still lack a comparison

Consider a randomized trial conducted in three prespecified eligibility strata. The investigator uses different treatment probabilities because the treatment is increasingly difficult to offer:

Stratum	Number of units	Treatment probability	Expected treated units
Standard eligibility	80	0.50	40
Elevated risk	15	0.05	0.75
Logistical exclusion	5	0	0

The first stratum supplies an ordinary randomized comparison. Treatment remains possible in the elevated-risk stratum, but it is rare: the probability that none of its 15 units is treated is $0.95^{15} \approx 0.46$. A logistical rule makes treatment impossible in the final stratum under the experiment as conducted.

Suppose the scientific target is the finite-population average treatment effect over all 100 trial units. Two different problems arise.

First, the treatment effects of the five logistically excluded units enter the target, but their treated outcomes can never be exposed by this design. This is an **identification** problem. Repeating the same experiment indefinitely would not reveal the missing comparison.

Second, a treated elevated-risk unit can appear, but there may be only one—or none—in the realized experiment. An inverse-probability estimator asks any one such unit to represent about $1/0.05 = 20$ similar units. This is a **precision and finite-sample stability** problem. The comparison exists in the design, but the realized experiment may contain little information about it.

The distinction between zero and merely small probability is the central distinction of this lecture. It also determines which remedies can be honest: no amount of weight regularization turns a structurally impossible assignment into randomized evidence.

Positivity in a finite randomized design

Return to the finite population of N units used in L01–L03. Potential outcomes $Y_i(1)$ and $Y_i(0)$, covariates X_i, and the target

$$\tau = \frac{1}{N}\sum_{i=1}^{N}\{Y_i(1) - Y_i(0)\}$$

are fixed. The assignment vector $\mathbf{Z}$ is random under a known design. For each unit and treatment state, define the marginal design probability

$$\pi_i(z) = \mathbb{P}_Z(Z_i = z), \qquad z \in \{0, 1\}.$$

For a binary treatment, $\pi_i(1) = \pi_i$ and $\pi_i(0) = 1 - \pi_i$. Design positivity for the finite-population ATE means

$$\pi_i(0) > 0 \quad \text{and} \quad \pi_i(1) > 0 \qquad \text{for every } i. \tag{0.1}$$

The condition is relative to the estimand. If unit i contributes both potential outcomes to τ, the design must give both outcomes a chance to be observed. If the target deliberately excludes unit i, its assignment probabilities need not identify that restricted target.

This is a condition on marginal exposure probabilities, not on every complete assignment vector. Under complete randomization, most of the 2^N binary assignment vectors have probability zero because the treated count is fixed. Nevertheless, every unit can appear in either arm, so Equation 0.1 holds whenever both arm sizes are positive.

i Positivity and overlap are related, but not identical

Positivity is a property of a treatment or exposure mechanism relative to a target: every state needed for the target has positive probability. In a randomized experiment, this is known from the design.
Population overlap usually describes the same support requirement through covariates. If assignment depends on X, the treated and control populations overlap at x when both treatment probabilities are positive there. In an observational study this becomes $0 < e(x) < 1$, where $e(x) = \mathbb{P}(Z = 1 \mid X = x)$.
Realized covariate overlap describes the sample actually obtained. Even when the design has positivity, random assignment can leave a small stratum with only treated or only control units. That realization may make estimation unstable, but it does not retroactively change the design probabilities. Conversely, seeing nearby treated and control units in a sample does not prove that an observational assignment mechanism is unconfounded.

In practice, "weak positivity" and "weak overlap" are often used for probabilities close to zero or one. Strictly speaking, positivity is binary: the probability is positive or it is not. The adjective *weak* signals that a formally positive design may contain so little information in some regions that its estimators behave poorly.

A stronger quantitative condition is often called **uniform** or **strong positivity**: for some fixed $\varepsilon > 0$,

$$\varepsilon \leq \pi_i \leq 1 - \varepsilon \qquad \text{for every target unit } i.$$

Ordinary positivity permits probabilities arbitrarily close to zero or one; strong positivity rules them out.

What the inverse probability is doing

The basic identity is already present in the Horvitz–Thompson estimator:

$$\mathbb{E}_Z\left\{\frac{\mathbf{1}\{Z_i = z\}Y_i^{\text{obs}}}{\pi_i(z)}\right\} = Y_i(z), \tag{0.1}$$

provided $\pi_i(z) > 0$. The indicator reveals $Y_i(z)$ only with probability $\pi_i(z)$; division by that known probability restores one full unit of that potential outcome in design expectation.

L03 used the same identity to correct prediction errors rather than outcome levels. Let $g_z(X_i)$ be a prediction fixed before assignment and define the fixed residual

$$r_i(z) = Y_i(z) - g_z(X_i).$$

Under independent Bernoulli assignment with unit-specific probabilities π_i, consider

$$\hat{\tau}(g) = \frac{1}{N}\sum_{i=1}^{N}\left[g_1(X_i) - g_0(X_i) + \frac{Z_i r_i(1)}{\pi_i} - \frac{(1-Z_i)r_i(0)}{1-\pi_i}\right]. \tag{0.2}$$

The prediction supplies a treatment contrast for every unit. Randomization then corrects its errors. When $0 < \pi_i < 1$,

$$\mathbb{E}_Z\left\{\frac{Z_i r_i(1)}{\pi_i}\right\} = r_i(1), \qquad \mathbb{E}_Z\left\{\frac{(1-Z_i)r_i(0)}{1-\pi_i}\right\} = r_i(0),$$

so $\mathbb{E}_Z\{\hat{\tau}(g)\} = \tau$ for any fixed predictions. The inverse probability is not a generic penalty for unlikely observations. It undoes the probability with which the design exposes the corresponding residual.

If $\pi_i = 0$, the treated identity and residual correction do not become "infinitely large." They cease to be defined, because $Z_i Y_i(1)$ is always zero and the design never reveals $Y_i(1)$. A model can still supply $g_1(X_i)$, but randomization cannot check or correct that model at unit i. The resulting claim is an extrapolation claim rather than a design-unbiased one.

Weak probabilities enter the variance

The same representation shows what happens before probability reaches zero. Subtracting the design expectation of the ith summand in Equation 0.2 gives

$$(Z_i - \pi_i)\left\{\frac{r_i(1)}{\pi_i} + \frac{r_i(0)}{1-\pi_i}\right\}. \tag{0.3}$$

Under independent Bernoulli assignment, contributions from different units are independent and $\mathrm{Var}_Z(Z_i) = \pi_i(1-\pi_i)$. Therefore

$$\mathrm{Var}_Z\{\hat{\tau}(g)\} = \frac{1}{N^2}\sum_{i=1}^{N} \pi_i(1-\pi_i)\left\{\frac{r_i(1)}{\pi_i} + \frac{r_i(0)}{1-\pi_i}\right\}^2. \tag{0.4}$$

The formula makes two mechanisms visible.

1. A small π_i magnifies the treated residual $r_i(1)$; a value near one magnifies the control residual $r_i(0)$.
2. Better fixed predictions can reduce the residuals, but cannot change which assignments the design permits.

The formula is exact for independent Bernoulli assignment. Under complete randomization, rerandomization, or another dependent design, cross-unit covariances also enter. The support conclusion is unchanged: weighting can correct unequal positive probabilities, but it cannot manufacture a treatment state with probability zero.

Four responses, four different claims

Once support is inadequate, there is no method that simultaneously preserves the original target, relies only on the original randomized evidence, retains a point conclusion, and enjoys stable precision. Something must change.

Response	What changes?	What can be claimed?
Clip or cap inverse weights	The estimator	A more stable but generally biased point estimate
Trim the population	The target units	A point estimate for an explicitly restricted population
Weight toward equipoise	The target weights	A point estimate for a smooth, design-supported population
Bound unsupported outcomes	The form of the conclusion and the assumptions	An identified set for the original target

A fifth response is often best: redesign the experiment, recruit different units, or collect more data in the poorly represented strata. Statistical adjustments are not automatically preferable to fixing the source of the information problem.

Clipping: regularize an estimator

Begin with the unaugmented Horvitz–Thompson estimator,

$$\hat{\tau}_{\mathrm{HT}} = \frac{1}{N}\sum_{i=1}^{N}\left\{\frac{Z_i Y_i^{\mathrm{obs}}}{\pi_i} - \frac{(1-Z_i)Y_i^{\mathrm{obs}}}{1-\pi_i}\right\}.$$

Suppose all probabilities are positive but some inverse probabilities are uncomfortably large. Choose a cap $C > 1$ and replace each inverse probability by

$$c_{i1} = \min\left(\frac{1}{\pi_i}, C\right), \qquad c_{i0} = \min\left(\frac{1}{1-\pi_i}, C\right). \tag{0.1}$$

This operation is called weight **clipping**, **truncation**, or **Winsorization**. The clipped estimator is

$$\hat{\tau}_C = \frac{1}{N}\sum_{i=1}^{N}\{Z_i c_{i1} Y_i^{\mathrm{obs}} - (1-Z_i)c_{i0}Y_i^{\mathrm{obs}}\}. \tag{0.2}$$

Clipping limits how much any rare assignment can contribute. It also breaks the cancellation that made Horvitz–Thompson unbiased. Taking the design expectation gives

$$\mathbb{E}_Z(\hat{\tau}_C) = \frac{1}{N}\sum_{i=1}^{N}\left[\min(1, C\pi_i)Y_i(1) - \min\{1, C(1-\pi_i)\}Y_i(0)\right]. \tag{0.3}$$

Whenever $\pi_i < 1/C$, the treated potential outcome is attenuated by $C\pi_i$; whenever $1-\pi_i < 1/C$, the control potential outcome is attenuated by $C(1-\pi_i)$. These arm-specific factors generally do not define one common target population. Thus clipping is not automatically unbiased for a neatly reweighted causal estimand.

For example, if $\pi_i = 0.02$ and $C = 10$, the treated contribution appears in expectation with multiplier $C\pi_i = 0.2$, not one. The cap removes 80 percent of that unit's treated mean contribution in exchange for limiting its realized weight to ten.

Normalizing the clipped treated and control weights to sum to one does not restore the original ATE. It makes each arm represent its own tilted population, and the two tilts need not agree. The practical gain is variance control; the price is design bias and increased dependence on outcome patterns in the clipped region.

Clipping can still be useful as an explicit regularization or sensitivity analysis. A credible report varies C and shows how the estimate, the largest weight, and an effective-sample-size diagnostic change. It should not present the clipped estimate as though the original randomization identity

still held. Cole and Hernán and Lee, Lessler, and Stuart discuss this practical use of truncated inverse weights in observational analyses (2008; 2011).

Clipping also cannot repair a structural zero. If $\pi_i = 0$, writing down a finite substitute for $1/\pi_i$ does not create a treated observation for unit i.

Trimming: change the target explicitly

A more transparent response is to name the population the design can reasonably represent. Let $A \subseteq \{1, \dots, N\}$ be a set of units defined using design probabilities and pretreatment information, before outcomes are examined, and require $0 < \pi_i < 1$ for every $i \in A$. Write $N_A = |A|$. The finite-population effect in this restricted target is

$$\tau_A = \frac{1}{N_A} \sum_{i \in A} \{Y_i(1) - Y_i(0)\}. \tag{0.1}$$

For fixed predictions $g_z(X_i)$, a design-unbiased estimator is

$$\hat{\tau}_A(g) = \frac{1}{N_A} \sum_{i \in A} \left[g_1(X_i) - g_0(X_i) + \frac{Z_i r_i(1)}{\pi_i} - \frac{(1 - Z_i) r_i(0)}{1 - \pi_i} \right]. \tag{0.2}$$

This estimator does not pretend to recover the effects of excluded units. It estimates a different, explicitly stated estimand. If the five logistically excluded units in the opening example are excluded, positivity may hold for the remaining 95 units. Excluding the elevated-risk stratum as well may produce a more stable estimate for 80 standard-eligibility units, but it changes the question again.

Under Bernoulli assignment, the exact residual variance of Equation 0.2 is

$$\mathrm{Var}_Z\{\hat{\tau}_A(g)\} = \frac{1}{N_A^2} \sum_{i \in A} v_i, \tag{0.3}$$

where

$$v_i = \pi_i(1 - \pi_i) \left\{ \frac{r_i(1)}{\pi_i} + \frac{r_i(0)}{1 - \pi_i} \right\}^2. \tag{0.4}$$

Removing a unit with a large v_i reduces the numerator but also reduces N_A, which increases the normalization applied to every retained unit. Trimming is therefore a **precision–relevance tradeoff**, not a rule to delete every difficult observation.

There is an important practical obstacle: v_i contains both potential outcomes and is not known before assignment. A feasible rule must replace it with a prespecified proxy or an estimate based on pretreatment covariates and appropriately separated outcome information. Once the same data choose the target and estimate its effect, inference must account for that adaptivity.

The superpopulation counterpart

The modern trimming literature is usually written for an i.i.d. superpopulation. Let X be pretreatment covariates,

$$e(x) = \mathbb{P}(Z = 1 \mid X = x), \qquad m_z(x) = \mathbb{E}\{Y(z) \mid X = x\},$$

and let

$$\sigma_z^2(x) = \mathrm{Var}\{Y(z) \mid X = x\}.$$

In an observational study, interpreting these quantities causally also requires consistency and conditional exchangeability; overlap alone is not enough. In a randomized design whose probabilities depend on X, the assignment probabilities are known and exchangeability follows from the design.

For a covariate set $\mathcal{A}$ with positive population probability, the target is

$$\tau_{\mathcal{A}} = \mathbb{E}\{Y(1) - Y(0) \mid X \in \mathcal{A}\}. \tag{0.5}$$

Crump, Hotz, Imbens, and Mitnik study the choice of such target populations when overlap is limited (2009). A central conditional residual-variance score is

$$k(x) = \frac{\sigma_1^2(x)}{e(x)} + \frac{\sigma_0^2(x)}{1 - e(x)}. \tag{0.6}$$

For a retained covariate set $\mathcal{A}$, the corresponding residual criterion is

$$V(\mathcal{A}) = \frac{\mathbb{E}[\mathbf{1}\{X \in \mathcal{A}\}k(X)]}{\mathbb{P}(X \in \mathcal{A})^2}. \tag{0.7}$$

Its minimizer has a threshold form: retain covariate values with $k(x) \leq \gamma$ for an appropriate γ. Under equal homoskedastic conditional variances,

$$k(x) = \frac{\sigma^2}{e(x)\{1 - e(x)\}},$$

so this rule reduces to a symmetric propensity-score region such as $\alpha \leq e(x) \leq 1-\alpha$. A conventional cutoff like 0.1 is not a universal optimum; the appropriate set depends on the probability and outcome structure.

Khan and Ugander extend this logic to heteroscedastic outcomes and develop doubly robust, sample-adaptive trimming procedures (2025). Trimming only on $e(x)$ can discard a rare but highly predictable region or retain a noisy region near one-half. The more refined score recognizes that weak overlap and residual uncertainty jointly determine precision.

The criterion in Equation 0.7 conditions on the covariates and focuses on residual variation. The full superpopulation variance also contains heterogeneity in $m_1(X) - m_0(X)$ within the target. One must specify the inferential regime before calling a trimming rule "optimal."

A numerical precision–relevance tradeoff

Suppose 90 percent of a source population has $e(X) = 0.5$ and 10 percent has $e(X) = 0.01$. Let both conditional residual variances equal one. The full-population residual criterion has numerator

$$0.9(4) + 0.1(101.01) \approx 13.70.$$

If the rare-treatment region is removed, the numerator is $0.9(4)$ and the retained mass is 0.9, giving

$$V(\mathcal{A}) = \frac{0.9(4)}{0.9^2} \approx 4.44.$$

The precision gain is large. Yet the new estimand omits the ten percent of the population for whom treatment was rare. If those units are the policy population of greatest concern, a precise answer about the other 90 percent does not answer the original question.

Equipoise: change the target smoothly

Hard trimming draws a boundary: a unit just inside the cutoff belongs fully to the target, while a nearly identical unit just outside does not belong at all. A smoother alternative is to give greater target weight to units for whom both treatment states were genuinely available.

In the finite randomized population, let $h_i \geq 0$ be a prespecified target weight and define

$$\tau_h = \frac{\sum_{i=1}^{N} h_i\{Y_i(1) - Y_i(0)\}}{\sum_{i=1}^{N} h_i}. \tag{0.1}$$

Choose

$$h_i = \pi_i(1 - \pi_i). \tag{0.2}$$

This weight is largest at $\pi_i = 1/2$, shrinks as the design makes one arm rare, and equals zero when one arm is impossible. A Horvitz–Thompson estimator for this target is

$$\hat{\tau}_h = \frac{1}{\sum_i h_i} \sum_{i:\, 0<\pi_i<1} h_i \left\{ \frac{Z_i Y_i^{\text{obs}}}{\pi_i} - \frac{(1 - Z_i)Y_i^{\text{obs}}}{1 - \pi_i} \right\}. \tag{0.3}$$

For units with positive probabilities, the observed arm weights simplify:

$$\frac{h_i}{\pi_i} = 1 - \pi_i \quad \text{when } Z_i = 1, \qquad \frac{h_i}{1 - \pi_i} = \pi_i \quad \text{when } Z_i = 0. \tag{0.4}$$

Units with $\pi_i \in \{0, 1\}$ have $h_i = 0$ and are omitted from the displayed sum so that no $0/0$ expression is formed. The weights on supported units are bounded. More importantly, the stabilization is not hidden clipping: both arms represent the same explicitly defined target τ_h. A structurally unsupported unit receives zero target weight rather than a fabricated inverse weight.

This target is often described as the **population in equipoise**: units for whom either treatment is plausible under the assignment mechanism. It can be more principled than an arbitrary hard cutoff because membership changes smoothly and the estimand emphasizes the part of the population that supports the comparison.

The interpretation requires care in a designed experiment. The investigator chooses π_i, so changing randomization probabilities changes the equipoise-weighted estimand. Under ordinary equal-probability randomization, every h_i is the same and $\tau_h = \tau$. The target becomes distinctive only when treatment probabilities vary across units or strata.

In a superpopulation, a nonnegative tilting function $h(x)$ defines

$$\tau_h = \frac{\mathbb{E}[h(X)\{Y(1) - Y(0)\}]}{\mathbb{E}\{h(X)\}}. \tag{0.5}$$

Li, Morgan, and Zaslavsky choose $h(x) = e(x)\{1 - e(x)\}$, yielding treated weights $1 - e(x)$ and control weights $e(x)$ (2018). The target density is proportional to $e(x)\{1 - e(x)\}f_X(x)$, not to $e(x)\{1-e(x)\}$ alone. It emphasizes covariate regions with treatment uncertainty while still reflecting how common those regions are in the source population.

What changes in an observational study?

The weighting algebra looks similar, but the source of validity is different. In a randomized experiment, assignment probabilities are selected and known, and design expectations hold without an outcome model. In an observational study:

- $e(x)$ must be estimated;
- conditional exchangeability is a substantive assumption, not a consequence of design;
- unmeasured confounding can remain even when propensity distributions look well overlapped; and
- high-dimensional positivity is difficult to assess from one-dimensional propensity histograms.

Thus good observed overlap is useful for estimation but does not itself make an observational comparison causal. D'Amour et al. explain why overlap becomes especially demanding with many covariates (2021).

When the target cannot be changed

Clipping regularizes a point estimator. Trimming and equipoise weighting name a different target population. Sometimes the original target is scientifically nonnegotiable. The five logistically excluded units in the opening example may be precisely the people a policy decision must address.

If the design never exposes one of their potential outcomes, a point effect for the full target requires extrapolation. **Partial identification** offers a different response: retain the original target, state additional restrictions that are scientifically defensible, and report every effect value compatible with the design, the observed-data law, and those restrictions.

Definition 0.0.1 (Identified set)**.** For an estimand θ and an assumption class $\mathcal{M}$, the **identified set** is the set of values of θ attained by full-data distributions—or finite science tables in a design-based analysis—that produce the same observed-data law and satisfy $\mathcal{M}$.

A bound is **sharp** when every value inside it is attainable, or can be approached arbitrarily closely, by some admissible full-data world. An identified set is a population object. Its width can remain positive even with infinite repeated data; sampling uncertainty about its estimated endpoints is a separate issue.

Manski's accounting: bound what the design cannot expose

Assume the outcomes are known to lie in

$$Y_i(z) \in [\ell, u] \qquad \text{for every } i, z. \tag{0.1}$$

For arm z, separate units into

$$\mathcal{J}_z^+ = \{i : \pi_i(z) > 0\}, \qquad \mathcal{J}_z^0 = \{i : \pi_i(z) = 0\}.$$

The design can expose $Y_i(z)$ for every $i \in \mathcal{J}_z^+$ in some assignment. It can never expose $Y_i(z)$ for $i \in \mathcal{J}_z^0$. Two finite science tables that agree on all supported potential outcomes but differ on the unsupported ones generate the same observed-data distribution under the design.

Let

$$\mu_z = \frac{1}{N} \sum_{i=1}^{N} Y_i(z).$$

Boundedness yields

$$\mu_z^- = \frac{1}{N} \left\{ \sum_{i \in \mathcal{J}_z^+} Y_i(z) + |\mathcal{J}_z^0| \ell \right\}, \qquad \mu_z^+ = \frac{1}{N} \left\{ \sum_{i \in \mathcal{J}_z^+} Y_i(z) + |\mathcal{J}_z^0| u \right\}. \tag{0.2}$$

Combining the two arm intervals gives the ATE bounds

$$\tau \in \left[\mu_1^- - \mu_0^+, \mu_1^+ - \mu_0^- \right]. \tag{0.3}$$

The lower effect pairs the smallest permissible treated mean with the largest permissible control mean; the upper effect does the reverse. Under only Equation 0.1, these bounds are sharp because each unsupported potential outcome can be set independently to an endpoint of its allowable range.

The supported sums in Equation 0.2 are not all observed in one realization, but they are design-estimable. For example,

$$\widehat{\mu}_z^- = \frac{1}{N} \left\{ \sum_{i \in \mathcal{J}_z^+} \frac{\mathbf{1}\{Z_i = z\} Y_i^{\text{obs}}}{\pi_i(z)} + |\mathcal{J}_z^0| \ell \right\} \tag{0.4}$$

is design-unbiased for μ_z^-. Replacing ℓ by u estimates the upper endpoint. This is where identification and inference separate: the population endpoints are determined by assumptions and the design, while a single randomized experiment estimates those endpoints with sampling error.

In the opening example, suppose only the five logistically excluded units lack treated support, control is supported for everyone, and outcomes lie in $[0, 100]$. Their unknown treated outcomes contribute between 0 and $5 \times 100/100 = 5$ points to the population treated mean. Even perfect knowledge about the other 95 units leaves an ATE identified interval of width five. More repetitions improve estimation of the supported contribution but do not narrow this design-imposed gap.

The familiar observational form

Manski's worst-case logic applies without a randomized design as well (1990). Let $\rho = \mathbb{P}(Z = 1)$ and write

$$m_1^{\text{obs}} = \mathbb{E}(Y^{\text{obs}} \mid Z = 1), \qquad m_0^{\text{obs}} = \mathbb{E}(Y^{\text{obs}} \mid Z = 0).$$

Consistency identifies the treated group's mean of $Y(1)$ and the control group's mean of $Y(0)$, but not the two missing counterfactual means. Under $Y(z) \in [\ell, u]$,

$$\mu_1 \in \left[\rho m_1^{\text{obs}} + (1-\rho)\ell, \rho m_1^{\text{obs}} + (1-\rho)u\right], \tag{0.5}$$

and

$$\mu_0 \in \left[(1-\rho) m_0^{\text{obs}} + \rho\ell, (1-\rho) m_0^{\text{obs}} + \rho u\right]. \tag{0.6}$$

The ATE bounds again combine the smallest treated mean with the largest control mean, and vice versa. Their width is not a small-sample defect. More observational data estimate the same endpoints more precisely; they do not identify the missing counterfactual means without additional assumptions.

Lee bounds: use randomized selection structure

Bounded outcomes can produce wide intervals. Credible structure can narrow them. Lee's bounds provide a particularly instructive randomized example (2009).

Consider a randomized job-training program. Let $S_i(z) \in \{0, 1\}$ indicate whether unit i's wage would be observed under assignment z—for example, whether the person would be employed. Let $Y_i(z)$ be the wage defined when $S_i(z) = 1$. The observed selection indicator is

$$S_i = Z_i S_i(1) + (1 - Z_i) S_i(0),$$

and a wage is observed only when $S_i = 1$. Although treatment is randomized, comparing wages among employed treated and employed control units need not be a causal comparison: treatment may change who becomes employed.

Assume random assignment and **monotone selection**,

$$S_i(1) \geq S_i(0) \qquad \text{for every } i. \tag{0.1}$$

Monotonicity rules out a unit who would be employed under control but not under treatment. It leaves three principal strata: always employed, employed only when treated, and never employed. The target is the effect among the always-employed stratum,

$$\tau_{\mathrm{AO}} = \mathbb{E}\{Y(1) - Y(0) \mid S(1) = S(0) = 1\}.$$

Let

$$s_z = \mathbb{P}(S = 1 \mid Z = z),$$

and suppose $0 < s_0 \leq s_1$. Randomization identifies both selection rates. Under monotonicity, every employed control unit is always employed. Among employed treated units, the fraction

$$q = \frac{s_0}{s_1} \tag{0.2}$$

is always employed; the remaining fraction $1 - q$ is employed only because of treatment. The data identify this mixture proportion but not which treated wages belong to which stratum.

Let $Q_1(v)$ denote the quantile function of wages among employed treated units. The smallest possible treated mean for the always-employed group assigns that group the bottom fraction q of treated wages. The largest assigns it the top fraction q:

$$\mu_{1,L}^{\mathrm{AO}} = \frac{1}{q}\int_0^q Q_1(v)\,dv, \qquad \mu_{1,U}^{\mathrm{AO}} = \frac{1}{q}\int_{1-q}^1 Q_1(v)\,dv. \tag{0.3}$$

The employed-control mean

$$\mu_0^{\mathrm{AO}} = \mathbb{E}(Y^{\mathrm{obs}} \mid Z = 0, S = 1)$$

identifies the control mean for the always-employed stratum. Therefore

$$\tau_{\mathrm{AO}} \in \left[\mu_{1,L}^{\mathrm{AO}} - \mu_0^{\mathrm{AO}}, \mu_{1,U}^{\mathrm{AO}} - \mu_0^{\mathrm{AO}}\right]. \tag{0.4}$$

If $s_1 = 0.8$ and $s_0 = 0.6$, then $q = 0.75$. The lower bound removes the top 25 percent of employed treated wages and averages the bottom 75 percent; the upper bound removes the bottom 25 percent and averages the top 75 percent. With ties at a cutoff, fractional trimming may be required.

This outcome trimming has a different purpose from Crump-style covariate trimming. Crump-style trimming changes the target population to improve support and precision. Lee trimming constructs extreme allocations within a latent mixture to bound a fixed principal-stratum effect.

The inequality $s_1 \geq s_0$ in the observed data is necessary for the stated direction of monotonicity, but it does not prove individual-level monotonicity. That substantive assumption is what narrows the identified set. In return, the Lee construction does not require a bounded wage scale.

Smoothness: controlled borrowing across support

Boundedness treats a response just outside support as unrelated to a response just inside. A parametric model takes the opposite extreme: after its form and parameters are fixed, it selects one extrapolated response. A smoothness restriction can occupy the middle ground.

Consider a target population with covariates X and a randomized **behavior policy** that chooses arm z with known probability $b_z(x)$. The policy may never assign an arm in some covariate regions. Let

$$\mathcal{O}_z = \{x : b_z(x) > 0\}$$

be the support for arm z, and define the conditional potential-outcome mean

$$m_z(x) = \mathbb{E}\{Y(z) \mid X = x\}.$$

Randomization identifies $m_z(x)$ on $\mathcal{O}_z$, but not outside it. Suppose a new target policy would use arm z outside $\mathcal{O}_z$. Off-policy evaluation then requires information that the behavior design did not directly collect.

Choose a distance $d(x, x')$ on the covariate space and a constant $L_z \geq 0$. Assume

$$|m_z(x) - m_z(x')| \leq L_z d(x, x') \tag{0.1}$$

for covariate values relevant to the target. This constrains the conditional mean, not every noisy individual outcome. The identified function on $\mathcal{O}_z$ must itself satisfy this restriction; otherwise the assumption class is empty. Every supported point x' implies

$$m_z(x') - L_z d(x, x') \leq m_z(x) \leq m_z(x') + L_z d(x, x').$$

All such restrictions must hold simultaneously. The tight pointwise envelopes outside support are

$$\underline{m}_z(x) = \sup_{x' \in \mathcal{O}_z} \{m_z(x') - L_z d(x, x')\}, \tag{0.2}$$

and

$$\overline{m}_z(x) = \inf_{x' \in \mathcal{O}_z} \{m_z(x') + L_z d(x, x')\}. \tag{0.3}$$

If outcomes are also bounded in $[\ell, u]$, truncate the lower envelope at ℓ and the upper envelope at u. Averaging the completed envelopes under the target policy and target covariate distribution produces bounds on its value. Khan, Saveski, and Ugander develop sharp smoothness-based bounds for off-policy evaluation beyond overlap (2024).

A one-dimensional example

Let X be uniform on $[0, 1]$. Suppose arm 1 is supported only on $[0, 0.6]$, where

$$m_1(x) = 0.4 + 0.5x.$$

Assume $Y(1) \in [0, 1]$ and that m_1 is 0.5-Lipschitz. At the support boundary, $m_1(0.6) = 0.7$. For $x > 0.6$, the sharp envelopes are

$$\underline{m}_1(x) = 1 - 0.5x, \qquad \overline{m}_1(x) = 0.4 + 0.5x.$$

At $x = 0.9$, smoothness restricts the missing mean to $[0.55, 0.85]$ rather than the boundedness-only interval $[0, 1]$.

Integration makes the gain for the population arm mean concrete. The supported contribution is

$$\int_0^{0.6} (0.4 + 0.5x)\, dx = 0.33.$$

The lower and upper unsupported contributions are

$$\int_{0.6}^1 (1 - 0.5x)\, dx = 0.24, \qquad \int_{0.6}^1 (0.4 + 0.5x)\, dx = 0.32.$$

Thus

$$\mathbb{E}\{Y(1)\} \in [0.57, 0.65].$$

Boundedness alone gives $[0.33, 0.73]$. Smoothness narrows the answer without selecting one extrapolated curve.

The gain is purchased with substantive choices.

- The metric d declares which covariate changes make units similar. Rescaling a coordinate changes the restriction.
- The constant L_z limits how quickly the mean may change. The unsupported data cannot verify that limit where it matters most.
- In high dimensions, supported points may be far away under any credible metric, so the envelopes remain wide and their estimation can be slow.

A persuasive analysis therefore reports bounds over defensible choices of d and L_z rather than treating them as innocuous tuning parameters.

Identification is not inference

An identified interval $[L, U]$ states what the population design or observed-data law and assumptions imply. In a realized sample, both endpoints must be estimated. The plug-in interval $[\widehat{L}, \widehat{U}]$ is not automatically a 95 percent confidence interval.

Two inferential goals are common:

1. construct a confidence region that covers the **entire identified set** $[L, U]$ with a stated probability;
2. construct a confidence interval that covers the unknown scalar effect, knowing only that it lies somewhere in $[L, U]$.

The first goal is stronger and generally produces a wider interval. The second can use the fact that one scalar cannot simultaneously lie near both endpoints. Either way, sampling uncertainty in the estimated endpoints—and their dependence—must be handled. Imbens and Manski give foundational results for this distinction (2004).

When both endpoint estimators are regular and asymptotically normal, a Bonferroni construction can conservatively expand the lower endpoint downward and the upper endpoint upward by $z_{1-\alpha/2}$ one-sided standard errors. Their joint covariance can yield a less conservative critical value. This familiar calculation is not automatic for endpoints produced by max–min operations, moment selection, or nonregular nuisance estimates. For smoothness bounds in particular, nuisance estimation and envelope optimization can converge more slowly than $n^{-1/2}$, so ordinary Wald reasoning needs particular care (Khan, Saveski, and Ugander 2024).

Keep three uncertainties separate:

- **Identification uncertainty:** $U - L$ remains positive even with infinite data under the maintained assumptions.
- **Sampling uncertainty:** L and U are estimated from one finite experiment or sample.
- **Sensitivity uncertainty:** the set changes when the outcome bounds, monotonicity direction, covariate metric, or smoothness constant changes.

Calling all three "the confidence interval" obscures which uncertainty comes from random sampling and which comes from scientific assumptions.

What to carry forward

Positivity is not a ritual condition attached to a weighting formula. It says that the design can expose every comparison required by the target estimand. Overlap describes how that support appears across covariate regions; realized overlap can still be poor by chance even under a positive design.

When probabilities are small but positive, inverse weighting remains centered on the design-based target, but residual noise is amplified. Clipping can reduce that amplification by accepting design bias. Trimming and equipoise weighting can yield stable, design-supported point estimates by naming a different target. Those target changes may be scientifically appropriate, but they should never be hidden as mere data cleaning.

When the original target must remain and a comparison has zero probability, partial identification changes the form of the answer. Boundedness, monotonicity, or smoothness can rule out some missing science tables without pretending that the data selected one of them. Stronger assumptions narrow the set; a better design can remove the gap.

Advanced material

Why a variance-optimal trimming rule has a threshold form

The threshold structure in Equation 0.7 follows from a local calculation. Write

$$q = \mathbb{P}(X \in \mathcal{A}), \qquad K = \mathbb{E}[\mathbf{1}\{X \in \mathcal{A}\}k(X)], \qquad V(\mathcal{A}) = \frac{K}{q^2}.$$

Consider adding an infinitesimal excluded covariate cell of mass ε and difficulty score k_0. The new criterion is

$$V_\varepsilon = \frac{K + \varepsilon k_0}{(q+\varepsilon)^2}.$$

Differentiating at zero gives

$$\left.\frac{dV_\varepsilon}{d\varepsilon}\right|_{\varepsilon=0} = \frac{k_0 q - 2K}{q^3}. \tag{0.1}$$

Adding the cell improves precision exactly when

$$k_0 < \frac{2K}{q} = 2\mathbb{E}\{k(X) \mid X \in \mathcal{A}\}.$$

At an optimum, retained cells therefore lie below a common threshold and excluded cells lie above it, apart from ties. The threshold satisfies

$$\gamma = 2\mathbb{E}\{k(X) \mid k(X) \le \gamma\}.$$

This argument concerns the residual, covariate-conditional criterion. For a superpopulation ATE, the full influence-function variance also contains

$$\frac{\mathbb{E}[\mathbf{1}\{X \in \mathcal{A}\}\{m_1(X) - m_0(X) - \tau_{\mathcal{A}}\}^2]}{\mathbb{P}(X \in \mathcal{A})^2}.$$

Treatment-effect heterogeneity can therefore alter which population is variance-optimal.

Adaptive trimming changes the inferential problem

Suppose an estimated score $\hat{k}$ and threshold $\hat{\gamma}$ define

$$\widehat{\mathcal{A}} = \{x : \hat{k}(x) \leq \hat{\gamma}\}.$$

For sampled covariates $X_1, \ldots, X_n$, the realized target is

$$\tau^{\text{sample}}_{\widehat{\mathcal{A}}} = \frac{\sum_{i=1}^{n} \mathbf{1}\{X_i \in \widehat{\mathcal{A}}\}\{m_1(X_i) - m_0(X_i)\}}{\sum_{i=1}^{n} \mathbf{1}\{X_i \in \widehat{\mathcal{A}}\}}.$$

This differs from the effect for a fixed population rule $\overline{\mathcal{A}}$ defined by population quantities. The two are close only under conditions on boundary estimation and the probability mass near that boundary. Khan and Ugander develop cross-fitted, doubly robust inference for the realized sample-adaptive target, including simultaneous intervals over prespecified trimming procedures (2025). That result does not automatically become inference for a fixed limiting target. Data-adaptive target selection is part of the inferential question, not preprocessing to forget.

Extrapolation as a bias–bias–variance tradeoff

Partial identification is not the only way to make extrapolation assumptions visible. Many regression and weighting estimators predict at a target covariate value x^* through

$$\widehat{m}(x^*) = \sum_{i=1}^{n} w_i Y_i.$$

If the weights are nonnegative and sum to one, the weighted covariates remain in the convex hull of the observed covariates. When x^* lies outside that hull, nonnegative weights cannot reproduce it exactly. Negative weights can reduce covariate imbalance, but then prediction depends on the outcome model continuing beyond the observed convex support and the weights may amplify noise.

Arbour, Parikh, Niknam, Stuart, Rudolph, and Feller formalize this as a bias–bias–variance tradeoff (2026). To isolate the geometry, one affine-normalized schematic variant of their regularized objective is

$$\min_{w} \left\| \sum_{i=1}^{n} w_i X_i - x^* \right\|_2^2 + \lambda \|w\|_2^2 + \gamma \sum_{i=1}^{n} (-w_i)_+ \|X_i\|^\alpha \quad \text{subject to} \quad \sum_{i=1}^{n} w_i = 1. \tag{0.2}$$

The first term measures imbalance: how far the weighted covariates remain from the target. The second is a variance proxy that penalizes dispersed weights. The third penalizes negative-weight extrapolation, where $(-w_i)_+ = \max(-w_i, 0)$. The parameters λ and γ regulate two different risks. The affine constraint shown here makes the convex-hull interpretation explicit; equivalent normalization can instead be built into the features, and the precise regularization depends on the outcome-model class and feature scaling.

This framework is not partial identification: it produces a regularized point estimate rather than a sharp set. Its connection to this lecture is conceptual. Poor support forces the analyst to negotiate among target fidelity, model extrapolation, and precision. A sensitivity analysis over γ can show how much the answer depends on permitting negative influence rather than concealing that dependence in a single fitted model.

Further reading

- Crump et al. (2009) formulate limited overlap as a target-population and optimal-trimming problem. Yang and Ding (2018) study inference when trimming uses an estimated propensity score.
- Khan and Ugander (2025) extend trimming to heteroscedastic outcomes and sample-adaptive, doubly robust inference. Li, Morgan, and Zaslavsky (2018) develop balancing weights and the overlap-weighted population in equipoise.
- Manski (1990) gives foundational nonparametric treatment-effect bounds. Imbens and Manski (2004) distinguish confidence regions for an identified set from confidence intervals for a partially identified parameter.
- Lee (2009) derives sharp treatment-effect bounds under sample selection using random assignment, monotonicity, and outcome-tail trimming.
- Khan, Saveski, and Ugander (2024) obtain sharp smoothness-based bounds for off-policy evaluation beyond overlap.
- Cole and Hernán (2008) discuss practical inverse-probability weight construction and truncation. Petersen et al. (2012) organize diagnostics and responses to positivity violations, and D'Amour et al. (2021) analyze overlap in high-dimensional covariate spaces.
- Arbour et al. (2026) recast extrapolation as a regularized bias–bias–variance tradeoff and propose sensitivity analysis for dependence on parametric modeling assumptions.

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L05: Interference

Exposure mappings, testing, and experimental design

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- See why treating one person can change somebody else's outcome, and expand the potential-outcome table accordingly.
- Specify which comparisons matter and which assignments reveal them.
- Test for interference by preserving the comparisons that a null hypothesis makes observable.
- Design experiments to create informative exposures, balancing their frequency against dependence and possible bias.

A treatment changes an environment

Suppose a messaging platform changes its interface to encourage users to send more messages. A treated user sees the new interface. An untreated user may nevertheless receive more messages because their friends see it. The experiment changes both a person's own interface and the social environment in which that interface is used. Comparing treated and untreated users need not answer what would happen if the platform changed the interface for everyone.

The preceding lectures treated the potential outcomes of a unit as fixed once that unit's treatment was specified. We now retain the finite-population, design-based framework but enlarge the intervention being specified. The central difficulty is not that randomization stops working. It is that randomizing individual treatment statuses creates a distribution of social environments, and that distribution may differ substantially from the environment named by the scientific question.

What SUTVA lets us leave out

The **stable unit treatment value assumption**, or SUTVA, combines two requirements. First, a unit's outcome does not depend on other units' treatment assignments: there is no interference. Second, each treatment label describes an intervention sufficiently well that there are no outcome-relevant, hidden versions of that treatment. The first requirement is about which units' assignments matter; the second is about whether the assignments themselves adequately describe what is done.

For a finite population of N units, let $\mathbf{z} = (z_1, \ldots, z_N) \in \{0,1\}^N$ be a possible assignment vector. The potential outcome $Y_i(\mathbf{z})$ describes unit i under that entire intervention. All of these functions are fixed. The design draws the random vector $\mathbf{Z}$, and consistency gives

$$Y_i^{\text{obs}} = Y_i(\mathbf{Z}).$$

The no-interference restriction is

$$Y_i(\mathbf{z}) = Y_i(\mathbf{z}') \quad \text{whenever } z_i = z_i'. \tag{0.1}$$

Under this restriction, the assignments to the other $N-1$ units can be suppressed and we recover the two entries $Y_i(0)$ and $Y_i(1)$. Without it, a binary-treatment science table has N rows and 2^N assignment columns. The table is large because the intervention is large, not because we have changed the definition of a potential outcome.

Consider two users. A hypothetical science table under SUTVA could be

Unit	Own control, $Y_i(0)$	Own treatment, $Y_i(1)$
User 1	2	3
User 2	4	7

Now allow messages from the other person to affect the outcome. For this example, specify the complete response functions as

$$Y_1(z_1, z_2) = 2 + z_1 + 2z_2, \qquad Y_2(z_1, z_2) = 4 + 3z_2 + z_1.$$

User 1 gains one unit from their own treatment and two from the other user's treatment; user 2 gains three from their own treatment and one from the other user's treatment. Transposing the expanded science table makes all four assignments easy to inspect:

Assignment (z_1, z_2)	$Y_1(z_1, z_2)$	$Y_2(z_1, z_2)$
$(0,0)$	2	4
$(1,0)$	3	5
$(0,1)$	4	7
$(1,1)$	5	8

If the realized assignment is $(1,0)$, we observe the entire second row of this transposed table: outcomes 3 and 5. We do not observe one person's outcome from the all-treated world and the other's from the all-control world. This is the first change in how we should read experimental data under interference.

The other component of SUTVA matters too. "Use the new recommendation algorithm" is incomplete if its training data change with the experiment's allocation. One can expand the intervention to include the training protocol and thereby obtain well-defined potential outcomes, but that expansion does not restore no interference. A precise intervention and the absence of spillovers are separate requirements.

Four mechanisms that create interference

The mechanisms below produce different scientific questions. They also suggest different restrictions on $Y_i(\mathbf{z})$. The displayed functions are illustrative models developed for these notes, not claims that the cited applications obey an exact equation.

Social interaction. A reminder can affect its recipient and, through conversation, the recipient's friends. Bond et al. (2012) studied political mobilization through an online social network. The distinction between receiving a message and being connected to its recipients is central to this setting. Let A_{ij} indicate a pretreatment social connection from j to i. A simple model is

$$Y_i(\mathbf{z}) = a_i + b_i z_i + \sum_{j \neq i} A_{ij} c_{ij} z_j. \tag{0.2}$$

Here a_i is the baseline outcome, b_i the own-treatment contribution, and c_{ij} the contribution of treating a connected person. These coefficients are fixed, and the equation is a restriction on the science table. It is not a regression fitted to the observed cross-section. Social reinforcement could also generate interactions such as $z_j z_k$, so additivity is a substantive simplification.

Markets. Two sellers can compete for the same buyer even if they never communicate. A treatment that makes one listing more attractive can divert demand from another. Holtz et al. (2024) use a pricing meta-experiment on Airbnb to study this kind of interference. For a small illustration, let two sellers divide one unit of demand, and let $\delta > 0$ measure the increase in a seller's attractiveness under treatment:

$$Y_1(z_1, z_2) = \frac{\exp(\delta z_1)}{\exp(\delta z_1) + \exp(\delta z_2)}, \qquad Y_2(z_1, z_2) = 1 - Y_1(z_1, z_2).$$

A treated seller outperforms an untreated competitor. Yet changing both sellers' treatment leaves both market shares at one half. An individual advantage can therefore coexist with zero change in average market share under universal adoption. For revenues or welfare, total demand might also change; that would require a different response function. Calling a treatment successful requires specifying which market outcome and which policy comparison matter.

Spatial movement. A local restriction can shift activity across a border. Chang et al. (2023) study spillovers in mobility under California's county-level COVID-19 restrictions using a regression-discontinuity design. Their application is observational; it illustrates the mechanism, while our subsequent analysis concerns randomized assignments. A stylized county outcome could depend on both its own policy and neighboring policies:

$$Y_i(\mathbf{z}) = a_i - b_i z_i + \sum_{j \neq i} w_{ij} z_j, \qquad w_{ij} \geq 0.$$

Own restrictions reduce local activity, whereas restrictions elsewhere can redirect activity toward county i. The weights might reflect travel connections rather than geographic adjacency. Restricting one county and restricting an entire region are distinct interventions even for a resident whose own county has the same policy in both cases.

Shared learning and symbiosis bias. A recommendation algorithm learns from the behavior it induces. If two experimental algorithms share those data, one can benefit from exploration performed by the other. Brennan et al. (2025) study the resulting discrepancy between mixed-algorithm experiments and separate deployment. Let $D_t(\mathbf{z})$ denote the training data accumulated before period t under allocation $\mathbf{z}$, and let $\theta_{a,t}(\mathbf{z})$ be the fitted state of algorithm $a \in \{0, 1\}$. A schematic representation is

$$\theta_{a,t}(\mathbf{z}) = \mathcal{A}_a\{D_t(\mathbf{z})\}, \qquad Y_{it}(\mathbf{z}) = g_i\{z_i, \theta_{z_i,t}(\mathbf{z})\}.$$

The symbol $\mathcal{A}_a$ denotes the algorithm's training procedure. Although user i interacts with one algorithm, its trained state depends on other users' assignments. Separating training data can reduce this channel, but also changes the quantity of data available to each algorithm. That intervention may itself differ from full deployment. The system being evaluated includes the learning process, not just a fixed scoring rule.

These examples are not exhausted by a friendship graph. A graph used in design should represent a plausible channel of causal influence: competition, mobility, shared learning, or social interaction. A convenient data structure alone is not evidence that all interference follows its edges.

Which treatment effect?

Universal adoption and individual treatment

For a decision about deployment to everyone, the natural finite-population target is

$$\tau_{\mathrm{G}} = \frac{1}{N}\sum_{i=1}^{N}\{Y_i(\mathbf{1}_N) - Y_i(\mathbf{0}_N)\}. \tag{0.3}$$

The vectors $\mathbf{1}_N$ and $\mathbf{0}_N$ assign everyone to treatment and control, respectively. We call τ_{G} the **global average treatment effect** (GATE); "total treatment effect" is another name used for this deployment contrast. It includes the consequences of treating the unit and of changing its environment. In the two-user table, it equals $\{(5-2)+(8-4)\}/2 = 3.5$.

Now independently treat each unit with probability $p \in (0, 1)$. The ordinary individual-treatment HT contrast is

$$\hat{\tau}_{\mathrm{own}} = \frac{1}{N}\sum_i \left\{ \frac{Z_i Y_i^{\mathrm{obs}}}{p} - \frac{(1-Z_i)Y_i^{\mathrm{obs}}}{1-p} \right\}.$$

Its expectation can still be calculated exactly. Conditioning first on the other units' assignments, independence gives

$$\mathbb{E}_Z(\hat{\tau}_{\mathrm{own}}) = \frac{1}{N}\sum_i \mathbb{E}_{Z_{-i}}\{Y_i(1, \mathbf{Z}_{-i}) - Y_i(0, \mathbf{Z}_{-i})\} \equiv \tau_D(p). \tag{0.4}$$

Here $\mathbf{Z}_{-i}$ is the vector with unit i removed, and $Y_i(a, \mathbf{Z}_{-i})$ places a in position i. The expectation averages over independent Bernoulli-p assignments to the others. Thus $\tau_D(p)$ changes one's own treatment while averaging over the same distribution of other people's treatments on both sides. It is a design-specific average direct effect, not generally the universal-adoption contrast.

In the two-user example, $\tau_D(p) = (1 + 3)/2 = 2$ for every p. The difference from 3.5 is the average spillover contribution, 1.5. In other response functions, own-treatment effects interact with others' treatments and $\tau_D(p)$ itself changes with p. Neither randomization nor inverse-probability weighting says these two estimands must coincide.

This calculation uses independent assignment. Under complete randomization, conditioning on a unit being treated leaves fewer treatments available for its peers than conditioning on it being a control. One must use the actual conditional assignment laws. The design is part of the interpretation of an individual-treatment contrast under interference.

Exposure mappings compress the science table

A messaging feature may depend on the assignments of a person's close contacts but not on assignments elsewhere. Or only the number of treated contacts may matter, rather than their identities. These are different restrictions, and **exposure mappings** give a language for stating them (Aronow and Samii 2017).

For each unit, an exposure mapping is a specified function

$$f_i : \{0, 1\}^N \longrightarrow \mathcal{E}_i, \qquad E_i = f_i(\mathbf{Z}),$$

where $\mathcal{E}_i$ is the set of possible exposure labels. The substantive assumption is that assignments receiving the same label produce the same outcome:

$$f_i(\mathbf{z}) = f_i(\mathbf{z}') \quad \Longrightarrow \quad Y_i(\mathbf{z}) = Y_i(\mathbf{z}'). \tag{0.5}$$

We can then write the common outcome as $y_i(e)$ for exposure e. Lowercase $y_i(e)$ distinguishes an exposure-indexed outcome from $Y_i(\mathbf{z})$, the original assignment-indexed function. The mapping is an assumption about equality among cells of the science table. Computing it from observed assignments does not establish that the equality holds.

Let G be a fixed, pretreatment, undirected interference graph. Write $\mathcal{N}_i$ for the **closed neighborhood** of i: the unit itself and its neighbors. Its size is $d_i = |\mathcal{N}_i|$. A local-interference model says that $Y_i(\mathbf{z})$ depends only on the subvector $\mathbf{z}_{\mathcal{N}_i}$. This permits arbitrary interactions among neighbors' assignments. A more restrictive mapping uses own treatment and the fraction of other neighbors treated:

$$f_i(\mathbf{z}) = (z_i, q_i(\mathbf{z})), \qquad q_i(\mathbf{z}) = \frac{1}{d_i - 1} \sum_{j \in \mathcal{N}_i \setminus \{i\}} z_j,$$

with $q_i = 0$ for an isolated unit. Equal fractions now imply equal outcomes even when different neighbors are treated. A close friend and a distant acquaintance become interchangeable for this purpose. That is additional modeling, not a consequence of locality.

For GATE estimation, two exposure conditions have a special role. Unit i is **network exposed to treatment** if its observed outcome equals $Y_i(\mathbf{1}_N)$, and **network exposed to control** if it equals $Y_i(\mathbf{0}_N)$. Under the local-interference model, sufficient observable conditions are

$$E_i = e_1 : \ z_j = 1 \text{ for every } j \in \mathcal{N}_i, \qquad E_i = e_0 : \ z_j = 0 \text{ for every } j \in \mathcal{N}_i. \tag{0.6}$$

These are full-neighborhood exposures (Ugander et al. 2013). They allow a local part of a mixed experiment to reveal an outcome from a global counterfactual. The outcome may vary arbitrarily among mixed neighborhoods; we need not pool all mixed assignments into one outcome-equivalent category. Threshold or fractional exposure rules can make the extreme labels more common, but identifying them with global outcomes then requires stronger substantive assumptions.

There are other ways to restrict the science table. An exposure model equates outcomes under different assignments. A low-order response model instead restricts how treatment indicators combine algebraically, potentially allowing every assignment to have a distinct outcome. L06 develops that alternative. Neither approach obtains its assumptions from randomization alone.

Moving SUTVA to clusters

Sometimes a plausible grouping is available before specifying a network: households, schools, or geographically isolated villages. Let $\mathcal{C} = \{C_1, \dots, C_K\}$ be a partition and $c(i) \in \{1, \dots, K\}$ the index of the cluster containing unit i. **Partial interference** assumes that

$$\mathbf{z}_{C_{c(i)}} = \mathbf{z}'_{C_{c(i)}} \quad \Longrightarrow \quad Y_i(\mathbf{z}) = Y_i(\mathbf{z}'). \tag{0.7}$$

Assignments within the cluster can interact arbitrarily; assignments outside it do not affect i. Hudgens and Halloran (2008) develop effects and designs in this setting. Karrer et al.'s network-experimentation paper, first circulated in 2020, calls this SUTVA for clusters (Karrer et al. 2021). It is a restriction on the interference structure, not an automatic consequence of assigning treatments by cluster. Candogan, Chen, and Niazadeh (2024) also use no interference across clusters when optimizing correlations among cluster assignments. This connects cluster SUTVA to an established literature on partial interference and to subsequent work on experimental design.

If each cluster receives a single treatment W_k, the allowed unit assignments satisfy $Z_i = W_{c(i)}$. On this restricted set of assignments, partial interference implies that the outcome depends only on $W_{c(i)}$. Cluster totals can then be analyzed as potential outcomes indexed by one cluster treatment, with size weighting if the target averages over individuals.

There is a useful distinction between requiring partial interference for every conceivable assignment and requiring this invariance only on the design's cluster-constant assignments. For example, suppose units 3 and 4 form another cluster and $Y_1(\mathbf{z}) = z_1 + z_3 - z_4$. Cross-cluster interference exists when units 3 and 4 receive different treatments. Under a design that always assigns them together, their effects cancel. The restricted experiment therefore has a cluster-level invariance that the unrestricted science table lacks. This example illustrates a distinction between assumption domains; it is not a new definition attributed to Karrer et al.

Cluster SUTVA can be operationally useful without committing to a detailed exposure mapping within each cluster. Mathematically, however, it can itself be expressed as a mapping that retains

the cluster's treatment vector. Exposure mappings and cluster SUTVA are therefore overlapping ways to express restrictions, not mutually exclusive causal frameworks.

From observed exposures to an estimator

The inverse probability is now an exposure probability

Suppose the mapping is adequate for exposures e_1 and e_0, so that $y_i(e_a) = Y_i(a\mathbf{1}_N)$ for $a \in \{0, 1\}$. Define the marginal and joint exposure probabilities under the chosen design by

$$\pi_i(e) = \mathbb{P}_Z(E_i = e), \qquad \pi_{ij}(e, e') = \mathbb{P}_Z(E_i = e, E_j = e').$$

These probabilities average over assignments, holding the graph and potential outcomes fixed. They are not estimated observational propensity scores. A unit can have treatment probability one half and an exceedingly small probability of experiencing an all-treated neighborhood.

If both required exposures have positive probability for every unit, define

$$\hat{\mu}_{\mathrm{HT}}(e) = \frac{1}{N} \sum_i \frac{\mathbf{1}\{E_i = e\} Y_i^{\mathrm{obs}}}{\pi_i(e)}, \qquad \hat{\tau}_{\mathrm{G,HT}} = \hat{\mu}_{\mathrm{HT}}(e_1) - \hat{\mu}_{\mathrm{HT}}(e_0). \tag{0.8}$$

Proposition 0.0.1 (Design unbiasedness under an adequate exposure model)**.** *If exposure e_a reveals $Y_i(a\mathbf{1}_N)$ and $\pi_i(e_a) > 0$ for every i and $a \in \{0, 1\}$, then $\mathbb{E}_Z(\hat{\tau}_{\mathrm{G,HT}}) = \tau_{\mathrm{G}}$.*

Proof. On the event $E_i = e_a$, consistency and the exposure assumption replace the observed outcome by the fixed number $Y_i(a\mathbf{1}_N)$. Consequently,

$$\mathbb{E}_Z \left\{ \frac{\mathbf{1}\{E_i = e_a\} Y_i^{\mathrm{obs}}}{\pi_i(e_a)} \right\} = \frac{\pi_i(e_a) Y_i(a\mathbf{1}_N)}{\pi_i(e_a)} = Y_i(a\mathbf{1}_N).$$

Average over units and subtract the control expression. No independence among exposures is used. □

The familiar HT cancellation survives interference. What changes is the event that makes a particular potential outcome observable. Weighting cannot repair a misspecified exposure model: if different assignments with label e_1 produce different outcomes, the weighted average targets their design-weighted mean, which need not equal the all-treated outcome.

A Hájek version divides the weighted outcome sum by the weighted count of units in that exposure. As in L02, this can improve stability, but exact design unbiasedness is generally lost and a zero denominator must be handled. Choosing normalization does not resolve an incorrect mapping or zero exposure probability.

Positivity and joint observability

Under independent Bernoulli-p unit assignment and full-neighborhood exposure,

$$\pi_i(e_1) = p^{d_i}, \qquad \pi_i(e_0) = (1-p)^{d_i}. \tag{0.9}$$

Each of the d_i assignments must agree. At $p = 1/2$, a closed neighborhood of size 11 receives either particular extreme exposure with probability 1/2048. A large experiment can contain many treated units but very few observations of the outcomes needed for its deployment target. This is L04's weak-positivity problem created by the combination of network structure, estimand, and design.

The pairwise probabilities matter for inference. Write $s_{e_1} = 1$ and $s_{e_0} = -1$. Expanding the variance of Equation 0.8 gives

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{G,HT}}) = \frac{1}{N^2} \sum_{i,j} \sum_{e,e' \in \{e_1, e_0\}} s_e s_{e'} \frac{\pi_{ij}(e,e') - \pi_i(e)\pi_j(e')}{\pi_i(e)\pi_j(e')} y_i(e) y_j(e'). \tag{0.10}$$

For $i = j$, interpret the joint probability in the usual way: it equals $\pi_i(e)$ if $e = e'$ and zero otherwise. The formula follows by writing the estimator as a sum of weighted indicators and using the covariance of each pair. It displays both rare-exposure weights and dependence among exposures.

Even when each marginal probability is positive, some joint probabilities are zero. If two closed neighborhoods overlap, one cannot simultaneously be all treated and the other all control. The corresponding product of potential outcomes is never jointly observed. It cannot be estimated by dividing an observed product by its joint probability. Conservative variance procedures handle these unidentified terms with bounds; normal approximations additionally require an appropriate sequence of growing experiments and restrictions on dependence (Aronow and Samii 2017). An ordinary independent-observation standard error ignores precisely the structure that made interference relevant.

Testing whether interference is present

An assumption can be useful without being beyond scrutiny. Testing interference requires care because "no interference" still permits each unit to have an unknown direct treatment effect. We must preserve that uncertainty while finding assignments under which the relevant outcomes can be imputed.

Make the null sharp on a conditional experiment

The global sharp null $Y_i(\mathbf{z}) = Y_i(\mathbf{z}')$ for every $i, \mathbf{z}, \mathbf{z}'$ rules out all treatment effects. Rejecting it does not specifically establish interference: ordinary direct effects suffice. The no-interference null in Equation 0.1 is weaker. It supplies no value for a treated person's outcome under their own control assignment.

The conditional-testing approach of Athey, Eckles, and Imbens (2018) makes the null sharp for selected **focal units** by restricting the assignments considered. Choose a set F of focal units

before seeing treatment or outcomes, for example using only the pretreatment graph. Condition on their realized own assignments $\mathbf{Z}_F$. Under no interference, their outcomes are then fixed for every reassignment of the remaining, auxiliary units.

To see a statistic one could use, let $q_i(\mathbf{z})$ be the treated-neighbor fraction defined earlier, and let $\overline{Y}_F^{\text{obs}}$ be the mean focal outcome. Consider

$$T(\mathbf{z}) = \sum_{i \in F} (Y_i^{\text{obs}} - \overline{Y}_F^{\text{obs}}) \left[q_i(\mathbf{z}) - \mathbb{E}_Z\{q_i(\mathbf{Z}) \mid \mathbf{Z}_F\} \right].$$

The outcome factors stay fixed. The exposure factors vary as auxiliary treatments are reassigned. Positive values indicate that focal outcomes tend to be larger where randomized neighbor exposure is larger. A two-sided test could use $|T|$.

The reference distribution must be the actual conditional design, $\mathbb{P}_Z(\mathbf{Z}_{-F} \mid \mathbf{Z}_F)$. Under independent Bernoulli assignment, auxiliary assignments remain independent with probability p. Under complete randomization, the number of auxiliary treatments is fixed by the original treatment budget and the observed focal assignments. Simply flipping independent coins would analyze the wrong experiment in the latter case.

Conditional on $\mathbf{Z}_F$, the observed assignment and a fresh draw from this conditional law have the same distribution under the null, while every outcome entering the statistic is known. Ranking the observed statistic in this distribution therefore gives a valid randomization p-value, with discreteness handled conservatively. For Monte Carlo sampling, including the observed allocation in the usual plus-one construction avoids zero estimated p-values.

This argument does not require the chosen neighbor statistic to describe the actual interference correctly. Under the null there is no interference to describe. A poor graph or statistic reduces power, whereas an incorrect conditional assignment law can invalidate the test. Choosing focal units after observing assignments is possible, but requires conditioning on the selection mechanism as well; selecting favorable focal units and then pretending they were fixed is not justified.

Randomize the randomization mechanism

A different strategy deliberately creates different local environments at the same marginal treatment probability. Saveski et al. (2017) compare individual and cluster randomization within a randomized meta-experiment. The outer randomization allocates groups to the two experimental strategies; the inner randomization assigns treatment according to each strategy. Under no interference, both strategies target the same average direct effect. Their discrepancy can reveal interference without requiring an exact spillover model.

For a concrete design-based version, let $R_i \in \{0, 1\}$ record the strategy to which unit i is assigned, with known probability $\rho_i(r) = \mathbb{P}(R_i = r)$. Suppose each strategy gives that unit conditional treatment probability p. An HT contrast for strategy r is

$$\hat{\tau}_r = \frac{1}{N} \sum_i \frac{\mathbf{1}\{R_i = r\}}{\rho_i(r)} \left\{ \frac{Z_i Y_i^{\text{obs}}}{p} - \frac{(1 - Z_i) Y_i^{\text{obs}}}{1 - p} \right\}.$$

Under no interference, iterated expectation over the outer and inner assignments gives $\mathbb{E}(\hat{\tau}_r) = N^{-1}\sum_i\{Y_i(1) - Y_i(0)\}$ for both strategies. Hence $\hat{\tau}_1 - \hat{\tau}_0$ is centered at zero. This calculation explains the role of randomizing strategy membership: the comparison is not confounded by choosing different populations for the two designs. Its variance must incorporate both randomizations and their covariances.

Under interference, cluster assignment makes own treatment more correlated with neighbors' treatments. In the two-user example, assigning both users together produces the global contrast 3.5, whereas independent assignment produces the direct-effect contrast 2. A strategy comparison can detect this discrepancy. For a real network with edges crossing clusters, neither strategy need identify GATE. The test compares the effects revealed by the two designs; it does not certify one as unbiased for universal deployment. Nonrejection likewise does not establish no interference, because the designs can miss relevant channels or have little power.

Use randomized increases in treatment allocation

Platforms often expand a treatment gradually. A control unit that remains untreated can experience an increasingly treated environment. Han, Li, Mao, and Wu (2023) use increasing-allocation experiments to test for cross-unit interference. The useful variation is in other units' randomized treatment histories, rather than a raw comparison of outcomes at different dates.

Consider two periods with marginal treatment probabilities $p_1 < p_2$. Draw independent $U_i \sim \text{Uniform}(0,1)$ and set $Z_{it} = \mathbf{1}\{U_i \le p_t\}$. Each unit's treatment history is one of $(0,0)$, $(0,1)$, or $(1,1)$, with probabilities $1-p_2$, p_2-p_1, and p_1. The history $(1,0)$ is impossible. Write $Y_{it}(\mathbf{z}_{1:2})$ for the potential outcome when the full two-period assignment matrix is specified. The no-cross-unit-interference null says that this outcome depends only on unit i's own treatment history. It can still depend on time and on that own history.

A conditional experiment can hold focal units' entire treatment histories fixed and randomize the auxiliary histories. For example, preselect a focal set F using the graph and condition on the histories of **every unit in this original set**. Within the statistic, use only focal units whose own assignment is unchanged, and form $\Delta Y_i = Y_{i2}^{\text{obs}} - Y_{i1}^{\text{obs}}$. These differences are fixed under the null after conditioning on focal histories. A statistic can associate ΔY_i with the change in treated-neighbor fraction, $q_i(\mathbf{Z}_2) - q_i(\mathbf{Z}_1)$. Focal units that switch treatment remain conditioned upon even though their outcomes do not enter this statistic; they do not move into the auxiliary pool.

Conditional on the counts of histories among units outside the original F, iid rollout assignment makes those auxiliary histories exchangeable. Permuting **whole histories only among units outside** F preserves both the rollout and this conditional law. Recompute neighbor exposures after each permutation and compare the observed statistic with the resulting distribution. Separately permuting the two treatment columns could create forbidden $(1,0)$ histories and would not reproduce the design.

Differencing can help remove persistent unit differences from the statistic; its validity here does not require all time trends to be common because focal outcomes are held fixed under the null. Tests that instead exchange periods, or interpret a changing treatment-control contrast as evidence by itself, require additional restrictions. The paper also develops a more powerful procedure under a

time-fixed-effect model. That additional model is consequential: seasonality, changing treatment effects, or carryover can otherwise resemble an allocation effect.

The three strategies use distinct sources of variation. Conditional tests hold own treatment fixed while varying others' assignments. Meta-experiments vary the mechanism assigning treatment. Rollouts vary treatment histories across time. In each case, valid inference follows from an explicit assignment or invariance argument; the visual appearance of spillovers is only a starting point.

Designing for informative exposures

Testing can reveal that individual randomization answers an incomplete policy question. The next design question is how to make the desired exposures occur more often while retaining enough independent comparisons.

Graph cluster randomization

For a fixed partition $\mathcal{C}$, assign cluster treatments $W_1, \dots, W_K$ independently as Bernoulli-p, and set $Z_i = W_{c(i)}$. Let $k_i(\mathcal{C})$ be the number of distinct clusters intersecting the closed neighborhood $\mathcal{N}_i$. Then full-neighborhood exposure probabilities are

$$\pi_i(e_1 \mid \mathcal{C}) = p^{k_i(\mathcal{C})}, \qquad \pi_i(e_0 \mid \mathcal{C}) = (1-p)^{k_i(\mathcal{C})}. \tag{0.11}$$

Only the assignments of these clusters must agree. The number of independent choices in the exponent can be much smaller than the number of units. This is the central mechanism of graph cluster randomization (GCR) (Ugander et al. 2013). It does not require there to be no edges across clusters: an exposure estimator can account for those edges when calculating the events and their probabilities.

Take a path of six units, $1-2-3-4-5-6$, and $p = 1/2$. Under individual randomization, each endpoint has closed-neighborhood size two and each interior unit size three. Group the path into clusters $\{1,2,3\}$ and $\{4,5,6\}$.

Unit	Closed-neighborhood size d_i	Individual $\pi_i(e_1)$	Clusters touched k_i	Cluster $\pi_i(e_1)$
1 and 6	2	1/4	1	1/2
2 and 5	3	1/8	1	1/2
3 and 4	3	1/8	2	1/4

Interior units away from the cut benefit most. Units 3 and 4 remain near a cluster boundary and require both clusters to agree. The calculation makes clear why counting the number of cut edges alone need not describe every unit's statistical difficulty.

Larger clusters can increase these marginal probabilities but also increase dependence. With one cluster containing the entire graph, every unit receives an extreme exposure, yet there is only one

randomized treatment coin. The observed data are either the whole all-treated population or the whole all-control population. HT remains unbiased across repeated experiments, but the two worlds are never observed together. Exposure frequency and precision are different objectives.

For unequal clusters, one must also distinguish averaging people from averaging clusters. The unweighted mean of cluster means targets an average cluster, giving a person in a small cluster more weight. An estimator of Equation 0.3 must preserve the intended individual weights.

Randomize the clustering too

In a fixed clustering, some units repeatedly occupy favorable interiors and others unfavorable boundaries. Randomized graph cluster randomization (RGCR) draws a partition from a specified distribution Q, then randomizes cluster treatments. Ugander and Yin (2023) study this design and randomized graph-decomposition algorithms, with variance guarantees under graph-structure assumptions.

The two-stage exposure probability is

$$\pi_i(e_1) = \mathbb{E}_{\mathcal{C}\sim Q}\{p^{k_i(\mathcal{C})}\}, \qquad \pi_i(e_0) = \mathbb{E}_{\mathcal{C}\sim Q}\{(1-p)^{k_i(\mathcal{C})}\}. \tag{0.12}$$

The expectation is over the random partition. For example, if a unit's neighborhood touches one cluster half the time and three clusters half the time, its treatment-exposure probability at $p = 1/2$ is $(1/2)(1/2) + (1/2)(1/8) = 5/16$. It is not permanently assigned the unfavorable probability $1/8$.

One simple partition construction gives units independent continuous random priorities and assigns each unit to the highest-priority vertex in its closed neighborhood; equal chosen vertices define a cluster. This is the 1-hop-max construction. Its purpose is to make cluster membership vary through local random choices while remaining tied to graph proximity. The resulting clusters are then independently treated or controlled.

An exposure estimator using the marginal probabilities in Equation 0.12 is unbiased over **both** randomizations. It need not be unbiased conditional on the realized partition. Using conditional probabilities instead defines another estimator, with conditional unbiasedness when those probabilities are positive. The analyst must match the probability in the denominator to the randomization distribution used for the claimed result.

Monte Carlo computation can approximate the marginal probabilities by repeatedly drawing partitions. This is design simulation and does not require outcomes. However, replacing a probability by a noisy estimate inside its reciprocal does not preserve exact unbiasedness. Rare exposures require accurate probability calculations and enough simulations; increasing the number of outcome units does not automatically eliminate probability-computation error.

A clustering is optimal for an objective

Community detection seeks a descriptive partition of a network. Experimental design asks which partition makes an estimator accurate for a specified causal target. These problems can overlap, but their objectives need not agree.

Viviano et al. (2026) study clustering for the particular assignment-weighted contrast

$$\hat{\tau}_{\mathcal{C}} = \frac{2}{N}\sum_i (2Z_i - 1)Y_i^{\text{obs}}, \tag{0.13}$$

under independent fair cluster assignment. The fixed denominator is N, not a realized treatment-arm count. This estimator weights own treatment, rather than requiring full-neighborhood exposure, and can be biased for GATE when clusters exchange spillovers. Their framework chooses clusters by a worst-case bias-variance criterion over a restricted class of potential outcomes, leading to a penalized graph-cut optimization. The result is therefore about an estimator, target, design class, and response class together. It is not a claim that a single partition is optimal for all analyses.

An additive calculation shows why this criterion has a different form from maximizing exposure probabilities. Use Equation 0.2, and abbreviate $\gamma_{ij} = A_{ij}c_{ij}$. Then

$$\tau_{\text{G}} = \frac{1}{N}\sum_i \left(b_i + \sum_{j \neq i} \gamma_{ij} \right).$$

Under fair cluster assignment, $\mathbb{E}_Z(2Z_i - 1) = 0$. Also,

$$2\mathbb{E}_Z\{(2Z_i - 1)Z_j\} = \begin{cases} 1, & c(j) = c(i), \\ 0, & c(j) \neq c(i). \end{cases}$$

For a same-cluster unit, $Z_j = Z_i$; for a different-cluster unit, the two fair assignments are independent. Substituting into the estimator gives

$$\mathbb{E}_Z(\hat{\tau}_{\mathcal{C}}) - \tau_{\text{G}} = -\frac{1}{N}\sum_i \sum_{j : c(j) \neq c(i)} \gamma_{ij}. \tag{0.14}$$

The estimator captures within-cluster contributions because they move with own treatment. Across-cluster contributions cancel from its expected treatment-control comparison. For nonnegative spillovers it underestimates GATE by the omitted contribution. With mixed signs, cancellation can make bias smaller or reverse its direction.

Variance pushes the partition in the opposite direction. Even in the simple case $Y_i(\mathbf{z}) = a_i$ with no effects, the estimator is random. Writing $A_k = \sum_{i \in C_k} a_i$,

$$\text{Var}_Z(\hat{\tau}_{\mathcal{C}}) = \frac{4}{N^2}\sum_{k=1}^{K} A_k^2.$$

If all baseline outcomes equal a constant a, this is $4a^2N^{-2}\sum_k |C_k|^2$. Combining two clusters of sizes u and v increases the sum of squared sizes by $2uv$, even if it eliminates a cut. A design criterion must weigh the expected importance of spillovers against this loss of independent variation. Covariate adjustment can change the variance calculation by replacing predictable baseline components with residuals.

The preceding calculations are a pedagogical special case. In the causal-clustering analysis, bounds on spillover strength and outcome magnitudes, along with local asymptotic restrictions, justify a corresponding worst-case optimization. Tuning the relative bias and variance penalty encodes how large relevant spillovers could be. Pilot evidence or subject knowledge can inform it, and sensitivity across plausible penalties is more informative than treating the choice as a purely graph-theoretic constant.

Changing to the exposure HT estimator changes the problem. Under a correct mapping, its bias is zero and the design concern becomes its joint exposure probabilities and variance. A partition that is attractive for Equation 0.13 need not be attractive for Equation 0.8. This is a concrete instance of designing an experiment and its analysis together.

Advanced material: designs built around incompatible exposures

Clustering begins by asking which units should share an assignment. Another starting point is to ask which outcomes needed for the estimand cannot be observed together. This makes the estimand itself part of the design's combinatorial structure.

Fix two desired neighborhood configurations for each unit, indexed by $a = 0, 1$, and let their potential outcomes be $y_i(e_{ia})$. The target is $N^{-1} \sum_i \{y_i(e_{i1}) - y_i(e_{i0})\}$. For GATE these configurations are all treated and all control. For a direct effect at untreated neighbors, they differ only in the focal unit's own assignment.

Two units conflict if **some** pair of their desired configurations cannot be realized by a single assignment vector. The conflict graph places an edge between such units, including a self-loop because one unit cannot receive both distinct configurations simultaneously. Kandiros, Pipis, Daskalakis, and Harshaw (2026) use this graph to construct a design matched to the contrast.

On the path $1 - 2 - 3 - 4$, units 1 and 3 are not adjacent. Their closed neighborhoods nevertheless overlap at unit 2. Giving unit 1 its all-treated exposure requires $Z_2 = 1$, while giving unit 3 its all-control exposure requires $Z_2 = 0$. They conflict for GATE. In general, the GATE conflict graph connects vertices at graph distance at most two. For the direct-effect contrast that holds all neighbors untreated, units 1 and 3 can receive either desired configuration together: both require unit 2 to be untreated. For this contrast, only neighbors in the original graph conflict. The same physical network creates different observability constraints for different estimands.

The conflict-graph design first generates sparse requests for desired exposures, then resolves conflicting requests using a suitable priority ordering. Accepted requests force the necessary local treatment assignments. Its estimator uses the event that a request is accepted, not every accidental occurrence of the same exposure. If B_{ia} is that recorded event and $q_{ia} = \mathbb{P}_Z(B_{ia}) > 0$, the form is

$$\hat{\tau}_{\mathrm{CG}} = \frac{1}{N} \sum_i \left\{ \frac{\mathbf{1}\{B_{i1}\} Y_i^{\mathrm{obs}}}{q_{i1}} - \frac{\mathbf{1}\{B_{i0}\} Y_i^{\mathrm{obs}}}{q_{i0}} \right\}.$$

Accepted request B_{ia} guarantees the desired outcome, so the same HT cancellation proves unbiasedness. Restricting the counted events makes their dependence more tractable. Under bounded average squared target outcomes, the paper obtains a variance bound of order $\lambda(H)/N$, where $\lambda(H)$ is the largest eigenvalue of the conflict graph's adjacency matrix, including its self-loops. This

guarantee belongs to the proposed design and modified estimator together; arbitrary prioritization or the ordinary exposure HT estimator does not inherit it automatically.

For intuition about the rate, a bounded-size set of conflicts around each unit allows $\lambda(H)$ to remain bounded and hence permits a $1/N$ variance bound. A complete conflict graph has $\lambda(H) = N$, so this bound no longer tends to zero. Merely increasing the number of outcomes does not create independent counterfactual comparisons when all desired exposures compete with one another.

Returning to the deployment question

Interference changes the relationship between what is assigned, what is observed, and what is wanted. The two-user science table already contained the essential issue: a person observed in a mixed assignment need not reveal their outcome under universal treatment or universal control.

An exposure assumption states when a local assignment reveals a global counterfactual. Exposure probabilities then turn this statement into an estimator, and joint probabilities govern its uncertainty. Tests can probe aspects of the assumption by designing conditional comparisons with imputable outcomes. Finally, experimental designs can make informative exposures more common, reduce the bias of a simpler estimator, or coordinate otherwise incompatible comparisons. Those are distinct statistical objectives, even when all three lead us to use a graph.

For a new application, the order of reasoning is therefore consequential: describe the mechanism and the deployment decision; state the target contrast; decide which restrictions on the science table are defensible; and choose an estimator and design that exploit those restrictions. L06 continues with restrictions on the form of the response function, where mixed assignments may be informative even without exact exposure equivalence.

Further reading

Hudgens and Halloran (2008) develop direct, indirect, total, and overall effects under partial interference, emphasizing that an allocation policy is itself an intervention. Aronow and Samii (2017) provide the design-exposure-estimand framework and the associated issues in variance estimation.

Athey, Eckles, and Imbens (2018), Saveski et al. (2017), and Han et al. (2023) give complementary routes to testing. Reading them together clarifies the distinction between sharpening a null by conditioning and changing the experiment to make its implications easier to detect.

Ugander et al. (2013) and Ugander and Yin (2023) develop fixed and randomized graph cluster designs. Eckles, Karrer, and Ugander (2017) analyze design and estimation as tools for reducing interference bias under response structures that need not satisfy a simple local exposure model. Karrer et al. (2021) discuss the practical interplay of clustering, regression adjustment, and experimentation at scale.

Viviano et al. (2026) and Kandiros et al. (2026) illustrate two different ways to turn a causal objective into a design problem: optimize a bias-variance criterion for a specified estimator, or coordinate estimand-specific observability constraints. Their guarantees should be compared with their estimator and response assumptions kept in view.

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L06: Estimation with Structural Models

Learning a global treatment effect from low-order interactions

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Use a restriction on how treatments interact to learn about a full rollout from a partially treated network.
- Derive an estimator first for additive spillovers, then for low-order interactions, using the experiment's known assignment probabilities.
- Connect this construction to least squares and explain which treatment contrasts the design can recover.
- Examine precision, model error, clustered designs, and covariate adjustment.
- Contrast low-order assumptions with monotonicity, which restricts the direction of spillovers rather than their interaction order.

Learning about a rollout before everyone is treated

Suppose a platform is testing a feature that makes it easier for users to send messages. A user's activity may increase because they receive the feature, because their friends receive it, or because a particular combination of friends receives it. The policy question concerns a full rollout: how would average activity change if everyone received the feature? The experiment, however, treats only a random subset of users.

L05 approached this problem by describing which assignments give a unit the same outcome as the all-treated or all-control assignment. Under neighborhood interference, observing every member of a unit's neighborhood in treatment provides its all-treated outcome. With independent treatment probability p, this happens with probability p^{d_i}, where d_i is the number of units in that neighborhood. These observations can be scarce even when each unit has a reasonable chance of treatment.

There is another source of information. If each friend's contribution is additive, a neighborhood containing a mixture of treated and untreated users still reveals something about the full rollout. We need not wait until all friends receive treatment simultaneously to learn their combined contribution. If interactions occur only among pairs of friends, we need information about pairs, but not an unrestricted description of every possible neighborhood assignment.

This lecture develops that idea through the low-order outcome framework of Cortez-Rodriguez, Eichhorn, and Yu (2023). Its central move is to restrict how many treatment indicators can interact

in an outcome. That restriction makes a global treatment effect a linear function of a manageable collection of coefficients. The assignment design then supplies the equations needed to estimate that linear function.

The reasoning remains design-based. The network, potential outcomes, and coefficients are fixed. Treatment assignment is random. A structural model is used to restrict the fixed potential-outcome schedule; it does not introduce random regression errors or a population of interchangeable units.

A model of interactions, rather than a model of identical units

Setup and target

There are N experimental units. Write $\mathbf{z} = (z_1, \ldots, z_N) \in \{0,1\}^N$ for a possible assignment and $\mathbf{Z}$ for the random assignment. Unit i has potential outcome $Y_i(\mathbf{z})$ and observed outcome $Y_i^{\text{obs}} = Y_i(\mathbf{Z})$.

Let $\mathcal{N}_i$ be the known set of units whose assignments can affect unit i's outcome, including i itself. We call it a closed interference neighborhood and set $d_i = |\mathcal{N}_i|$. For a directed network, membership $j \in \mathcal{N}_i$ means that j can influence i; the reverse need not hold. Neighborhood interference means that assignments outside $\mathcal{N}_i$ do not affect Y_i. A neighborhood may be a conservative superset of the actual influencing units. Missing an influencing unit is a different matter: that can invalidate the model.

The target is the **global average treatment effect** (GATE), also called the total treatment effect (TTE):

$$\tau_{\text{G}} = \frac{1}{N}\sum_{i=1}^{N}\{Y_i(\mathbf{1}_N) - Y_i(\mathbf{0}_N)\}. \tag{0.1}$$

Here $\mathbf{1}_N$ and $\mathbf{0}_N$ assign everyone to treatment and control, respectively. This effect combines direct and spillover effects. It does not hold neighbors' treatments fixed while changing only a unit's own treatment.

Initially, assignments are independent, with $Z_j \sim \text{Bernoulli}(\pi_j)$ and known $0 < \pi_j < 1$. We use $\mathbb{E}_Z$ and Var_Z for expectation and variance under this design, keeping the entire potential-outcome schedule fixed. Common-probability assignment is the special case $\pi_j = p$.

Every binary response surface has a polynomial representation

For a two-person neighborhood $\mathcal{N}_i = \{j, k\}$, where one of j, k is i itself, the outcome has four possible values. They can always be written as

$$Y_i(z_j, z_k) = c_{i,\emptyset} + c_{i,\{j\}}z_j + c_{i,\{k\}}z_k + c_{i,\{j,k\}}z_j z_k. \tag{0.2}$$

The intercept is the outcome when neither person is treated. The first-order coefficient $c_{i,\{j\}}$ is the effect of treating j when k is untreated. The pair coefficient is the additional effect of treating both people beyond the sum of their effects when treated separately. Explicitly,

$$
\begin{aligned}
c_{i,\emptyset} &= Y_i(0,0),\\
c_{i,\{j\}} &= Y_i(1,0) - Y_i(0,0),\\
c_{i,\{k\}} &= Y_i(0,1) - Y_i(0,0),\\
c_{i,\{j,k\}} &= Y_i(1,1) - Y_i(1,0) - Y_i(0,1) + Y_i(0,0).
\end{aligned}
$$

The interaction coefficient is a difference in the effect of j between the two treatment states of k. It can be positive, negative, or zero. Thus a polynomial here is not an approximation to a smooth curve on a continuous domain. With binary inputs, it is an exact way to organize the entries of the science table.

More generally, every function on a binary neighborhood has a unique multilinear representation

$$
Y_i(\mathbf{z}) = \sum_{S \subseteq \mathcal{N}_i} c_{i,S} \prod_{j \in S} z_j. \tag{0.3}
$$

The product over the empty set is defined to equal one. Multilinear means that no indicator appears with a power greater than one: for binary treatment, $z_j^2 = z_j$, so higher powers add no information. There are 2^{d_i} coefficients, matching the number of possible local assignments. The representation itself therefore imposes no restriction beyond neighborhood interference.

To see uniqueness, evaluate the function at the assignment treating exactly the members of some subset $T \subseteq \mathcal{N}_i$. Its value is $\sum_{S \subseteq T} c_{i,S}$. Starting with $T = \emptyset$, then the singletons, pairs, and successively larger sets, each new equation identifies one new coefficient after subtracting the coefficients already determined.

Low order is the substantive restriction

The **order-β model** imposes

$$
c_{i,S} = 0 \quad \text{whenever } |S| > \beta. \tag{0.4}
$$

Define $\mathcal{S}_i^{\beta} = \{S \subseteq \mathcal{N}_i : |S| \leq \beta\}$, including the empty set. The model becomes

$$
Y_i(\mathbf{z}) = \sum_{S \in \mathcal{S}_i^{\beta}} c_{i,S} \prod_{j \in S} z_j. \tag{0.5}
$$

For $\beta = 1$, all influences are additive. A unit can have many influencing neighbors, and their effects can all differ. For $\beta = 2$, any pair of assignments can interact, but the contribution of that pair cannot depend on a third assignment. Neither specification requires common coefficients across units. The effect of j on i may differ from its effect on k, and direct effects may differ throughout the network.

Low order consequently differs from low degree in the network. A user with hundreds of friends can satisfy an additive model. Conversely, an outcome that changes only when all three members of a small neighborhood receive treatment requires a third-order term. It also differs from assuming that

only the fraction of treated neighbors matters: the polynomial allows different friends to matter differently.

The number of retained coefficients for unit i is $K_i = \sum_{r=0}^{\min(\beta, d_i)} \binom{d_i}{r}$. For fixed small β, this grows polynomially with neighborhood size, rather than as 2^{d_i}. But choosing a small order is a scientific assumption, not merely a way to reduce computation. A threshold or saturation response need not have small degree just because its verbal description is simple.

The global effect is a sum of coefficients

At the all-treated assignment, every product in Equation 0.5 equals one. At the all-control assignment, only the empty-set term remains. Therefore,

$$Y_i(\mathbf{1}_N) - Y_i(\mathbf{0}_N) = \sum_{\substack{S \in \mathcal{S}_i^\beta \\ S \neq \emptyset}} c_{i,S}, \qquad \tau_{\mathrm{G}} = \frac{1}{N} \sum_i \sum_{\substack{S \in \mathcal{S}_i^\beta \\ S \neq \emptyset}} c_{i,S}. \tag{0.6}$$

The intercept cancels, but every nonempty interaction enters the global contrast. We have converted the causal target into a linear function of the coefficients. We next ask whether the experiment can estimate that function without learning each response surface accurately.

Start with additive spillovers

Randomization isolates a neighbor's contribution

Under $\beta = 1$, write the response as

$$Y_i(\mathbf{Z}) = c_{i,\emptyset} + \sum_{k \in \mathcal{N}_i} c_{i,\{k\}} Z_k.$$

To learn the contribution of a particular neighbor j, multiply the observed outcome by the centered treatment indicator $Z_j - \pi_j$. Independence makes this indicator uncorrelated with the intercept and with every other neighbor's assignment. The one surviving term is

$$\begin{aligned} \mathbb{E}_Z[Y_i^{\mathrm{obs}}(Z_j - \pi_j)] &= c_{i,\emptyset} \mathbb{E}_Z[Z_j - \pi_j] + \sum_{k \in \mathcal{N}_i} c_{i,\{k\}} \mathbb{E}_Z[Z_k(Z_j - \pi_j)] \\ &= c_{i,\{j\}} \pi_j (1 - \pi_j). \end{aligned} \tag{0.7}$$

Dividing by the treatment variance gives an unbiased estimator of $c_{i,\{j\}}$. Summing these estimates over j and then averaging over outcome units yields

$$\hat{\tau}_{\mathrm{SNIPE}(1)} = \frac{1}{N} \sum_i Y_i^{\mathrm{obs}} \sum_{j \in \mathcal{N}_i} \frac{Z_j - \pi_j}{\pi_j (1 - \pi_j)}. \tag{0.8}$$

SNIPE stands for Structured Neighborhood Interference Polynomial Estimator (Cortez-Rodriguez, Eichhorn, and Yu 2023). Each outcome receives a weight determined by the assignments of all units that can influence it. For additive interference, the weight accumulates the evidence supplied by each neighbor's randomized assignment.

This expression has the familiar Horvitz–Thompson estimator as a special case. If $\mathcal{N}_i = \{i\}$, then

$$\frac{Z_i - \pi_i}{\pi_i(1-\pi_i)} = \frac{Z_i}{\pi_i} - \frac{1-Z_i}{1-\pi_i}.$$

Under interference, observing Y_i gives information about effects associated with several assignments, not just Z_i. The extra terms use that information under the additive restriction.

An explicit four-assignment calculation

Consider one outcome with neighborhood $\{1,2\}$ and response $Y_i(z_1,z_2) = 10 + 2z_1 + 3z_2$. The global contrast for this unit is 5. Under independent fair coins, each of the four assignments occurs with probability 1/4. Let ℓ_i denote the inner sum of weights in Equation 0.8. The contribution from this outcome is $Y_i^{\text{obs}}\ell_i$.

(Z_1, Z_2)	Y_i^{obs}	$\ell_i = 4(Z_1 - 1/2) + 4(Z_2 - 1/2)$	$Y_i^{\text{obs}}\ell_i$
$(0,0)$	10	-4	-40
$(1,0)$	12	0	0
$(0,1)$	13	0	0
$(1,1)$	15	4	60

The average of the last column is 5. A single contribution can nevertheless be far from 5. Unbiasedness does not mean that one observed outcome estimates all of its coefficients precisely. Precision, if available, comes from averaging contributions across many units while controlling their dependence.

The intercept of 10 contributes random noise although it cancels from the target. Subtracting a known baseline would leave the expectation unchanged and could improve precision. We return to this observation when introducing covariate adjustment.

With two neighbors and fair coins, the mixed assignments happen to receive zero weight. This cancellation is special. With three neighbors and fair coins, the weight is $4\sum_{j\in\mathcal{N}_i} Z_j - 6$, which equals -2 or 2 for mixed assignments containing one or two treated units. Those assignments directly contribute to the estimator.

From coefficient estimation to the pseudoinverse

The known design replaces a missing regression sample

The additive calculation used independence to isolate one coefficient. A matrix formulation will handle interactions and more general designs with the same underlying logic.

Choose an ordering of $\mathcal{S}_i^\beta$, putting the empty set first. Let $\mathbf{v}_i(\mathbf{z})$ be the K_i-dimensional column vector whose entry indexed by S is $\prod_{j\in S} z_j$. Let $\mathbf{c}_i$ collect the coefficients in the same order. For example, with two neighbors and $\beta = 2$,

$$\mathbf{v}_i(\mathbf{z}) = (1, z_1, z_2, z_1 z_2)^\top, \qquad Y_i(\mathbf{z}) = \mathbf{v}_i(\mathbf{z})^\top \mathbf{c}_i.$$

The vector $\mathbf{a}_i = \mathbf{v}_i(\mathbf{1}_N) - \mathbf{v}_i(\mathbf{0}_N)$ has first entry zero and all remaining entries one. It specifies the desired coefficient sum: the unit's global contrast is $\mathbf{a}_i^\top \mathbf{c}_i$.

Now imagine, only as a thought experiment, repeating the assignment many times for the same unit with its response function unchanged. With enough repeated assignments, ordinary least squares would regress the repeated outcomes on the corresponding feature vectors. The population normal equations would be

$$\underbrace{\mathbb{E}_Z[\mathbf{v}_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})^\top]}_{\mathbf{M}_i} \mathbf{c}_i = \mathbb{E}_Z[\mathbf{v}_i(\mathbf{Z}) Y_i^{\text{obs}}]. \tag{0.9}$$

The $K_i \times K_i$ matrix $\mathbf{M}_i$ is the **design moment matrix**. Its entries depend on assignment probabilities, not unknown outcome coefficients. For Bernoulli assignment,

$$(\mathbf{M}_i)_{S,T} = \mathbb{E}_Z\left[\prod_{j\in S\cup T} Z_j\right] = \prod_{j\in S\cup T} \pi_j. \tag{0.10}$$

The union appears because an indicator present in both products still equals itself when squared. The first entry, indexed by two empty sets, equals one.

In the actual experiment we have one outcome per unit, not a sample of repeated outcomes for each unit. We therefore cannot run a separate ordinary regression and identify all of its coefficients from that one row. Nor can we silently pool units: their coefficient vectors are allowed to differ.

What we can do is use the known left-hand matrix in Equation 0.9 and an unbiased, one-observation estimate of its right-hand side. When $\mathbf{M}_i$ is invertible,

$$\hat{\mathbf{c}}_i = \mathbf{M}_i^{-1} \mathbf{v}_i(\mathbf{Z}) Y_i^{\text{obs}}, \qquad \mathbb{E}_Z[\hat{\mathbf{c}}_i] = \mathbf{c}_i. \tag{0.11}$$

The apparent abundance of unbiased coefficient estimates is not an abundance of information in one realized assignment. All entries of $\hat{\mathbf{c}}_i$ use the same outcome and can be very noisy and strongly dependent. Their expectations recover the coefficients across assignments.

What singularity means

Independent Bernoulli assignment with every probability strictly between zero and one gives positive probability to every local assignment. The distinct multilinear features are linearly independent on that support, so $\mathbf{M}_i$ is positive definite. Other designs can make it singular. For example, always assigning two neighbors together makes their two treatment indicators identical in every experiment.

The Moore–Penrose pseudoinverse, written $\mathbf{M}_i^\dagger$, inverts the nonzero eigenvalues of this symmetric matrix and leaves its zero eigenvalues at zero. The associated estimator is

$$\hat{\tau}_{\mathrm{PI}} = \frac{1}{N}\sum_i Y_i^{\mathrm{obs}} \ell_i(\mathbf{Z}), \qquad \ell_i(\mathbf{Z}) = \mathbf{a}_i^\top \mathbf{M}_i^\dagger \mathbf{v}_i(\mathbf{Z}). \tag{0.12}$$

PI denotes pseudoinverse. This is the pseudoinverse of an expectation under the design, not the pseudoinverse of the single realized outer product $\mathbf{v}_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})^\top$.

Proposition 0.0.1 (When the pseudoinverse estimates the desired contrast)**.** *Suppose the order-β response model holds. If $\mathbf{a}_i$ lies in the column space of $\mathbf{M}_i$ for every i, then $\mathbb{E}_Z[\hat{\tau}_{\mathrm{PI}}] = \tau_{\mathrm{G}}$. Because $\mathbf{M}_i$ is symmetric, its column and row spaces coincide.*

Proof. For each unit, substitute $Y_i^{\mathrm{obs}} = \mathbf{v}_i(\mathbf{Z})^\top \mathbf{c}_i$ to obtain

$$\begin{aligned}\mathbb{E}_Z[Y_i^{\mathrm{obs}} \ell_i(\mathbf{Z})] &= \mathbf{a}_i^\top \mathbf{M}_i^\dagger \mathbb{E}_Z[\mathbf{v}_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})^\top]\mathbf{c}_i \\ &= \mathbf{a}_i^\top \mathbf{M}_i^\dagger \mathbf{M}_i \mathbf{c}_i.\end{aligned}$$

The matrix $\mathbf{P}_i = \mathbf{M}_i^\dagger \mathbf{M}_i$ is the orthogonal projection onto the column space of $\mathbf{M}_i$. The stated condition gives $\mathbf{a}_i^\top \mathbf{P}_i = \mathbf{a}_i^\top$, so the expectation is $\mathbf{a}_i^\top \mathbf{c}_i$. Average these equalities over units. □

Singularity can thus prevent estimation of individual coefficients without preventing estimation of their sum. If two neighbors always move together, their separate additive effects cannot be distinguished, but their combined effect may remain estimable. The target contrast, not the full coefficient vector, determines the required support condition.

The condition is also necessary for unbiased estimation of every possible coefficient vector in this unrestricted unit-specific model. If it fails, there is a vector $\mathbf{b}_i$ in the null space of $\mathbf{M}_i$ for which $\mathbf{a}_i^\top \mathbf{b}_i \neq 0$. Since

$$0 = \mathbf{b}_i^\top \mathbf{M}_i \mathbf{b}_i = \mathbb{E}_Z[(\mathbf{v}_i(\mathbf{Z})^\top \mathbf{b}_i)^2],$$

adding $\mathbf{b}_i$ to the coefficients changes no outcome on the design's support. It does change the desired global contrast. The experiment cannot distinguish these two response schedules. Applying a pseudoinverse does not resolve this identification failure; it selects the projection that is visible under the design.

The geometry behind the weights

The estimation problem can also be expressed without estimating coefficients. We want a random weight $w_i(\mathbf{Z})$ that reproduces the global contrast for every allowed polynomial response. That requires the coordinatewise moment equations

$$\mathbb{E}_Z[w_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})] = \mathbf{a}_i. \tag{0.13}$$

The first equation says the weight has mean zero, so a constant outcome makes no contribution in expectation. Every other equation says that a permitted treatment product receives the correct coefficient of one in the target.

When the estimability condition holds, these equations are feasible. Among weights satisfying them, ℓ_i in Equation 0.12 has minimum $\mathbb{E}_Z[w_i^2]$. Indeed, if $w_i = \ell_i + r_i$, the moment equations give $\mathbb{E}_Z[r_i\mathbf{v}_i] = 0$. Since ℓ_i is a linear combination of these features, $\mathbb{E}_Z[r_i\ell_i] = 0$, and

$$\mathbb{E}_Z[w_i^2] = \mathbb{E}_Z[\ell_i^2] + \mathbb{E}_Z[r_i^2].$$

This is a least-squares projection argument: the additional component of a weight orthogonal to the allowed response features cannot help reproduce their contrast. Minimum squared weight is useful for controlling noise, but it is not a universal minimum-variance claim for the whole estimator. Outcomes differ across assignments, and contributions from different units can covary.

Higher-order interactions under independent assignment

Centering provides an orthogonal basis

The raw monomial matrix can be large. Independent assignment gives a more transparent route to the same estimator. For a subset S of the neighborhood, define the centered product

$$u_S(\mathbf{Z}) = \prod_{j\in S}(Z_j - \pi_j), \qquad u_\emptyset(\mathbf{Z}) = 1.$$

If $S \neq T$, some indicator occurs in exactly one of the products. Its mean-zero factor and independence imply $\mathbb{E}_Z[u_S u_T] = 0$. Moreover, $\mathbb{E}_Z[u_S^2] = \prod_{j\in S}\pi_j(1-\pi_j)$. These centered products span the same order-β polynomial space as the raw products. Subtracting constants from the inputs changes coefficients, but not the maximum degree.

We can therefore write $Y_i(\mathbf{z}) = \sum_{S\in\mathcal{S}_i^\beta} b_{i,S}u_S(\mathbf{z})$, where the $b_{i,S}$ are coefficients in this centered basis. Orthogonality gives

$$\mathbb{E}_Z\left[Y_i^{\text{obs}}\frac{u_S(\mathbf{Z})}{\prod_{j\in S}\pi_j(1-\pi_j)}\right] = b_{i,S}.$$

The global contrast of the basis function u_S is

$$\Delta_S = u_S(\mathbf{1}_N) - u_S(\mathbf{0}_N) = \prod_{j \in S}(1-\pi_j) - \prod_{j \in S}(-\pi_j).$$

For the empty set this difference is zero. Combining the unbiased estimates of the centered coefficients with their global contrasts gives

$$\hat{\tau}_{\mathrm{SNIPE}(\beta)} = \frac{1}{N}\sum_i Y_i^{\mathrm{obs}} \sum_{\substack{S \in \mathcal{S}_i^\beta \\ S \neq \emptyset}} \Delta_S \prod_{j \in S} \frac{Z_j - \pi_j}{\pi_j(1-\pi_j)}. \tag{0.14}$$

This explicit formula equals Equation 0.12 under independent Bernoulli assignment. The centered basis diagonalizes the design inner product, doing the work that matrix inversion performs in the raw basis.

A pair interaction and a revealing cancellation

Consider again a two-person neighborhood, now with $Y_i(z_1, z_2) = 10 + 2z_1 + 3z_2 + 4z_1z_2$. Its global contrast is 9. Take common treatment probability $p = 1/4$ and define $q_j = (Z_j - p)/\{p(1-p)\}$. Each singleton has $\Delta_{\{j\}} = 1$, while $\Delta_{\{1,2\}} = (1-p)^2 - p^2 = 1 - 2p = 1/2$. The order-two weight is

$$\ell_i = q_1 + q_2 + \tfrac{1}{2}q_1q_2.$$

The first two terms alone recover the additive contribution 5 plus $2p \cdot 4 = 2$ from the pair interaction. To check the latter statement, $\mathbb{E}_Z[Z_1Z_2q_1] = p$ and $\mathbb{E}_Z[Z_1Z_2q_2] = p$. The pair term supplies the missing amount: $\mathbb{E}_Z[Z_1Z_2q_1q_2] = 1$, so its contribution is $(1/2) \cdot 4 = 2$. The expectation of the full contribution is therefore $5 + 2 + 2 = 9$.

This calculation also explains why the additive estimator is generally biased when pair interactions are present. It estimates only the part of the pair effect that appears when the other member is treated at the experiment's assignment probability. The full rollout requires that other member to be treated with certainty.

There is an instructive exception. At $p = 1/2$, the pair contrast $\Delta_{\{1,2\}}$ is zero, so the order-two and order-one SNIPE weights coincide. Pair effects still enter the GATE. They are already captured through the first-order centered coefficients: changing from raw to centered products moves part of a raw pair term into the linear terms. For the example above, the additive weight at $p = 1/2$ recovers $5 + 2p \cdot 4 = 9$.

This cancellation does not make arbitrary high-order interactions harmless. For a raw triple term $\gamma z_1 z_2 z_3$, the order-one weight at common probability p has expectation $3p^2\gamma$, rather than γ. At $p = 1/2$ it recovers only $3\gamma/4$. Correctness depends jointly on the target, the response model, and the design.

Precision requires information about shared assignments

Interaction order controls a local difficulty

The centered representation makes the squared size of a unit's weight explicit:

$$\mathbb{E}_Z[\ell_i^2] = \sum_{\substack{S \in \mathcal{S}_i^\beta \\ S \neq \emptyset}} \frac{\Delta_S^2}{\prod_{j \in S} \pi_j (1 - \pi_j)}. \tag{0.15}$$

Products of small treatment probabilities still matter. However, the largest product now involves at most β units, rather than the entire neighborhood. For fixed β and probabilities bounded away from zero and one, the number of terms grows polynomially with d_i. This is the mechanism by which low order can improve precision relative to requiring full neighborhood exposure. It does not guarantee small variance for every network or every outcome schedule.

For additive interference with common p, the formula simplifies to $\mathbb{E}_Z[\ell_i^2] = d_i / \{p(1-p)\}$. Each extra influencing assignment adds to the random fluctuation of the weight. A high-degree outcome can be informative about many treatment contributions, but that information is not costless.

Shared neighbors generate covariance

Let $T_i = Y_i^{\text{obs}} \ell_i(\mathbf{Z})$ be unit i's contribution. The exact variance is

$$\operatorname{Var}_Z(\hat{\tau}_{\text{PI}}) = \frac{1}{N^2} \sum_{i,k} \operatorname{Cov}_Z(T_i, T_k). \tag{0.16}$$

Under independent unit assignment and neighborhood interference, T_i and T_k are independent when $\mathcal{N}_i \cap \mathcal{N}_k = \emptyset$. When their neighborhoods overlap, one treatment coin can affect both weighted outcomes. The number of units is therefore not the number of independent pieces of information.

A simple bound exposes this distinction. Suppose $|Y_i(\mathbf{z})| \leq B$ for all units and assignments, let $\mathbb{E}_Z[\ell_i^2] \leq A$, and suppose each neighborhood overlaps with at most D neighborhoods, counting itself. By Cauchy–Schwarz, $|\operatorname{Cov}_Z(T_i, T_k)| \leq B^2 A$. There are at most ND potentially nonzero covariance terms, so

$$\operatorname{Var}_Z(\hat{\tau}_{\text{PI}}) \leq \frac{B^2 A D}{N}. \tag{0.17}$$

This deliberately elementary bound separates three difficulties: outcome magnitude, the size of the weights, and dependence across contributions. It also shows why bounding neighborhood size alone is insufficient in a directed network. A single influential unit may appear in many small neighborhoods, creating widespread dependence.

Cortez-Rodriguez, Eichhorn, and Yu (2023) give sharper variance bounds and asymptotic normality results under their network and outcome regularity conditions. Such results require a sequence of experiments in which dependence and influential contributions remain controlled. Exact unbiasedness by itself supplies neither a central limit theorem nor a reliable standard error.

Inference is a further problem

An empirical variance treating the T_i as independent drops the covariance terms in Equation 0.16. A usual regression standard error does not recover those terms either. The relevant dependence comes from shared assignments, and uncertainty calculations must reflect it.

Even with a known design, exact variance estimation is difficult. The variance contains products of unknown potential-outcome coefficients, while each unit reveals only one outcome. Unbiased estimation of each coefficient does not imply that multiplying two estimated coefficients gives an unbiased estimate of their product. Conservative bounds and dependence-aware asymptotic variance estimators address different parts of this difficulty. A normal approximation should be paired with a justified variance procedure and the assumptions under which the approximation is valid.

For a small network, exact enumeration is useful for checking an implementation under a specified response schedule. It is a simulation exercise: the missing potential outcomes are supplied by the analyst. It does not turn the variance under the real, unknown schedule into an observed quantity.

Changing the design while retaining the model

The moment equations did not require independent treatment assignment. What changes under a different design is $\mathbf{M}_i$, its column space, and the covariance between contributions. Eichhorn, Khan, Ugander, and Yu (2026) develop this general-design perspective, with particular attention to clustered experiments.

If two additive neighbors always receive the same cluster assignment W, then

$$Y_i = c_{i,\emptyset} + (c_{i,\{1\}} + c_{i,\{2\}})W.$$

The separate effects are indistinguishable, but the sum is the global contrast and is identified when $0 < \mathbb{P}_Z(W = 1) < 1$. The moment matrix based on $(1, Z_1, Z_2)$ is singular; its null direction changes the two coefficients by opposite amounts and leaves their sum unchanged. This is exactly the favorable case in Proposition 0.0.1.

More generally, under independent Bernoulli assignment of clusters, every configuration of the clusters touching a neighborhood has positive probability. If a polynomial vanishes on this support, it vanishes at both the all-treated and all-control cluster assignments, so its global contrast is zero. The desired contrast is therefore estimable even when the original unit-level coefficient vector is not.

Complete randomization of a fixed number of clusters can behave differently. For an extreme example, take a two-unit neighborhood and always treat exactly one of its members. The two additive schedules $Y_i(\mathbf{z}) = 0$ and $Y_i(\mathbf{z}) = z_1 + z_2 - 1$ agree at both supported assignments. Their all-treated-minus-all-control contrasts are 0 and 2. No estimator can distinguish them under this design. A restriction on total treatment count can remove precisely the variation needed to learn a global rollout effect.

Clustering also changes precision. It can reduce the number of independently varying treatment components relevant to an outcome, while correlating the weighted outcomes of more units. The cited work combines these effects in design-specific bounds and uses the bounds to compare candidate

clusterings. The lesson is to choose the design and analysis together: a low-order model does not eliminate the reasons to cluster, and clustering does not eliminate the value of outcome structure.

Covariate adjustment should target the estimator's variance

The additive example showed that a large baseline outcome can create noise without changing the GATE. Suppose X_i is a fixed vector of pretreatment covariates predictive of that baseline. Since the first coordinate of Equation 0.13 gives $\mathbb{E}_Z[\ell_i] = 0$, any fixed coefficient vector θ yields the adjusted estimator

$$\hat{\tau}(\theta) = \frac{1}{N}\sum_i \ell_i(\mathbf{Z})\{Y_i^{\text{obs}} - X_i^\top \theta\}. \tag{0.18}$$

Its expectation is still τ_{G} whenever the original estimator is unbiased. The adjustment subtracts a random quantity with known mean zero. No correct linear model relating outcomes to X_i is required for this fixed-θ statement.

To see how to choose θ, define the random covariate vector $\mathbf{H} = N^{-1}\sum_i \ell_i X_i$, with $\mathbb{E}_Z[\mathbf{H}] = 0$. Then $\hat{\tau}(\theta) = \hat{\tau}(0) - \theta^\top \mathbf{H}$ and

$$\text{Var}_Z\{\hat{\tau}(\theta)\} = \text{Var}_Z\{\hat{\tau}(0)\} - 2\theta^\top \text{Cov}_Z\{\mathbf{H}, \hat{\tau}(0)\} + \theta^\top \text{Var}_Z(\mathbf{H})\theta. \tag{0.19}$$

If $\text{Var}_Z(\mathbf{H})$ is invertible, the oracle choice is its inverse times $\text{Cov}_Z\{\mathbf{H}, \hat{\tau}(0)\}$. This choice minimizes the variance of the final effect estimator. Ordinary regression of observed outcomes on covariates solves a different objective, and can neglect cross-unit covariance induced by interference.

Wang and Li (2026) make this difference central. Their variance-improvement-maximized (VIM) procedure estimates the change in variance from adjusting and chooses coefficients to maximize that improvement. A useful cancellation makes this feasible: the variance of the unadjusted estimator contains quadratic terms in unknown response coefficients, but those terms cancel when subtracting the adjusted variance. The remaining improvement can be estimated using the coefficient-estimation machinery already developed.

Their no-harm guarantee concerns the asymptotic variance of this particular procedure under the stated sparsity, moment, and coefficient-estimation conditions. It is not a guarantee for arbitrary regression adjustment or for every realized sample. Fixed coefficients preserve exact unbiasedness; fitting $\widehat{\theta}$ using the same experimental outcomes introduces additional randomness and requires an asymptotic argument. The paper also develops an asymptotically conservative variance estimator for inference.

In practice, choose covariates before examining treatment effects, use the dependence-aware adjustment procedure rather than substituting a familiar regression by analogy, and inspect whether the network conditions supporting its approximation are plausible. Randomly splitting outcome units does not automatically produce independent training and evaluation data, because their outcomes can depend on the same treatment coins.

Other structural information: monotone interference

Restricting directions instead of interaction order

An intervention may plausibly benefit from wider adoption without producing an additive response. A messaging feature might become useful only when a critical mass of friends uses it. A vaccination may become more protective as more neighbors are vaccinated. In such settings, the sign of a spillover can be more defensible than a low-order polynomial model.

One possible restriction is coordinatewise monotonicity: if $\mathbf{z} \leq \mathbf{z}'$ in every coordinate, then $Y_i(\mathbf{z}) \leq Y_i(\mathbf{z}')$. For spillovers alone, one might impose this ordering only while holding z_i fixed. The sign must match the outcome: beneficial vaccination could make disease incidence nonincreasing. These are substantive restrictions on potential outcomes, not consequences of randomization.

Monotonicity and low order are distinct. The function $\prod_{j \in \mathcal{N}_i} z_j$ is monotone and can have arbitrarily high order. An additive model with one positive and one negative coefficient is low order but not coordinatewise nondecreasing. Nor does monotonicity force every raw polynomial coefficient to be nonnegative: the function $z_1 + z_2 - z_1 z_2$, representing whether at least one person is treated, is monotone and has a negative pair coefficient.

What monotonicity can provide

For a simple illustration, define the marginal-treatment estimator

$$\hat{\tau}_{\text{marg}} = \frac{1}{N} \sum_i \left\{ \frac{Z_i Y_i^{\text{obs}}}{\pi_i} - \frac{(1 - Z_i) Y_i^{\text{obs}}}{1 - \pi_i} \right\}.$$

Its expectation compares the outcome conditional on own treatment to that conditional on own control. Under full coordinatewise nondecreasing outcomes, every outcome observed when $Z_i = 1$ is at most $Y_i(\mathbf{1}_N)$, and every outcome observed when $Z_i = 0$ is at least $Y_i(\mathbf{0}_N)$. Consequently,

$$\mathbb{E}_Z[\hat{\tau}_{\text{marg}}] \leq \tau_{\text{G}}. \tag{0.20}$$

This is a directional bias statement. It is neither a point-identification result for τ_{G} nor a claim that a realized estimate is a lower confidence bound. Establishing uncertainty around a bound needs further inference.

Several papers use structural inequalities for different purposes:

- Choi (2017) develops confidence intervals for attributable effects, comparing outcomes under the realized assignment with outcomes under no treatment. The restrictions resemble nonnegative treatment effects and do not require partial interference. That attributable-effect target differs from the all-treated-versus-all-control GATE.
- Eckles, Karrer, and Ugander (2017) use monotonicity and related conditions to study bias reduction through correlated assignment and exposure-based analysis, allowing potentially global interference. Their conditions explain when moving experimental exposure closer to a global policy comparison reduces bias; they do not make every clustering universally preferable.

- Pouget-Abadie and colleagues (2018) study comparing clustered designs when the corresponding estimators are biased in a common direction. If both expected estimates lie below the target, the larger expected estimate is closer to it. Their experiment-of-experiments design addresses how to compare clusterings experimentally. Here the relevant monotonicity is a property of bias across a family of designs, which should not be conflated with one generic unit-level inequality.
- Huang, Li, and Toulis (2025) develop finite-sample-valid randomization tests for monotone spillover contrasts using appropriate conditioning and network constructions. An inequality null does not reveal every missing potential outcome as a sharp null does; valid testing needs that additional construction. Such tests concern directional hypotheses, not unrestricted estimation of the GATE.

Submodularity is another possible restriction. Viewing an outcome as a function of the treated set, it says that the incremental effect of adding a given treated unit is no larger when the already-treated set is larger. It formalizes diminishing returns. It can coexist with monotonicity but is not implied by it, and it does not impose low polynomial order. The appropriate structure depends on the mechanism one is prepared to defend.

The central lesson of the lecture is that a partially treated network can inform a global rollout through several different kinds of structure. Low order turns the rollout contrast into a linear coefficient functional, and randomization supplies unbiased estimating equations for that functional. Monotonicity instead supplies inequalities. In either case, the experiment's design determines what variation is available, and the structural assumption determines what that variation can tell us about unobserved assignments.

Advanced material: model error and the cost of extrapolation

An exact bias decomposition

Suppose a chosen order-β model leaves a remainder:

$$Y_i(\mathbf{z}) = \mathbf{v}_i(\mathbf{z})^\top \mathbf{c}_i + R_i(\mathbf{z}).$$

Assume the contrast is estimable for the retained polynomial space. The polynomial part remains unbiased, so the bias of the estimator for the actual GATE is exactly

$$\mathbb{E}_Z[\hat{\tau}_{\mathrm{PI}}] - \tau_{\mathrm{G}} = \frac{1}{N}\sum_i \left[\mathbb{E}_Z\{\ell_i(\mathbf{Z})R_i(\mathbf{Z})\} - \{R_i(\mathbf{1}_N) - R_i(\mathbf{0}_N)\}\right]. \tag{0.21}$$

Two features matter: how the omitted response aligns with the estimator's weight over the experimental distribution, and how much it contributes at the two policy endpoints. A model can fit typical experimental assignments well while misrepresenting the full rollout. With low treatment probability, a high-order interaction may rarely activate in the experiment yet contribute substantially when everyone is treated.

For example, if $|R_i(\mathbf{z})| \le \varepsilon_i$ uniformly over all assignments, Cauchy–Schwarz yields the bound

$$|\operatorname{Bias}_Z(\hat{\tau}_{\mathrm{PI}})| \le \frac{1}{N}\sum_i \varepsilon_i \left\{\sqrt{\mathbb{E}_Z[\ell_i^2]} + 2\right\}.$$

The uniform qualification is doing work: accuracy only on assignments that occur frequently does not control the endpoint remainder. Large weights also amplify model error over the experimental distribution.

If the contrast is not estimable under the design, there is an additional polynomial bias term $N^{-1}\sum_i \mathbf{a}_i^\top(\mathbf{P}_i - \mathbf{I})\mathbf{c}_i$, where $\mathbf{P}_i = \mathbf{M}_i^\dagger \mathbf{M}_i$ and $\mathbf{I}$ is the identity matrix. This separates lack of design support from misspecifying the outcome class. Either can obstruct the target, and they call for different responses.

Choosing an order and extending the experiment

Increasing β includes more response schedules but can increase the number and size of weights. A defensible analysis should connect the chosen order to a mechanism, examine sensitivity to plausible higher-order effects, and consider whether the design can improve the relevant treatment variation. Selecting the order after searching for a favorable estimated effect is not covered by the fixed-model unbiasedness argument.

The downstream literature also changes what data are collected. Under a common Bernoulli treatment probability p, the design mean of an order-β response is a univariate polynomial in p:

$$\frac{1}{N}\sum_i \mathbb{E}_{Z,p}[Y_i(\mathbf{Z})] = \frac{1}{N}\sum_i \sum_{S\in\mathcal{S}_i^\beta} c_{i,S}\, p^{|S|}.$$

Its value at one minus its value at zero is the GATE. Varying treatment saturation across rollout stages can therefore replace some detailed network information with observations at several values of p. But extrapolating a polynomial from a narrow range of probabilities to a full rollout can be unstable. Repeated stages also require a stable response interpretation, with time and carryover effects addressed by the design and assumptions.

Cortez-Rodriguez, Eichhorn, and Yu (2025) study a two-stage rollout that first selects a subpopulation and then rolls out treatment within it. Their analysis connects bias to cut edges and variance to clustering properties including covariate balance. Alongside clustered pseudoinverse estimation and VIM covariate adjustment, this is a concrete development beyond the original SNIPE construction: improve the information available to the estimator, rather than expecting an outcome restriction alone to solve every precision problem.

Further reading

Cortez-Rodriguez, Eichhorn, and Yu (2023) are the main source for this lecture. The centered-basis derivation above is a way to see the estimator's mechanics; the paper supplies the broader variance and asymptotic analysis. Eichhorn and colleagues (2026) extend the construction to general designs and make design selection part of the analysis. Their preprint first appeared in 2024; the cited version was revised in 2026.

For subsequent work, Wang and Li (2026) distinguish predicting outcomes well from reducing the variance of an interference estimator, while Cortez-Rodriguez and colleagues (2025) develop the rollout-design direction. These are complementary routes: use covariates, change dependence through clustering, or observe more than one treatment saturation.

The monotonicity papers cited above ask different causal questions with different inequalities. Reading them together is useful precisely because monotonicity is not one transferable assumption with one universal conclusion. Track the target, the direction of the restriction, the experimental design, and whether the result concerns bias, a bound, or a hypothesis test.

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L07: Policy Evaluation and Panel Data

Stochastic seeding policies and low-rank counterfactuals

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Evaluate a rule for choosing treatments, using seeding strategies in networks as a concrete example.
- Use known assignment probabilities to learn about strategies other than the one that generated the data.
- Recast panel-data causal inference as a problem of missing untreated outcomes.
- Explain how shared temporal patterns make a low-rank model useful, and what additional assumptions make its predictions causal.
- Distinguish uncertainty from randomization from uncertainty that rests on an outcome model.

Evaluate a rule, not just a treatment label

A public-health campaign can contact a small number of people in each village and hope that information spreads. Suppose the budget permits contacting one person. One strategy chooses that person uniformly at random. Another chooses a random resident, asks them to name a friend, and contacts the friend. Both strategies deliver the same intervention to one person. They differ in which person is likely to receive it.

The policy question is about the resulting diffusion through the village: which strategy produces more adoption, at the same contact budget? Comparing outcomes of contacted and uncontacted people is not the relevant comparison. An uncontacted person who learns from a neighbor is part of the strategy's success. The causal intervention is the rule used to choose seeds, and its outcome includes the responses of people who were not seeded.

Chin, Eckles, and Ugander (2022) study precisely this evaluation problem. Their formulation connects network experiments to **policy evaluation**: learning the expected outcome under a specified rule for assigning an intervention. A policy can be stochastic, meaning that it assigns probabilities to several possible interventions rather than always choosing the same one.

This perspective continues L05. There we distinguished changing a person's own treatment from changing the surrounding treatment environment. Here we specify a distribution over those

environments and ask how well that distribution performs. The known randomization law will let us evaluate more than the experimental arm labels alone.

Potential outcomes for whole seed sets

Consider M villages, indexed by $g = 1, \ldots, M$. Village g has a fixed pretreatment graph with vertex set V_g, and $\mathcal{S}_g$ is its set of allowable seed sets. For example, with a budget of k distinct contacts, $\mathcal{S}_g = \{s \subseteq V_g : |s| = k\}$. Let $Y_g(s)$ be the number of adopters in that village at a specified follow-up time if seed set s were contacted.

All of these potential outcomes and graphs are fixed in the initial design-based argument. Randomness comes from selecting the seed sets. If diffusion itself is modeled as random, an additional expectation over that process must be stated; it is not silently included in the assignment expectation below.

Arbitrary interference within a village is allowed: $Y_g(s)$ summarizes the entire response to the selected set. Two other restrictions are important. First, one village's seed set must not change another village's outcome. Second, the strategy label must have no effect beyond the realized seed set. If two strategies select the same people but train them differently, their potential outcomes cannot be indexed by the set alone. The intervention would need to include the training protocol.

A policy p specifies known probabilities $p_g(s)$ over $\mathcal{S}_g$ for each village. Its finite-population value is

$$V_M(p) = \frac{1}{M} \sum_{g=1}^{M} \sum_{s \in \mathcal{S}_g} p_g(s) Y_g(s). \tag{0.1}$$

There are two averages here: first over the policy's possible seed sets in a particular village, then over the villages. This target is the average number of adopters per village, not the average treatment effect on a seeded person. If village sizes differ, average adoption rates or a resident-weighted target would be different choices. Specify that choice through the outcome and the village weights before estimating anything.

For two fixed policies A and B, write

$$\Delta_M(A, B) = V_M(p^A) - V_M(p^B) = \frac{1}{M} \sum_{g} \sum_{s} \{p_g^A(s) - p_g^B(s)\} Y_g(s). \tag{0.2}$$

The same potential outcome can receive positive weight, negative weight, or zero weight depending on how the strategies change its probability. Evaluating a stochastic policy is thus an average of causal interventions, not an attempt to declare each realized seed set a new experimental arm.

A four-person example

Take a star with one center c connected to three leaves a, b, d. Consider one seed and a deliberately simple response: the seed adopts, and adoption then spreads to its immediate neighbors in one transmission step. It does not continue to a second step. Consequently, seeding the center produces four adopters, while seeding any leaf produces two.

Policy B chooses a uniform seed. Policy A chooses a uniform resident and then a uniform neighbor of that resident. The center is named whenever one of the three leaves is chosen first. A particular leaf is named only when the center is chosen first and then selects that leaf.

Seed	Uniform policy p^B	One-hop policy p^A	Adopters $Y(s)$
Center c	1/4	3/4	4
Leaf a	1/4	1/12	2
Leaf b	1/4	1/12	2
Leaf d	1/4	1/12	2

The two values are

$$V(p^B) = \frac{1}{4}(4) + \frac{3}{4}(2) = \frac{5}{2}, \qquad V(p^A) = \frac{3}{4}(4) + \frac{1}{4}(2) = \frac{7}{2}.$$

The policy gain is one additional adopter, averaged over the random choice of seed. This is an illustrative science table, not an empirical finding about one-hop targeting. If deterministic adoption instead continued until everyone reachable from the seed had adopted, all four seeds would produce four adopters and the policy contrast would be zero. The outcome horizon and diffusion mechanism matter as much as the graph.

For a general undirected graph without isolated vertices, let $\mathcal{A}_g(u)$ be the neighbors of u, excluding u itself, and let $\deg_g(u) = |\mathcal{A}_g(u)|$. The one-hop probability of selecting vertex v is

$$p_g^A(\{v\}) = \frac{1}{|V_g|} \sum_{u:\, v \in \mathcal{A}_g(u)} \frac{1}{\deg_g(u)}. \tag{0.3}$$

Each possible nominator supplies one route to choosing v. This distribution is not generally proportional to v's degree: being connected to a person with few friends gives a larger nomination probability than being connected to someone with many friends. Isolated vertices require an explicit protocol for failed nominations.

For several seeds, the complete sampling algorithm matters. Sampling with replacement, conditioning all nominations on being distinct, and rejecting duplicates sequentially need not yield the same distribution over sets. Inverse-probability weights must use the probability of the actual seed set under the implemented algorithm. A verbal label such as "friend targeting" does not specify those probabilities.

Evaluate a policy using the experiment we actually ran

The design and the target need not be the same distribution

Suppose the experiment draws $S_g \sim q_g$ independently across villages and observes $Y_g^{\text{obs}} = Y_g(S_g)$. The distribution q_g is the known experimental design; p_g^A and p_g^B are the policies we wish to evaluate. This is often called off-policy evaluation when a target policy differs from the data-generating policy.

For now assume that $q_g(s) > 0$ whenever either target policy assigns positive probability to s. Define

$$w_g^a(s) = \frac{p_g^a(s)}{q_g(s)}, \qquad a \in \{A, B\}.$$

The ratio is defined only for $q_g(s) > 0$, which includes every realized seed set. This ratio says how much more often the target would choose a seed set than the experiment does. It weights the entire village response, including spillovers. It is not the inverse probability that an individual resident is treated.

The importance-weighted value and contrast estimators are

$$\widehat{V}_{\text{HT}}(p^a) = \frac{1}{M}\sum_g w_g^a(S_g) Y_g^{\text{obs}}, \qquad \widehat{\Delta}_{\text{HT}} = \frac{1}{M}\sum_g \{w_g^A(S_g) - w_g^B(S_g)\} Y_g^{\text{obs}}. \tag{0.4}$$

Here HT emphasizes the same design-based cancellation used in L02, although the weight now contains a target-policy probability in its numerator.

Proposition 0.0.1 (Unbiased evaluation of a fixed stochastic policy)**.** *Under the stated consistency and support conditions,* $\mathbb{E}_q\{\widehat{V}_{\text{HT}}(p^a)\} = V_M(p^a)$ *and* $\mathbb{E}_q(\widehat{\Delta}_{\text{HT}}) = \Delta_M(A, B)$*. The subscript q denotes randomness under the experimental seed-selection design, with graphs and potential outcomes held fixed.*

Proof. For each village,

$$\mathbb{E}_q[w_g^a(S_g) Y_g^{\text{obs}}] = \sum_{s: q_g(s) > 0} q_g(s) \frac{p_g^a(s)}{q_g(s)} Y_g(s) = \sum_s p_g^a(s) Y_g(s).$$

Support coverage justifies the last equality. Averaging over villages estimates the policy value; subtracting the two identities estimates the contrast. Independence across villages is useful for variance analysis but is not needed for this expectation calculation if the correct marginal design probabilities are used. □

On the star, suppose only uniform seeding was used. To evaluate one-hop seeding, the center receives weight 3 and each leaf weight 1/3. The weighted outcome is therefore 12 when the center is selected and 2/3 otherwise. Its expectation is $(1/4)12 + (3/4)(2/3) = 7/2$, the one-hop value. The counterfactual policy is evaluated without pretending that we observed the same village under a second seed.

The policy contrast uses contributions 8 at the center and $-4/3$ at a leaf. Their expectation is one. A single contribution is a noisy estimate; the unbiasedness statement concerns repetitions of the experiment, and precision requires enough independent villages.

Why retain seed identities when strategies were randomized?

An experiment might first randomize a strategy label $R_g \sim \text{Bernoulli}(\rho)$, with $R_g = 1$ selecting policy A and $R_g = 0$ selecting B, then draw the seed set from that policy. After averaging over the label, the seed-set distribution is

$$q_g(s) = \rho p_g^A(s) + (1-\rho)p_g^B(s), \qquad 0 < \rho < 1. \tag{0.5}$$

An ordinary arm-based HT contribution is

$$T_g^{\text{label}} = \left\{ \frac{R_g}{\rho} - \frac{1-R_g}{1-\rho} \right\} Y_g^{\text{obs}}.$$

Its expectation is the village-specific policy contrast. But a seed set often could have been generated by either strategy. The label retains random variation that does not matter for the outcome once the seed set is known. Bayes' rule gives

$$\mathbb{P}(R_g = 1 \mid S_g = s) = \frac{\rho p_g^A(s)}{q_g(s)}.$$

Because the outcome depends only on s, conditioning the label-based contribution on that set yields

$$\mathbb{E}_q[T_g^{\text{label}} \mid S_g] = \frac{p_g^A(S_g) - p_g^B(S_g)}{q_g(S_g)} Y_g^{\text{obs}}. \tag{0.6}$$

This is exactly the seed-set contribution in Equation 0.4. The law of total variance therefore proves that replacing the label contribution by this conditional expectation cannot increase its variance. For independent villages, the same comparison holds for their averages. This is a Rao–Blackwell argument relative to the specified label-HT estimator, not a universal efficiency ordering over every difference-in-means or normalized estimator.

The practical implication is simple: record which seeds were selected, not only which strategy was assigned. These details reveal how much the two policies agree about the realized intervention.

For the star with $\rho = 1/2$, the mixture chooses the center with probability 1/2 and each leaf with probability 1/6. The seed-set contrast contribution is 4 for the center and -2 for a leaf. Its mean is

one and its variance is $\frac{1}{2}(4^2) + \frac{1}{2}(-2)^2 - 1^2 = 9$. The label-HT contribution has the same mean but variance 39. This is an exact calculation for the illustrative schedule, not a general numerical efficiency claim.

Normalization, support, and uncertainty

As in L02, a normalized estimator may stabilize random total weights:

$$\hat{V}_{\mathrm{H}}(p^a) = \frac{\sum_g w_g^a(S_g) Y_g^{\mathrm{obs}}}{\sum_g w_g^a(S_g)}, \qquad \widehat{\Delta}_{\mathrm{H}} = \hat{V}_{\mathrm{H}}(p^A) - \hat{V}_{\mathrm{H}}(p^B). \tag{0.7}$$

Under full target support, each weight has expectation one, so the denominator fluctuates around M. The ratio need not be exactly unbiased; a zero denominator must also be handled. Normalization changes the estimator, not which unobserved seed sets the experiment can inform.

The two policy estimates use the same villages. Their covariance is part of the uncertainty of the contrast. To see this directly, define the signed contribution $D_g = \{w_g^A(S_g) - w_g^B(S_g)\} Y_g^{\mathrm{obs}}$. Under independent village assignments,

$$\mathrm{Var}_q(\widehat{\Delta}_{\mathrm{HT}}) = \frac{1}{M^2} \sum_g \left[\sum_{s: q_g(s) > 0} \frac{\{p_g^A(s) - p_g^B(s)\}^2 Y_g(s)^2}{q_g(s)} - \Delta_g^2 \right], \tag{0.8}$$

where $\Delta_g = \sum_s \{p_g^A(s) - p_g^B(s)\} Y_g(s)$. The difference of policy probabilities appears inside the square. Adding two separate value variances would discard the covariance that produces this expression.

For fixed, potentially heterogeneous villages, let $s_D^2 = (M-1)^{-1} \sum_g (D_g - \overline{D})^2$ with $\overline{D} = M^{-1} \sum_g D_g$. Independence gives

$$\mathbb{E}_q \left[\frac{s_D^2}{M} \right] = \mathrm{Var}_q(\widehat{\Delta}_{\mathrm{HT}}) + \frac{\sum_g \{\Delta_g - \Delta_M(A,B)\}^2}{M(M-1)}.$$

Thus the familiar sample-variance calculation is conservative in expectation in this setting; between-village contrast heterogeneity contributes an extra term. It is not an exact estimate of the assignment variance, nor does conservativeness in expectation alone justify a normal confidence interval. An appropriate many-village approximation and control of influential contributions are still needed. Counting thousands of residents does not create thousands of independent village-level experiments.

Under an explicitly stated iid-village sampling framework, full target support and finite second moments give the usual ratio expansion when the denominators converge to their positive expectations. The leading Hájek contrast fluctuation instead uses $w^A(Y - V_A) - w^B(Y - V_B)$, where V_A, V_B are the corresponding population policy values. This again requires the covariance between policy estimates. It also changes the source of uncertainty by treating villages as a sample from a larger population rather than conditioning on the particular graphs.

The mixture design in Equation 0.5 bounds w_g^A by $1/\rho$ and w_g^B by $1/(1-\rho)$. A policy evaluated from an unrelated design can have much larger weights. Inspect the distribution of signed contributions, not just the average weight. Rare seed sets with large outcomes can dominate precision.

There is a subtle support distinction. Estimating both policy values requires coverage of their combined support. Estimating only their unnormalized contrast requires coverage wherever $p_g^A(s) \neq p_g^B(s)$: unobserved sets with equal probabilities cancel. For example, with $p^A = (.4, .2, .4)$, $p^B = (.2, .4, .4)$, and $q = (.5, .5, 0)$, the identifiable contrast is $.2Y(s_1) - .2Y(s_2)$, although neither complete policy value is identified. Separate Hájek normalization would divide each observable weighted mean by a quantity converging to .6 across independent copies of this three-set setup, and generally would not recover that contrast. The weaker support argument applies to the signed, unnormalized estimator, not automatically to ratios.

Finally, evaluation of a fixed policy is not the same problem as learning which policy to deploy. Selecting a rule because it has the largest estimated value and reporting that estimate as if the rule were prespecified introduces selection bias. Independent evaluation villages or a suitable policy-learning analysis are needed. Splitting residents within a village does not, by itself, remove the dependence created by diffusion.

Policy evaluation when we observe a panel

The seeding problem asks what would happen under a different distribution over actions, using a known experiment to connect observed outcomes to that policy. Now change the information available. A city introduces a policy, a firm changes a practice, or a school adopts a program. We observe outcomes before and after adoption and observe other units over the same periods, but we do not necessarily know a credible randomization distribution for adoption. Can repeated observations supply the missing comparison?

A panel is a collection of units observed over time. Index units by $i = 1, \ldots, N$ and periods by $t = 1, \ldots, T$. Let $Z_{it} \in \{0, 1\}$ indicate treatment, and let Y_{it}^{obs} be the observed outcome. Initially, assume that the outcome depends on the unit's current treatment rather than its past treatment history or other units' assignments. We can then write potential outcomes $Y_{it}(0)$ and $Y_{it}(1)$, with $Y_{it}^{\text{obs}} = Y_{it}(Z_{it})$. We will return to treatment histories and spillovers before applying this formulation.

The object of interest is not necessarily the effect on all units in every period. Define the sets of untreated and treated cells by

$$\mathcal{O} = \{(i, t) : Z_{it} = 0\}, \qquad \mathcal{T} = \{(i, t) : Z_{it} = 1\}.$$

For a nonempty $\mathcal{T}$, consider the average effect on the units and periods actually treated:

$$\tau_{\mathcal{T}} = \frac{1}{|\mathcal{T}|} \sum_{(i,t) \in \mathcal{T}} \{Y_{it}(1) - Y_{it}(0)\}. \tag{0.9}$$

Each treated cell receives equal weight; an average over units, cohorts, or event times would be a different target and should be defined accordingly. All the $Y_{it}(1)$ appearing here are observed. The problem is to recover the untreated outcomes for these same cells.

This gives a useful rearrangement of the data. Form an N-by-T matrix of *untreated potential outcomes*. An untreated cell is observed, while a treated cell is missing. The observed treated outcome is not an observation of that missing entry: it belongs to a different potential-outcome matrix. The causal question becomes a matrix-completion problem, but changing its description does not supply an identification assumption.

What a before-and-after comparison needs

Suppose one unit adopts between periods 2 and 3. Its change in outcome mixes the policy's effect with anything that would have changed without the policy. Difference-in-differences uses the change among untreated comparison units to account for the latter.

Let $\overline{Y}_{\mathrm{tr},t}$ and $\overline{Y}_{\mathrm{co},t}$ denote the observed averages for the treated and control groups in period t. A two-period DID estimator is

$$\hat{\tau}_{\mathrm{DID}} = (\overline{Y}_{\mathrm{tr},3} - \overline{Y}_{\mathrm{tr},2}) - (\overline{Y}_{\mathrm{co},3} - \overline{Y}_{\mathrm{co},2}). \tag{0.10}$$

The identifying comparison concerns what would have happened *without* treatment: the mean untreated change in the treated group must equal the mean untreated change in the controls. Equality of their outcome values is not required. Conversely, similar pretreatment values alone do not imply that their untreated changes would agree.

Why might DID fail while the rest of the panel still be useful? Different units may respond differently to common changes: a national demand shift may help one industry and hurt another. The following deliberately noiseless example isolates that issue. There are two never-treated controls and one unit treated only in the final period:

Unit	Period 1	Period 2	Period 3: untreated outcome
Control A	1	2	3
Control B	2	1	0
Treated unit	4	5	?

Suppose the treated unit's untreated trajectory is twice A's trajectory plus B's trajectory. The first two entries satisfy this relation: $4 = 2(1) + 2$ and $5 = 2(2) + 1$. If the same relation continues, the missing entry is $2(3) + 0 = 6$. If the observed treated outcome is 8, the treatment effect is 2. Equal-weight controls instead average 1.5 in every period. DID therefore predicts no untreated change for the treated unit, imputes 5, and estimates an effect of 3. The problem is not a shortage of time observations in principle: it is the assumption that the unweighted control change is the right comparison.

The relation used to obtain 6 needs justification. It is not guaranteed because it fits two observations. We next express precisely what it means to assume that a small collection of stable patterns governs the panel.

Low rank means shared patterns with different responses

Imagine that untreated outcomes reflect a small number of common time patterns. One might represent national demand, another the relative cost of energy. Units have different sensitivities to these patterns. Let $\mathbf{u}_i \in \mathbb{R}^r$ collect unit i's sensitivities, called *loadings*, and let $\mathbf{f}_t \in \mathbb{R}^r$ collect the common factors in period t. The interaction component of the mean is

$$L_{it} = \mathbf{u}_i^\top \mathbf{f}_t = \sum_{k=1}^{r} u_{ik} f_{tk}. \tag{0.11}$$

The loadings do not receive a time index: their stability is part of the model. Factors may vary over time without following a linear or smooth trend. A single factor can increase, fall, and increase again, while units respond with different magnitudes and even different signs.

Stack the row vectors $\mathbf{u}_i^\top$ into an N-by-r matrix $\mathbf{U}$ and $\mathbf{f}_t^\top$ into a T-by-r matrix $\mathbf{F}$. Then

$$\mathbf{L} = \mathbf{U}\mathbf{F}^\top, \qquad \operatorname{rank}(\mathbf{L}) \le r. \tag{0.12}$$

Every row of $\mathbf{L}$ is a linear combination of the same r time patterns, and every column is a linear combination of the same r loading patterns across units. Low rank is therefore a restriction in both directions, not merely a claim that outcomes are highly correlated.

The factors need not have unique scientific names. Replacing $\mathbf{U}$ by $\mathbf{U}\mathbf{H}$ and $\mathbf{F}$ by $\mathbf{F}\mathbf{H}^{-\top}$, for an invertible r-by-r matrix $\mathbf{H}$, leaves $\mathbf{L}$ unchanged. Identifying a counterfactual mean can be possible without identifying a uniquely labeled set of latent mechanisms.

For policy evaluation, separate persistent unit differences and common time changes from the interaction component. A working model is

$$Y_{it}(0) = \underbrace{\alpha_i + \delta_t + L_{it}}_{M_{it}^0} + \varepsilon_{it}, \qquad \mathbf{L} = \mathbf{U}\mathbf{F}^\top. \tag{0.13}$$

Here M_{it}^0 is the systematic untreated mean and ε_{it} is a residual. The additive terms allow, for example, one city to have a persistently higher outcome while all cities experience a common period shock. The interaction terms allow the same shock to have different effects across cities. This is the interactive-fixed-effects perspective developed in the panel literature (J. Bai 2009).

Three distinctions matter. First, the *mean* is modeled as low dimensional; independent measurement noise will generally make the realized outcome matrix full rank. Second, the rank bound r applies to $\mathbf{L}$. The full mean matrix $\mathbf{M}^0$ has rank at most $r + 2$, since each additive component contributes at most one. Third, this is now an outcome-model argument. Without a randomized adoption design, the model and the way treated cells differ from untreated cells must do work that randomization previously did for us.

Here a mean refers to an expectation over residual variation, holding the systematic components fixed. Mean-zero residuals before conditioning on treatment need not remain mean zero among treated cells. A causal analysis needs restrictions on that selection as well as on the rank of the signal; we will make their role explicit in the estimator's error decomposition.

Which missing entries are determined?

Return to the noiseless example, now imposing rank two directly on its untreated mean matrix. The first two control rows are independent. Their first two coordinates determine the coefficients of the treated row:

$$a + 2b = 4, \qquad 2a + b = 5 \quad \Longrightarrow \quad a = 2, \quad b = 1.$$

Rank two says the third row stays in their span, which determines the final entry as 6. Both the structural restriction and the observed arrangement are essential.

For a general noiseless matrix, reorder rows into controls followed by treated units and columns into pre- and post-treatment periods. Write

$$\mathbf{M}^0 = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}. \tag{0.14}$$

We observe control outcomes before and after treatment, $\mathbf{A}$ and $\mathbf{B}$, and treated-unit outcomes before treatment, $\mathbf{C}$. The block $\mathbf{D}$ is missing.

Proposition 0.0.1 (Completing a noiseless block)**.** *If* $\operatorname{rank}(\mathbf{M}^0) = \operatorname{rank}(\mathbf{A}) = r_0$*, then*

$$\mathbf{D} = \mathbf{C}\mathbf{A}^\dagger\mathbf{B}, \tag{0.15}$$

where $\mathbf{A}^\dagger$ *is the Moore–Penrose pseudoinverse.*

Proof. Stack the two observed pretreatment blocks vertically to form $\mathbf{K} = \begin{pmatrix} \mathbf{A} \\ \mathbf{C} \end{pmatrix}$. Its columns have rank r_0 and span the columns of the complete matrix. Thus the post-treatment columns can be written $\begin{pmatrix} \mathbf{B} \\ \mathbf{D} \end{pmatrix} = \mathbf{K}\mathbf{H}$ for some coefficient matrix $\mathbf{H}$. Reading its upper and lower blocks gives $\mathbf{B} = \mathbf{A}\mathbf{H}$ and $\mathbf{D} = \mathbf{C}\mathbf{H}$. Also, $\operatorname{rank}(\mathbf{K}) = \operatorname{rank}(\mathbf{A})$ implies that each row of $\mathbf{C}$ lies in the row space of $\mathbf{A}$. Thus $\mathbf{C} = \mathbf{C}\mathbf{A}^\dagger\mathbf{A}$. Substituting,

$$\mathbf{C}\mathbf{A}^\dagger\mathbf{B} = \mathbf{C}\mathbf{A}^\dagger\mathbf{A}\mathbf{H} = \mathbf{C}\mathbf{H} = \mathbf{D}.$$

□

The rank condition says that the controls in the pretreatment periods reveal all the directions needed elsewhere in the matrix. It is a form of overlap in latent structure. A policy-responsive group cannot have an additional untreated pattern that is completely invisible in the observed comparison data.

Low rank alone does not supply that overlap. Suppose the three observed entries of a two-by-two matrix are zero and only its bottom-right entry is missing:

$$\begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix}.$$

Every value of d is compatible with rank at most one; restricting rank to exactly one still allows every nonzero d. Nothing observed tells us how large the missing entry is. Even when completion is unique, it may be unstable: for a rank-one matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $a \neq 0$, the missing value is $d = bc/a$. A small error in a nearly zero a can produce a large prediction error.

Estimating a mean matrix from noisy untreated cells

In a realistic panel, exact algebraic completion is too brittle. Untreated observations contain noise, rank is unknown, and adoption may create a staircase of missing cells rather than a single rectangle. We need a method that rewards fitting untreated observations while discouraging elaborate patterns supported only by noise.

Without a complexity restriction, minimizing squared errors over $\mathcal{O}$ reproduces every observed untreated value and leaves every missing entry arbitrary. A rank constraint would connect these entries, but choosing a rank and solving the resulting factorization problem can be difficult. The nuclear norm provides a convex alternative.

Let $\sigma_k(\mathbf{L})$ be the singular values of a matrix, the nonnegative strengths of its orthogonal row-column patterns. Its rank counts nonzero singular values; its *nuclear norm* adds their magnitudes:

$$\|\mathbf{L}\|_* = \sum_{k=1}^{\min(N,T)} \sigma_k(\mathbf{L}). \tag{0.16}$$

Penalizing this sum encourages weak patterns to vanish and strong patterns to shrink. It is not a penalty on the absolute values of individual cells. Changing a missing cell can change the singular values and therefore change the penalty even though that cell has no observed-data loss.

A matrix-completion estimator developed for causal panels by Athey et al. (2021) solves the following optimization problem. Write $\alpha = (\alpha_1, \dots, \alpha_N)$ and $\delta = (\delta_1, \dots, \delta_T)$ for the vectors of additive effects:

$$(\hat{\alpha}, \hat{\delta}, \widehat{\mathbf{L}}) \in \arg\min_{\alpha,\delta,\mathbf{L}} \left[\frac{1}{2|\mathcal{O}|} \sum_{(i,t)\in\mathcal{O}} \{Y_{it}^{\text{obs}} - \alpha_i - \delta_t - L_{it}\}^2 + \lambda \|\mathbf{L}\|_* \right], \tag{0.17}$$

where $\lambda \geq 0$ controls regularization. The factor of two merely sets the scale of λ. The prediction is $\widehat{M}_{it}^0 = \hat{\alpha}_i + \hat{\delta}_t + \widehat{L}_{it}$. An additive normalization, such as $\sum_i \alpha_i = 0$, can fix the redundancy between unit and time intercepts; the fitted sum is the object we need.

The additive effects are unpenalized so that a large persistent difference between units need not be shrunk merely to achieve a simpler interaction matrix. Crucially, the summation uses only untreated cells. Including treated outcomes in this same loss would invite the model to explain the policy's effect as part of the untreated signal. Doing that legitimately would require additional restrictions on the treatment-effect component.

What the penalty does computationally

The complete-data analogue makes the mechanism transparent. If a matrix $\mathbf{R}$ is fully observed and additive effects have been removed, consider

$$\min_{\mathbf{L}} \frac{1}{2}\|\mathbf{R} - \mathbf{L}\|_F^2 + \kappa\|\mathbf{L}\|_*, \tag{0.18}$$

where $\|\cdot\|_F^2$ adds squared entries and $\kappa \geq 0$. Write a singular-value decomposition $\mathbf{R} = \mathbf{P}\,\text{diag}(s_k)\mathbf{Q}^\top$. The optimizer retains these singular-vector directions and replaces each s_k by $(s_k - \kappa)_+ = \max(s_k - \kappa, 0)$. Along one such direction, the remaining optimization is simply $\min_{\ell\geq 0} \frac{1}{2}(s_k - \ell)^2 + \kappa\ell$: an

interior solution has $\ell = s_k - \kappa$, and otherwise the minimum is at zero. Thus weak directions disappear instead of being fitted exactly.

With missing entries we cannot take one decomposition of a complete observed matrix. Soft-impute algorithms repeatedly fill missing entries with current predictions and shrink the resulting singular values, updating until convergence (Mazumder, Hastie, and Tibshirani 2010). Missing entries are working predictions, not new observations. A one-time replacement of missing values by zero followed by an ordinary SVD does not generally solve the masked fitting problem.

Turning predictions into an effect estimate

Average the differences between observed treated outcomes and predicted untreated means:

$$\hat{\tau}_{\text{MC}} = \frac{1}{|\mathcal{T}|} \sum_{(i,t)\in\mathcal{T}} \{Y_{it}^{\text{obs}} - \widehat{M}_{it}^{0}\}. \tag{0.19}$$

This formula is simple, but its error exposes two different tasks. Under Equation 0.13,

$$\hat{\tau}_{\text{MC}} - \tau_{\mathcal{T}} = \frac{1}{|\mathcal{T}|} \sum_{(i,t)\in\mathcal{T}} (M_{it}^{0} - \widehat{M}_{it}^{0}) + \frac{1}{|\mathcal{T}|} \sum_{(i,t)\in\mathcal{T}} \varepsilon_{it}. \tag{0.20}$$

The first term is error in estimating untreated means. The second is the average untreated residual for the treated cells. Even knowing the mean matrix exactly would not reveal each unobserved realization of $Y_{it}(0)$. To estimate the realized finite-panel effect precisely, we need that residual average to be small or an uncertainty analysis that accounts for it.

A condition such as mean-zero residuals given the signal and the treatment pattern helps prevent selection on unexplained shocks; suitable independence or weak dependence then makes averaging informative. If cities adopt precisely when their unmodeled outcome shocks worsen, the residual average need not be zero. Better matrix fitting cannot repair that selection automatically. Alternatively, an analyst might target an average effect in conditional expectation under a stated model. That is a legitimate change of estimand, but not a reason to equate a predicted mean with an individual missing potential outcome.

Choosing complexity and checking the comparison

Larger λ more strongly favors fewer and smaller interaction patterns; too much regularization misses real heterogeneity, while too little can fit noise. Choose it using untreated outcomes only. A basic cross-validation procedure withholds some untreated cells, fits on the rest, and compares predictions with the withheld values over a grid of penalties. Refit the chosen model using all available untreated cells.

The arrangement of holdouts matters. Predicting randomly scattered entries with neighbors observed on both sides is easier than predicting a long post-adoption block. In addition to ordinary masking, examine pseudo-adoption dates or withheld time blocks that resemble the intended extrapolation, while leaving enough controls and periods to estimate the model. This is a diagnostic design choice, not a theorem that cross-validation identifies causal effects. If final control-period outcomes will be available for the retrospective analysis, they may legitimately inform its fit; forecasting before those periods occur is a different information problem.

Inspect sensitivity to penalty strength, donor units, the pretreatment window, and the treatment date. Large changes reveal dependence on particular extrapolations. Good held-out predictions support the model in regions where outcomes can be checked; they do not test whether a new unobserved shock affects only the treated cells.

Other ways to construct the missing comparison

Low rank connects two familiar approaches. A *synthetic control* predicts a treated unit's counterfactual from other units' contemporaneous outcomes. For a single treated unit $i_\star$, choose donor weights using pretreatment data, then predict $\hat{Y}_{i_\star t}(0) = \sum_{j \in \text{co}} w_j Y_{jt}^{\text{obs}}$ in post-treatment periods. The canonical construction uses nonnegative weights summing to one (Abadie, Diamond, and Hainmueller 2010). Under a factor model, a successful comparison reproduces the treated unit's relevant loadings. The restrictions also limit extrapolation: the toy example's coefficients $(2, 1)$ are not convex weights, so its algebraic completion is not a canonical synthetic control.

A *horizontal prediction* instead learns, among controls, how later outcomes relate to earlier outcomes and applies that relation to the treated units' earlier outcomes. These are different uses of the panel: synthetic control learns combinations across units that continue over time, while horizontal prediction learns combinations across times that continue across units. Matrix completion estimates a shared representation that links both directions.

Weighted differences rather than an entire completed matrix

Sometimes the target is one average effect, not every missing entry. Synthetic difference-in-differences combines unit and time weighting with the removal of additive differences (Arkhangelsky et al. 2021). Consider a common adoption date, control units co, treated units tr, pretreatment periods pre, and post-treatment periods post. Define the treated mean in period t as $\overline{Y}_{\text{tr},t}$, and choose control weights w_j and pretreatment time weights v_t, each nonnegative and summing to one.

For fixed such weights, the weighted DID contrast is

$$\begin{aligned} \hat{\tau}(w, v) = \frac{1}{T_{\text{post}}} \sum_{t \in \text{post}} \left(\overline{Y}_{\text{tr},t} - \sum_{j \in \text{co}} w_j Y_{jt}^{\text{obs}} \right) \\ - \sum_{t \in \text{pre}} v_t \left(\overline{Y}_{\text{tr},t} - \sum_{j \in \text{co}} w_j Y_{jt}^{\text{obs}} \right), \end{aligned} \tag{0.21}$$

where T_{post} is the number of post-treatment periods. It compares the treated-control difference after treatment with a weighted version of that same difference before treatment.

To see the benefit, hold weights fixed and ignore noise. Under the factor model, the remaining untreated-mean discrepancy is

$$\underbrace{\left(\overline{\mathbf{u}}_{\text{tr}} - \sum_{j \in \text{co}} w_j \mathbf{u}_j \right)^{\top}}_{\text{difference in factor loadings}} \quad \underbrace{\left(\bar{\mathbf{f}}_{\text{post}} - \sum_{t \in \text{pre}} v_t \mathbf{f}_t \right)}_{\text{difference in factor averages}}. \tag{0.22}$$

Here bars denote arithmetic averages over the indicated group or periods. Unit intercepts cancel when taking the weighted time difference; common time effects cancel in the unit comparison. The interaction discrepancy is a product. Matching loadings well or matching the relevant time-factor average well can reduce it; neither requires explicitly interpreting each latent factor.

The actual method learns weights from observed outcomes. Unit weights match treated and weighted-control pretreatment trajectories up to a constant, with regularization to discourage concentration. Time weights make a weighted pretreatment average resemble post-treatment averages among controls, again allowing a constant. Neither fit uses treated post-treatment outcomes as an untreated target. The learned weights then enter a weighted two-way fixed-effects regression, equivalently Equation 0.21.

The product in Equation 0.22 explains a mechanism, not an unconditional guarantee for data-fitted weights. Good pretreatment fit must generalize into the missing block; noisy adaptive weights and residual dependence require additional analysis. Nor are these weights assignment probabilities like those in the seeding experiment. They construct comparisons from outcomes under structural assumptions.

What must be checked before calling the result causal?

Panel structure supplies repeated measurements, not randomized adoption. The scientific justification should specify which untreated comparison can travel across units and time, and why treatment does not select residual changes left out of that comparison. The recovery assumptions also need enough untreated information in relevant periods and enough pretreatment information for relevant units. If everyone adopts and no post-adoption controls remain, unrestricted future factors cannot generally be learned from these data alone.

The indexing of potential outcomes must fit the policy. A treatment may have effects that accumulate, persist after withdrawal, or begin in anticipation. With irreversible adoption, one can instead index outcomes by adoption date g and compare $Y_{it}(g)$ with the never-adopted outcome $Y_{it}(\infty)$. No anticipation allows pre-adoption observations to inform the latter. This formulation permits duration-dependent treatment effects; it does not make a current-treatment-only model correct. Staggered adoption then produces different missing blocks for different cohorts. Specify the cohort-period effects being averaged, and do not use already-treated observations as untreated controls without additional assumptions.

The units must also be chosen with spillovers in mind. If a city's policy changes outcomes in neighboring control cities, their observed outcomes need not be observations of the counterfactual world being imputed. The interference issues of L05 have not disappeared; a useful panel method must either rule them out, model them, or redefine the comparison.

Finally, prediction and inference are separate achievements. A small matrix-reconstruction error does not by itself produce a confidence interval for a particular treated-cell average. A small error averaged over the whole panel can even hide a large error in a small target block. Uncertainty can include counterfactual prediction error, regularization bias, the unobserved residual average, and estimation of weights or factors. Resampling cells independently ignores serial dependence within units. Unit-level resampling is more appropriate under independent-unit models, but few treated units and dependence between units create further limitations. Placebo comparisons are informative

diagnostics; they are exact randomization tests only when a suitable assignment mechanism actually justifies those permutations.

For contrast, Athey and Imbens (2022) study panels from an explicitly design-based perspective in which adoption dates are randomly assigned. That returns uncertainty to the randomization distribution and provides a different justification from assuming a low-rank untreated mean. One should say which argument is being used rather than blending them.

Synthesis

Both halves of this lecture begin by specifying an intervention and its missing comparison. In seeding experiments, the policy is a distribution over seed sets, and known randomization can identify its value through reweighting. In a nonrandomized panel, repeated observations can reveal shared patterns from which to construct an untreated comparison, but only if those patterns remain informative where the counterfactuals are missing.

The common question is not whether an estimator has an appealing name. It is what connects the observed outcomes to the proposed intervention. Support in the experiment and informative factor variation in the panel both limit extrapolation, although they are different mathematical conditions. A useful analysis makes that connection explicit, then separates estimating the comparison from quantifying its uncertainty.

Further reading

Chin, Eckles, and Ugander (2022) develop the seeding-policy evaluation framework. The seed-set calculations above connect it to the change-of-distribution argument behind importance sampling and to Rao–Blackwellization. Their applications also illustrate why the relative performance of targeting rules is an empirical question, not a conclusion that follows from centrality alone.

Athey et al. (2021) connect causal-panel imputation to matrix completion and to prediction across units and across time. J. Bai (2009) supplies earlier foundations for panels with interactive fixed effects, while Mazumder, Hastie, and Tibshirani (2010) develops the spectral regularization algorithms.

Abadie, Diamond, and Hainmueller (2010) develop synthetic controls for comparative case studies. Arkhangelsky et al. (2021) combine unit and time weighting with difference-in-differences. Their inferential analysis, and Chernozhukov et al. (2023) on inference after low-rank estimation, show why uncertainty requires more than an accurate fitted matrix.

Athey and Imbens (2022) offer the contrasting design-based treatment of staggered adoption. Their separation of assignment assumptions from restrictions on anticipation and treatment history is useful even when one ultimately chooses a model-based analysis.

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L08: Advanced Estimation Topics

Doubly robust estimation, orthogonalization, and covariate weighting

Lecture brief: View the source `brief.md` · How these notes are made

> **Note**
>
> Lecture notes for this session have not yet been generated.

L09: AI for Causal Inference

A structured survey of a rapidly changing literature

Lecture brief: View the source `brief.md` · How these notes are made

> **Note**
>
> Lecture notes will be generated after a current literature search.

Notation

The conventions below are maintained in the course-wide `NOTATION.md`. They are extended as later lectures establish new notation.

Course notation

This file records notation intended to remain stable across lectures. Before introducing notation in a lecture, check the conventions here. Add notation when it becomes course-wide; leave symbols used only in a local derivation or example in that lecture.

Units, samples, and populations

- N is the number of units in the finite experimental population, indexed by $i = 1, \dots, N$.
- n is the sample size in explicitly stated i.i.d. superpopulation or observational-sampling arguments. Do not silently replace the finite experimental population indexed by N with an i.i.d. sample indexed by n.
- X_i is the vector of pretreatment covariates for unit i; $\mathbf{X}$ collects these vectors.
- A target population beyond the experimental units is described explicitly. Do not use "ATE" without making its target population clear.

Treatments, assignments, and exposures

- $Z_i \in \{0, 1\}$ is unit i's treatment assignment, with 1 denoting treatment and 0 control. The assignment vector is $\mathbf{Z} = (Z_1, \dots, Z_N)$.
- $N_1 = \sum_{i=1}^{N} Z_i$ and $N_0 = N - N_1$ are the realized treatment-arm sizes.
- $\mathcal{Z}$ is the support of the experimental design, the set of assignment vectors with positive probability.
- When units are partitioned into H design strata or blocks, $h = 1, \dots, H$ indexes the strata, N_h is the stratum size, and N_{1h} and N_{0h} are its treatment-arm sizes.

Potential outcomes and observed outcomes

- $Y_i(z)$ is the potential outcome for unit i under treatment level $z \in \{0, 1\}$.
- $Y_i^{\text{obs}} = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$ is the observed outcome when SUTVA and consistency hold.
- $\tau_i = Y_i(1) - Y_i(0)$ is unit i's treatment effect.

Estimands and estimators

- $\overline{Y}(z) = N^{-1} \sum_{i=1}^{N} Y_i(z)$ is the finite-population mean potential outcome under treatment level z.
- $\tau = N^{-1} \sum_{i=1}^{N} \tau_i = \overline{Y}(1) - \overline{Y}(0)$ is the finite-population average treatment effect. Qualify population and conditional versions with a subscript or argument, such as τ_P, $\tau(x)$, or τ_g.
- In a stated superpopulation, $\tau_P = \mathbb{E}\{Y(1) - Y(0)\}$ denotes the population ATE. For a measurable covariate subset A, $\tau_A = \mathbb{E}\{Y(1) - Y(0) \mid X \in A\}$ is the ATE in the restricted target population. For a nonnegative tilting function h, $\tau_h = \mathbb{E}[h(X)\{Y(1) - Y(0)\}]/\mathbb{E}\{h(X)\}$ is the ATE in the corresponding tilted target population.
- A hat denotes an estimator, a random function of the assignment and observed outcomes. Its value at the realized data is an estimate.
- $\hat{\tau}_{\mathrm{DM}}$ denotes the treated-minus-control difference in observed means.
- $\hat{\mu}_{z,\mathrm{HT}}$ and $\hat{\mu}_{z,\mathrm{H}}$ denote, respectively, the Horvitz–Thompson and Hájek estimators of $\overline{Y}(z)$. The corresponding treatment-effect estimators are $\hat{\tau}_{\mathrm{HT}} = \hat{\mu}_{1,\mathrm{HT}} - \hat{\mu}_{0,\mathrm{HT}}$ and $\hat{\tau}_{\mathrm{H}} = \hat{\mu}_{1,\mathrm{H}} - \hat{\mu}_{0,\mathrm{H}}$.
- $\hat{\mu}_{\lambda}$ denotes a member of the Trotter–Tukey affine normalization family, with Horvitz–Thompson at $\lambda = 0$ and Hájek at $\lambda = 1$. $\hat{\mu}_{\mathrm{AN}}$ denotes the adaptively normalized fixed-point estimator.
- $\overline{X} = N^{-1} \sum_{i=1}^{N} X_i$ is the finite-population covariate mean. Center covariates at $\overline{X}$ when writing regression-adjusted treatment effects.
- $\hat{\tau}_X = \overline{X}_1^{\mathrm{obs}} - \overline{X}_0^{\mathrm{obs}}$ denotes realized covariate mean imbalance.
- In superpopulation arguments, $m_z(x) = \mathbb{E}\{Y(z) \mid X = x\}$ denotes the conditional potential-outcome mean for arm z. In design-based arguments, the same symbol may denote a working outcome prediction; distinguish the population conditional mean, a function fixed before assignment, and a function fitted using realized outcomes.
- $\hat{\mu}_{z,\mathrm{AIPW}}$ denotes the prediction-plus-IPW-correction estimator of $\overline{Y}(z)$, and $\hat{\tau}_{\mathrm{AIPW}} = \hat{\mu}_{1,\mathrm{AIPW}} - \hat{\mu}_{0,\mathrm{AIPW}}$.

Assignment and sampling probabilities

- $\mathbb{P}_Z$, $\mathbb{E}_Z$, Var_Z, and Cov_Z refer to probability under the assignment design, holding the finite population and all potential outcomes fixed.
- $\pi_i = \mathbb{P}_Z(Z_i = 1)$ is unit i's marginal treatment probability.
- When both treatment states appear symmetrically, use $\pi_i(z) = \mathbb{P}_Z(Z_i = z)$, so that $\pi_i(1) = \pi_i$ and $\pi_i(0) = 1 - \pi_i$.
- Use p for a common Bernoulli assignment probability.
- In observational or superpopulation arguments, $e(x) = \mathbb{P}(Z = 1 \mid X = x)$ is the propensity score. Introduce the change of framework explicitly; do not use $e(x)$ as though it were the known finite-population design probability π_i.

Finite-population and superpopulation operators

- For $z \in \{0, 1\}$, $S_z^2 = (N-1)^{-1} \sum_{i=1}^{N} \{Y_i(z) - \overline{Y}(z)\}^2$ is the finite-population variance of potential outcomes.
- $S_\tau^2 = (N-1)^{-1} \sum_{i=1}^{N} (\tau_i - \tau)^2$ is the finite-population variance of unit-level treatment effects.

- Within stratum h, use τ_h, S^2_{zh}, and $S^2_{\tau h}$ for the stratum ATE and the corresponding finite-population variances.
- s^2_z is the usual sample variance of observed outcomes within realized arm z.
- Unsubscripted probabilistic expectations such as $\mathbb{E}$ refer to a stated superpopulation or statistical model, not to the assignment design.

Networks and interference

- Under interference, $Y_i(\mathbf{z})$ is unit i's potential outcome under the full assignment vector $\mathbf{z} \in \{0,1\}^N$, and $Y_i^{\text{obs}} = Y_i(\mathbf{Z})$. The two-outcome notation $Y_i(z)$ applies when other units' assignments do not affect unit i.
- $\mathcal{N}_i$ is the known interference neighborhood of unit i, including i unless explicitly stated otherwise; $d_i = |\mathcal{N}_i|$. For a directed interference network, $j \in \mathcal{N}_i$ means that treatment of j may affect the outcome of i.
- $\tau_{\text{G}} = N^{-1} \sum_i \{Y_i(\mathbf{1}_N) - Y_i(\mathbf{0}_N)\}$ is the global average treatment effect (GATE), also called the total treatment effect (TTE) in the interference literature. It is an average over units even when called a total effect.
- $E_i = f_i(\mathbf{Z})$ is the exposure induced by a prespecified mapping. When the mapping is sufficient, $y_i(e)$ denotes the common potential outcome under assignments with exposure label e. When used for exposure probabilities, $\pi_i(e) = \mathbb{P}_Z(E_i = e)$ and $\pi_{ij}(e, e') = \mathbb{P}_Z(E_i = e, E_j = e')$; the argument denotes an exposure label rather than a unit's own treatment state.
- $\mathcal{C} = \{C_1, \ldots, C_K\}$ is a partition into clusters, $c(i)$ is the index of unit i's cluster, and W_k is cluster k's treatment. Cluster-constant assignment means $Z_i = W_{c(i)}$. $k_i(\mathcal{C})$ counts the clusters intersecting $\mathcal{N}_i$.
- A low-order outcome model has maximum interaction order β and subset collection $\mathcal{S}_i^\beta = \{S \subseteq \mathcal{N}_i : |S| \le \beta\}$, including the empty set. The feature vector $\mathbf{v}_i(\mathbf{z})$ contains the monomials $\prod_{j \in S} z_j$ for these subsets, with the empty product equal to one.
- $\mathbf{c}_i$ collects the coefficients in $Y_i(\mathbf{z}) = \mathbf{v}_i(\mathbf{z})^\top \mathbf{c}_i$. $\mathbf{a}_i = \mathbf{v}_i(\mathbf{1}_N) - \mathbf{v}_i(\mathbf{0}_N)$ selects unit i's contribution to the global contrast.
- $\mathbf{M}_i = \mathbb{E}_Z[\mathbf{v}_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})^\top]$ is the design second-moment matrix, and $\mathbf{M}_i^\dagger$ is its Moore–Penrose pseudoinverse. Define any fitted or empirical matrix separately.

Stochastic policy evaluation

- In network seeding examples, M is the number of villages or independent networks, indexed by g. S_g is the selected seed set, and $Y_g(s)$ is the village-level potential outcome under seed set s.
- $p_g^a(s)$ is the seed-set probability under target policy a; $q_g(s)$ is its probability under the actual experimental design. $\mathbb{E}_q$ and Var_q hold the villages, graphs, and potential outcomes fixed and average over this design.
- $V_M(p) = M^{-1} \sum_g \sum_s p_g(s) Y_g(s)$ is a policy's value averaged over the fixed villages. $\Delta_M(A, B) = V_M(p^A) - V_M(p^B)$ is the policy-value contrast. State any alternative village or resident weights.

Panel data and matrices

- Panel units are indexed by $i = 1, \ldots, N$ and periods by $t = 1, \ldots, T$. Z_{it} denotes treatment. The notation $Y_{it}(z)$ presumes that a contemporaneous treatment indicator adequately describes the intervention; introduce treatment-history notation when anticipation or carryover matters.
- $\mathcal{O} = \{(i,t) : Z_{it} = 0\}$ is the set of observed untreated cells, and $\mathcal{T} = \{(i,t) : Z_{it} = 1\}$ is the treated set. The complete matrix of observed outcomes is not the matrix of untreated potential outcomes: entries of the latter are missing on $\mathcal{T}$.
- $\tau_{\mathcal{T}} = |\mathcal{T}|^{-1} \sum_{(i,t)\in\mathcal{T}} \{Y_{it}(1) - Y_{it}(0)\}$ averages effects over the specified treated cells.
- M^0_{it} is the systematic untreated outcome in $Y_{it}(0) = M^0_{it} + \varepsilon_{it}$. In a factor model, $M^0_{it} = \alpha_i + \delta_t + L_{it}$ with $\mathbf{L} = \mathbf{U}\mathbf{F}^\top$ and $L_{it} = \mathbf{u}_i^\top \mathbf{f}_t$. r denotes the rank bound on $\mathbf{L}$, not necessarily on the full mean matrix after adding fixed effects.
- $\|\mathbf{L}\|_* = \sum_j \sigma_j(\mathbf{L})$ is the nuclear norm, the sum of the singular values. Distinguish this penalty from a hard constraint on the number of nonzero singular values.

Probability, convergence, and norms

- $\mathbf{1}\{A\}$ is the indicator of event A. When a vector of ones is needed, write $\mathbf{1}_N$ for the length-N vector so it is not confused with the indicator notation.
- Always subscript probability, expectation, and variance when more than one source of randomness is in play.

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Problem Sets

The two course problem sets are completed individually.

Problem set	Assignment	Deadline
Problem Set 1	TBD	TBD
Problem Set 2	TBD	TBD

Projects

The three course projects are completed in groups of 2-4 students. Each project has three parts: a **proposal 1-pager**, an **in-class presentation**, and a **write-up**.

The proposal 1-pager is a one-page PDF uploaded to Canvas. It should state the project topic—including the paper selected for Project 1 or Project 2—list the group members, and give a one-paragraph summary of what the group intends to investigate.

Both proposal 1-pagers and write-ups are due at **11:59 p.m. ET** on the dates shown below.

Project	Proposal 1-pager due	In-class presentation	Write-up due
Project 1	September 11, 2026	Sep 30, 2026	Oct 7, 2026
Project 2	October 23, 2026	Nov 4, 2026	Nov 11, 2026
Project 3	November 20, 2026	Dec 9, 2026	Dec 16, 2026

Project 1

Choose a paper roughly corresponding to L01–L04, study its contribution closely, replicate its analysis, and consider possible extensions. The paper choice must be approved by the instructor before the group proceeds.

Possible papers

The following papers are starting points rather than a closed list. They span the material in L01–L04 and vary in theoretical, computational, and empirical emphasis. In the proposal 1-pager, identify a tractable result, simulation, or empirical analysis to replicate and a specific extension or sensitivity analysis to investigate.

- **Randomization inference beyond the sharp null (L01).** Caughey, Dafoe, Li, and Miratrix (2023), *Randomisation Inference beyond the Sharp Null: Bounded Null Hypotheses and Quantiles of Individual Treatment Effects*. Reproduce one of the proposed tests or bounds, then study its finite-sample behavior, computational demands, or sensitivity to the assignment design.
- **Estimated propensity scores and efficiency (L02).** Hirano, Imbens, and Ridder (2003), *Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score*. Reconstruct the paper's efficiency comparison and investigate when estimating rather than inserting a known propensity score helps in finite samples.

- **Adaptive normalization of inverse-probability weights (L02).** Khan and Ugander (2023), *Adaptive Normalization for IPW Estimation*. Replicate the comparisons among Horvitz–Thompson, Hájek, and adaptively normalized estimators, then alter the assignment design, outcome structure, or nuisance estimation method.
- **Regression adjustment in randomized experiments (L02–L03).** Lin (2013), *Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman's Critique*. Reproduce a central simulation or empirical comparison and examine the role of treatment–covariate interactions, imbalance, or unequal treatment allocation.
- **Rerandomization (L03).** Morgan and Rubin (2012), *Rerandomization to Improve Covariate Balance in Experiments*. Implement the design and reproduce its balance or precision gains, then vary the acceptance rule, balance metric, covariate dimension, or analysis procedure.
- **Algorithmic covariate balancing (L03).** Harshaw, Sävje, Spielman, and Zhang (2024), *Balancing Covariates in Randomized Experiments with the Gram–Schmidt Walk Design*. Implement a manageable part of the design and compare its balance–randomness tradeoff with complete randomization or rerandomization.
- **Limited overlap and optimal trimming (L04).** Crump, Hotz, Imbens, and Mitnik (2009), *Dealing with Limited Overlap in Estimation of Average Treatment Effects*. Reproduce the trimming rule and a key simulation, then examine how trimming changes both precision and the population whose effect is being estimated.
- **Smooth trimming and post-trimming inference (L04).** Yang and Ding (2018), *Asymptotic Inference of Causal Effects with Observational Studies Trimmed by the Estimated Propensity Scores*. Compare hard trimming with smooth weighting and study bootstrap inference or uncertainty from the design stage.
- **Heteroscedasticity-aware trimming (L04).** Khan and Ugander (2025), *Doubly Robust and Heteroscedasticity-Aware Sample Trimming for Causal Inference*. Replicate a simulation or application, then compare propensity-only and conditional-variance-aware trimming under alternative forms of heteroscedasticity.
- **Regularizing extrapolation (L04).** Arbour, Parikh, Niknam, Stuart, Rudolph, and Feller (2026), *Regularizing Extrapolation in Causal Inference*. Implement the proposed regularization and investigate the paper's bias–bias–variance tradeoff as overlap, dimension, or the outcome model changes.

Project 2

Choose a paper aligned with the material in L05–L07, study its contribution closely, replicate its analysis, and consider possible extensions. The paper choice must be approved by the instructor before the group proceeds.

Possible papers

The following papers are starting points rather than a closed list. They span interference, structural outcome models, policy evaluation, and panel data. As in Project 1, identify a tractable result, simulation, or empirical analysis to replicate and a specific extension or sensitivity analysis to investigate. If the original application uses unavailable data, a simulation or theoretical investigation can provide the basis for the project.

- **Testing interference (L05).** Athey, Eckles, and Imbens (2018), *Exact p-Values for Network Interference*. Implement one conditional randomization test and reproduce its size and power under a specified interference model, then investigate how the choice of focal units, test statistic, or network structure affects its ability to detect spillovers.
- **Randomizing graph clusters (L05).** Ugander and Yin (2023), *Randomized Graph Cluster Randomization*. Reproduce a comparison of fixed and randomized clustering, including the calculation or estimation of exposure probabilities, then investigate how graph structure, the clustering algorithm, or computational effort changes estimator variance and mean squared error.
- **Low-order interference models (L06).** Cortez-Rodriguez, Eichhorn, and Yu (2023), *Exploiting Neighborhood Interference with Low-Order Interactions under Unit Randomized Design*. Implement the low-order estimator and reproduce a simulation comparison, then study the bias–variance tradeoff when the assumed interaction order is too small or unnecessarily large.
- **Combining outcome models and experimental design (L05–L06).** Eichhorn, Khan, Ugander, and Yu (2026), *Low-order Outcomes and Clustered Designs: Combining Design and Analysis for Causal Inference under Network Interference*. Reproduce a pseudoinverse-estimator comparison across clusterings, then examine whether the proposed variance bounds select useful designs when the graph or outcome model changes.
- **Covariate adjustment under interference (L06).** Wang and Li (2026), *Covariate Adjustment Cannot Hurt: Treatment Effect Estimation under Interference with Low-Order Outcome Interactions*. Compare the proposed adjustment with unadjusted estimation and ordinary regression adjustment, then investigate finite-sample precision and interval coverage as covariate informativeness, network density, or model misspecification changes.
- **Monotonicity and attributable effects (L06).** Choi (2017), *Estimation of Monotone Treatment Effects in Network Experiments*. Implement a confidence-interval procedure for effects attributable to the realized treatment assignment, then study how its width and coverage depend on network information and the monotonicity assumption, including sensitivity to departures from that assumption.
- **Evaluating stochastic seeding policies (L07).** Chin, Eckles, and Ugander (2022), *Evaluating Stochastic Seeding Strategies in Networks*. Reproduce a comparison of seeding strategies using the proposed policy-value estimators, then vary the diffusion model, network structure, or experimental assignment rule to study precision and overlap between policies.
- **Matrix completion for causal panels (L07).** Athey, Bayati, Doudchenko, Imbens, and Khosravi (2021), *Matrix Completion Methods for Causal Panel Data Models*. Reproduce a

counterfactual-prediction comparison, then investigate how latent rank, the pattern of treatment adoption, or the choice of regularization affects treatment-effect estimates. Distinguish prediction accuracy on observed cells from accuracy on treated cells.

- **Synthetic difference-in-differences (L07).** Arkhangelsky, Athey, Hirshberg, Imbens, and Wager (2021), *Synthetic Difference-in-Differences*. Replicate a comparison with difference-in-differences and synthetic control, then investigate the role of unit and time weights as pretreatment history, latent trends, or the donor pool changes.

Project 3: Open-ended research question

Propose a research question and conduct an open-ended initial investigation of it.

Further instructions for the presentations and write-ups will be provided separately.